



Intestinal permeation enhancers for oral peptide delivery[☆]



Sam Maher^a, Randall J. Mersny^b, David J. Brayden^{c,*}

^a RCSI School of Pharmacy, Royal College of Surgeons in Ireland, St Stephen's Green, Dublin 2, Ireland

^b Department of Pharmacy and Pharmacology, University of Bath, UK

^c UCD School of Veterinary Medicine and Conway Institute, University College Dublin, Belfield, Dublin 4, Ireland

ARTICLE INFO

Article history:

Received 29 April 2016

Received in revised form 7 June 2016

Accepted 9 June 2016

Available online 16 June 2016

Keywords:

Oral peptide delivery

Intestinal permeation enhancers

Paracellular transport

Transcellular

Solid dose formulation

Surfactants

Emulsions

ABSTRACT

Intestinal permeation enhancers (PEs) are one of the most widely tested strategies to improve oral delivery of therapeutic peptides. This article assesses the intestinal permeation enhancement action of over 250 PEs that have been tested in intestinal delivery models. In depth analysis of pre-clinical data is presented for PEs as components of proprietary delivery systems that have progressed to clinical trials. Given the importance of co-presentation of sufficiently high concentrations of PE and peptide at the small intestinal epithelium, there is an emphasis on studies where PEs have been formulated with poorly permeable molecules in solid dosage forms and lipoidal dispersions.

© 2016 Elsevier B.V. All rights reserved.

Contents

1. Introduction	278
2. Therapeutic peptides	278
3. Barriers to translation of PE-based oral peptide technologies	279
4. Intestinal PEs	286
4.1. Paracellular PEs	287
4.1.1. Paracellular PEs emerging from the study of toxins	287
4.1.2. Paracellular PEs that bind claudins	288
4.1.3. Paracellular PEs that target E-cadherin and Ca ²⁺	288
4.1.4. Paracellular PEs that target occludin	289
4.1.5. Paracellular PEs and cytoskeletal reorganisation	289
4.2. Transcellular PEs	289
4.2.1. Soluble surfactant PEs	289
4.2.2. Insoluble surfactants	295
4.3. Peptide hydrophobisation	299
4.4. Non-surfactant PEs	301
4.5. Multiple modes of enhancement action	303
5. Safety and regulation of PEs	304
5.1. Transcellular enhancers and membrane perturbation	304
5.2. The bystander absorption argument	305
5.3. Are paracellular PEs safer than transcellular PEs?	306
6. PE developability classification system	307

[☆] This review is part of the *Advanced Drug Delivery Reviews* theme issue on "SI: Oral delivery of peptides".

* Corresponding author. Tel.: +353 17166013; fax: +353 17166204.

E-mail address: david.brayden@ucd.ie (D.J. Brayden).

7. Conclusions	307
Acknowledgements	307
References	307

1. Introduction

Growth in global peptide markets has spurred development of technologies that enable oral delivery of poorly permeable drugs. Initial delivery strategies focused on inclusion of candidate excipients that protected the peptide from intestinal degradation and transiently altered intestinal permeability [1]. The majority of oral peptide delivery technologies that are currently in clinical trials use formulations with established intestinal PEs that have a history of safe use in man [2]. Recent clinical data suggests that inclusion of PEs in oral formulations can safely assist absorption of selected potent peptides with a large therapeutic index. For example, primary endpoints were met in a Phase III trial of octreotide formulated in an oily suspension with a medium chain fatty acid salt, sodium caprylate (C_8) [3]. In parallel, a new generation of PEs with more specific mechanisms of action are in preclinical research, and may confer improved safety and efficacy over those currently in development. This article summarises the progress of ~250 PEs that have been tested in preclinical intestinal delivery models (Tables 1, S1). An in-depth review of pre-clinical data is presented for PEs in proprietary delivery systems that have progressed to clinical development. The review by Aguirre *et al.* (this Issue [4]) evaluates the performance of technologies in clinical trials, of which most are enteric-coated solid dosage forms containing PEs. We focus here on how PEs alter intestinal permeability and on innovations that may further assist translation of safety and efficacy outcomes from pre-clinical models to man.

2. Therapeutic peptides

A drug delivery system that facilitates oral peptide administration has long been desired. There are ~55 therapeutic peptides marketed as parenteral formulations (based on a ~9 kDa cut-off in molecular weight (MW)) (Table 2) and a further 140 in clinical development [5]. Compared to small molecules, peptides are attractive due to their specificity, potency, efficacy, and low toxicity. Clinical potential of unmodified injectable peptides can be hampered by a short plasma half-life ($t_{1/2}$) due to labile moieties and higher manufacturing costs relative to small molecules. A breakdown of marketed peptide products indicates that injection routes (61%) are the most common, followed by topical (11%), nasal (9%), oral (9%) and ophthalmic (4%), noting that bioavailability is typically low and variable from non-injectable routes [6].

Injection requirements are associated with lack of adherence to dosing regimens, hence the impetus towards long acting formulations that are administered less often. Thus, for glucagon-like-Peptide 1 (GLP-1) analogues, sub-cutaneous (s.c.) injection of exenatide has shifted from twice-a-day administration (Byetta®; Lilly, USA) to once weekly administration (e.g. Bydureon®, Lilly). This was achieved by development of a microsphere-based controlled release system [7], whereas competing approaches have attempted to improve stability and reduce recognition by the reticuloendothelial system by conjugating lipid moieties to amino acid residues or by fusing the analogue to albumen. Although needle fabrication technology has improved in the last 20 years, injections are still inconvenient in the longer term and can delay take-up and adherence to regimes necessitated by chronic diseases. In the case of type 2 diabetes (T2D), early initiation of insulin can slow the progressive destruction of pancreatic β -cells [8], but T2D patients frequently require dose adjustments related to peripheral hypoglycaemia [9]. Oral insulin may reduce such risks because it is absorbed via the portal vein and therefore imitates pancreatic secretion to the liver [10]. This

can also reduce two other side effects attributed to s.c. insulin in the periphery: weight gain and lipodystrophy [11].

An oral peptide dosage form would likely reduce costs associated with sterile manufacture of injectables, cold chain, needle disposal, and staff/patient training, but these savings would be offset against the requirement for higher doses compared to injection. A commercial driver for oral peptides is life cycle extension and increased revenue from branded medicines based around new patents. Development of oral delivery systems for approved injectable peptides has the benefit of known pharmacology for the active pharmaceutical ingredient (API), good safety profiles (at least for the injected route) and established analytical detection methods. The most clinically-advanced oral peptide formulations are being developed for diabetes (insulin, GLP-1 analogues), osteoporosis (salmon calcitonin, sCT; teriparatide (PTH 1–34)), and acromegaly (octreotide). Anti-diabetic peptides account for ~40% of peptides in commercial oral peptide delivery programmes and Table S2 details selected patents filed on oral insulin over the last 30 years. Synthesis of injectable anti-diabetic peptides with long plasma $t_{1/2}$ values is also contributing to investment in oral peptide delivery systems (e.g. $t_{1/2}$ = 160 h for the GLP-1 analogue, semaglutide, Novo-Nordisk, Denmark [12]), as they may yield better oral pharmacokinetic (PK) data than short-acting counterparts. Competition between GLP-1 analogues makes oral formulation a key battleground [5].

Development of non-injected dosage forms has had some commercial successes, including oral desmopressin (DDAVP®, Ferring, Switzerland), oral cyclosporin (Neoral®, Novartis, Switzerland) and nasal calcitonin (Miacalcin®, Novartis). The suitability of commercially available peptides for oral reformulation depends on their physico-chemical properties (MW, solubility), chemical complexity, therapeutic considerations (route/frequency of administration, therapeutic index) and cost-effectiveness. Peptides typically exhibit high aqueous solubility and low permeability, properties that unofficially place them in the Biopharmaceutics Classification system (BCS) Class III. Nevertheless, some peptides with cationic and anionic functional groups exhibit complicated pH-dependent solubility, where solubility is high in acidic conditions at pH values below their isoelectric point (pI), and is relatively low at pH values at and above their pI. Many basic molecules rely on acid/base phenomena for dissolution within the stomach and subsequent absorption across the duodenum and jejunum, so peptides with low intrinsic solubility are problematic. For example, insulin dissolves in dilute acid but not at neutral pH, which could manifest as poor dissolution in the small intestine. Peptides that have a MW >6000 Da do not have any appreciable intestinal permeability when delivered orally, this makes insulin (5808 Da) especially challenging, with difficulty decreasing in the order of teriparatide (4118 Da) > exenatide (4187 Da) > sCT (3532 Da) > octreotide (1019 Da). In addition, there is a correlation between MW and susceptibility to proteolysis [13].

An ideal oral candidate peptide should therefore have a low MW, high potency, enzymatic/chemical stability (e.g. cyclised peptides, D-substituted amino acids), a high therapeutic index and be of relatively low cost to synthesise. Desmopressin (MW 1069 Da) contains stable amino acids; it has an oral bioavailability (F) of only 0.17%, so high potency is its key attribute [14]. Prandial insulin is more challenging because it requires three relatively high mealtime doses to reach the required plasma levels per day. The s.c. insulin dose required for management of Type 1 diabetes (T1D) of 0.5–0.8 IU/kg per dose (1.2–1.9 mg); if normalised for an oral system designed for an oral F of 10%–20%, a dose level of 6–20 mg would be required. A recent oral insulin clinical study included 8 mg (240 IU) insulin three times daily [15], whereas

exenatide is injected at a dose of 10 µg and has been tested orally at 15-fold higher doses using the same technology [16].

3. Barriers to translation of PE-based oral peptide technologies

Peptides have poor oral bioavailability due to peptidase sensitivity and low intestinal permeability. They may be sensitive to gastric pepsin and acid-dependent destabilisation of disulphide bridges, hydrogen bonding and electrostatic interactions, although stomach-related breakdown can largely be overcome by enteric coating (e.g. Eudragit®, Evonik, Germany; Kollicoat® (BASF, Germany) [17]. Enteric coating excipients exhibit pH-dependent dissolution due to deprotonation of weakly acidic functional groups at high pH values. Oral peptide formulations that are enterically coated must be administered pre-prandially to avoid premature release in the stomach when buffered by food. Gastric emptying time is therefore a consideration for peptides like insulin that require absorption to coincide with ingestion of a meal. In the fasted state, capsule dosage forms are consistently found in the small intestine 1 h post administration [18]. The lag time between dose and food intake is an important therapeutic consideration for peptides that require prandial administration (e.g. insulin), but less so for peptides like exenatide and octreotide. Requirement for pre-prandial administration also raises concerns around adherence, when the dosage form must be administered in complex regimes over an indefinite period. Application of Eudragit® coatings without inclusion of excipients that address peptide degradation and poor permeability ultimately will not increase oral F [19,20].

Upon leaving the stomach the peptide is vulnerable to proteolytic degradation in the lumen, brush border membrane, and in the cytosol of small intestinal enterocytes. The pancreas can produce over 40 g of proteolytic enzymes [21] delivered in 2.5 L of pancreatic juice per day [22]. Large linear peptides including insulin, sCT, glucagon and secretin are sensitive to human intestinal fluid (HIF), while higher stability is noted for short and structurally-confined peptides with stable bonds (e.g. octreotide, cyclosporin and desmopressin) [13]. PEs can also have a dual benefit in inhibiting regional proteolysis, examples being sodium glycocholate [23] and ethylenediaminetetraacetic acid (EDTA) [24]. However, any PE that is a peptide may itself be sensitive to proteolysis, examples being *zonula occludens* toxin (ZoT) and the C-terminal fragment of *Clostridium perfringens* enterotoxin (C-CPE).

Many PEs are surfactants, so it is possible to protect the peptide in lipoidal dispersions including microemulsions (e.g. Macrolin™, Provalis, UK) (Section 4.2.2.3). A leading PE-based technology appears to offer peptidase inhibition by non-covalent complexation of the peptide with a carrier (e.g. sodium salcaprozate, Eligen®, Emisphere, USA) (Case 14: Eligen®). Inclusion of peptidase inhibitors like aprotinin can improve oral peptide delivery, however established excipients with similar properties are less risky in terms of toxicology. Acidifying organic acids including citric acid (CA) and tartaric acid lower the optimal pH for proteolysis and can benefit oral peptide formulation, since if they reach a pH of 1–2 units below the isoelectric point, they can improve solubility (e.g. insulin). If however, the pH remains at the isoelectric point for the peptide, the solubility of the peptide will be low, and it may be sensitive to secreted bicarbonate. Acidifiers can also interfere in the dissolution and enhancement action of anionic PEs, some of which exhibit low dissolution and poor enhancement action at pH values below their pKa. For example, the pKa of another lead PE, sodium caprate (C₁₀) (the sodium salt of the medium chain fatty acid, capric acid) is 6.5; should an acidifier decrease pH to 5.5, over 90% of the molecule will exist as an insoluble oil. Further, the hydrophilic-lipophilic balance (HLB) of capric acid is 4.8, which is lower than C₁₀ (HLB: 21.8) and ultimately well below the HLB considered optimal for permeation enhancement [2]. The reduction in luminal pH by co-encapsulated acidifiers can also decrease the dissolution of enteric-coated dosage forms, and efforts to overcome this include separate coating of granules prior to tableting to prevent such interactions. In addition, the cationic charge

imparted on many therapeutic peptides (e.g. sCT, pI = 10.1) in acidified conditions can increase entrapment in mucus by electrostatic complexation. Finally, some acidifiers chelate metals, which can reduce proteolysis due to removal of peptidase co-factors. Ca²⁺ is also an important component in epithelial tight junction (TJ) formation, and some studies suggested that CA can also increase intestinal permeability via chelation (Table 1), although this hypothesis was challenged recently in an *in vitro* insulin permeability study in rat tissue mucosae where the data suggested that the main role of CA in oral peptide formulations is to reduce peptidase activity [25].

Protease inhibitors in oral peptide dosage forms include soybean trypsin inhibitor (SBTI) [26], aprotinin [23], ovomucoids [27], EDTA [24], sodium glycocholate [28] and camostat mesilate [23]. Some improved absorption of peptides to an extent [23,28]; a combination of an enteric coating with aprotinin significantly improved oral peptide bioavailability in rats [20]. A safety argument against the use of inhibitors is that impaired dietary protein digestion may occur. However, inhibitors may provide localised protection where the dosage form dissolves and not throughout the GI tract. Nevertheless, agents like aprotinin target ubiquitous biological functions, which raises concerns regarding suitability for oral peptides. SBTI was included in oral exenatide formulations at concentrations as high as 125 mg/dose in clinical trials [26]. Despite the GRAS status of soy protein isolate [29], purified SBTI can cause pancreatic hyperplasia and carcinoma in rats [30,31] and systemic absorption is undesirable. To this end, retention in the intestine has focused on conjugation to non-absorbed polymers (e.g. chitosan-*aprotinin* [32], chitosan-EDTA [33]). It is noteworthy that pancreatic peptidases are responsible for only 20% of the degradation of ingested proteins, with the brush border enzymes accounting for the majority [34]. Therefore, inhibitors need also to access the brush border for optimum efficacy. One example was the protection at the brush border membrane achieved for metkephamid by inhibiting aminopeptidase N with puromycin, thereby improving oral F in rats from 0.5% to 3.5% [35]. Ovomucoids also inhibit intestinal serine proteases and are commonly isolated from egg white of avian species [27]. Despite successful peptidase inhibition, the apparent permeability coefficient (P_{app}) of insulin across rat jejunal mucosae was decreased by ovomucoids, so peptidase inhibition is not predictive of improved flux *per se* [36]. Combining peptidase inhibitors with PEs may therefore achieve improved permeability compared to either approach alone [37].

Mucous can decrease the rate and extent that peptides diffuse to the intestinal epithelium. The estimated (variable) mesh pore diameter of porcine jejunal mucous ranges between 200 and 2000 Å [38], much larger than the molecular radius of most candidate peptides for oral delivery (e.g. insulin <2 Å). Nevertheless, diffusivity (D) of peptides through mucous is affected by viscosity (ranging between 1000 and 10,000-fold greater than water at low shear [39]) and the MW of the peptide (approximating the dissolved peptide as a sphere) according to the Stoke-Einstein equation ($D = RT/6\eta\pi rN$). Poor diffusivity is directly related to residence time in the small intestinal lumen, which depends on the peptide surface charge at a given luminal pH, the potential for non-covalent bonding, and susceptibility to proteolysis [40,41]. Diffusivity measurements for cyclosporine and desmopressin were comparable at selected concentrations across porcine GI mucous, although they have comparable MW but different lipophilicity [42]. The capacity of mucous to reduce diffusivity may in part account for why mucolytics such N-acetylcysteine (NAC), either alone or in combination with non-ionic surfactant PEs (e.g. polyoxyethylene octyl phenyl ether (Triton® X-100, Dow Chemicals, USA) improve fluorescent dextran-4kDa (FD4) and sCT bioavailability in rat intestinal instillations [43,44]. Despite NAC having established safety in man, with approvals in respiratory and abdominal conditions, there is little interest in combining PEs with NAC or other mucolytics. In part, this relates to high variability in mucous production, the requirement for high concentrations of mucolytics in the formulation, and because disulphide bond reduction can also degrade certain peptides [45]. The capacity of mucous to complex

Table 1
Leading PEs tested in oral delivery of poorly permeable drugs and transport markers.

Enhancer	Mode	Actions	Model	Representative peptide/metric	Concentration & dose	Enhancement	Ref
C ₁₂ E ₉	Transcellular	Membrane fluidity	<i>In situ</i> (rat): intestinal loop	<i>In situ</i> : flux (fosfomycin)	<i>In situ</i> : 1%	<i>In situ</i> : —	[209]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (FD-10)	<i>In vitro</i> : 0.1%	<i>In vitro</i> : 520-fold	[488]
			<i>Ex vivo</i> (rabbit): Ussing	<i>Ex vivo</i> (rabbit): flux (insulin)	<i>Ex vivo</i> : 5%	<i>Ex vivo</i> : —	[489]
			<i>In vivo</i> (rabbit): rectal instillation	<i>In vivo</i> (rabbit): AUC (insulin)	<i>In vivo</i> : 1%	<i>In vivo</i> : —	[489]
			<i>In vivo</i> (dog): suppository	<i>In vivo</i> : RH (insulin)	<i>In vivo</i> : 1%	<i>In vivo</i> : RH = 55%	[211]
			<i>In vivo</i> (rat): gavage	<i>In vivo</i> : PK/PD (heparin)	<i>In vivo</i> : 500 mg/kg	<i>In vivo</i> : —	[203]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (PABA)	<i>In situ</i> : 1%	<i>In situ</i> : 19-fold	[490]
			<i>In vivo</i> (dog): suppository	<i>In vivo</i> : PK/PD, flux (insulin)	<i>In vivo</i> : 3% w/w	<i>In vivo</i> : —	[408]
			<i>In situ</i> (rat): rectal instillation	<i>In situ</i> : PK/PD (insulin)	<i>In situ</i> : 5%	<i>In situ</i> : 3-fold	[208]
			<i>In situ</i> (rat): rectal	<i>In situ</i> : PK/PD (calcitonin)	<i>In situ</i> : 0.5%	<i>In situ</i> : —	[212]
			<i>In vivo</i> (dog): suppository	<i>In vivo</i> : flux, PK/PD (insulin)	<i>In vivo</i> : 3% w/w	<i>In vivo</i> : —	[210]
			<i>In vivo</i> (rat): suppository	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 3%	<i>In vivo</i> : 118-fold	[491]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (mannitol)	<i>In vitro</i> : 1% w/v	<i>In vitro</i> : 34-fold	[283]
			<i>Ex vivo</i> (rat): ileum	<i>Ex vivo</i> : flux (LY)	<i>Ex vivo</i> : 0.1% v/v	<i>Ex vivo</i> : no effect	[492]
			<i>In situ</i> (rat): closed loop	<i>In situ</i> : F (rhodamine123)	<i>In situ</i> : 0.1% v/v	<i>In situ</i> : F = 22.82%	[492]
			<i>In situ</i> (rat): colon	<i>In situ</i> : F (insulin)	<i>In situ</i> : —	<i>In situ</i> : F = 0.25%	[287]
			<i>In situ</i> (rat): jejunal patches	<i>In situ</i> : AUC, F (erythropoietin)	<i>In situ</i> : 94 mg/kg	<i>In situ</i> : 21-fold	[284]
			<i>In situ</i> (rat): jejunal	<i>In situ</i> : AUC, F (erythropoietin)	<i>In situ</i> : 50 mg/kg	<i>In situ</i> : 12-fold	[288]
			<i>In situ</i> (rat): duodenal	<i>In situ</i> : F (lansoprazole)	<i>In situ</i> : 170 mg	<i>In situ</i> : F = 28.1%	[494]
			<i>In situ</i> (rat): duodenal	<i>In situ</i> : AUC (LMWH)	<i>In situ</i> : 30 mg/kg	<i>In situ</i> : 4-fold	[161]
Caprylocaproyl PEG 8 glycerides	Transcellular	Fluidic dispersion Membrane fluidity	<i>In situ</i> (rat): colon	<i>In situ</i> : F (gentamicin)	<i>In situ</i> : 1 mL/kg	<i>In situ</i> : F = 55.3%	[495]
			<i>In situ</i> (rat): ileal	<i>In situ</i> : AUC (vancomycin)	<i>In situ</i> : 1.06 g/kg	<i>In situ</i> : —	[495]
			<i>In situ</i> (rat): jejunal	<i>In situ</i> : AUC (LMWH)	<i>In situ</i> : 50 mg/kg	<i>In situ</i> : —	[279]
			<i>In situ</i> (rat): intraduodenal	<i>In situ</i> : AUC (LMWH)	<i>In situ</i> : 30 mg/kg	<i>In situ</i> : —	[161]
			<i>In vivo</i> (Dog): oral	<i>In vivo</i> : F (gentamicin)	<i>In vivo</i> : 0.6 mL	<i>In vivo</i> : F = 22.4%	[497]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (insulin)	<i>In vitro</i> : 5 mg	<i>In vitro</i> : no effect	[25]
			<i>In vivo</i> (rat): oral	<i>In vivo</i> : PK/PD, F (sCT)	<i>In vivo</i> : 10 mg	<i>In vivo</i> : F = 1.8%	[201]
			<i>Ex vivo</i> (rat): Ussing	<i>Ex vivo</i> : flux (FD-4)	<i>Ex vivo</i> : —	<i>Ex vivo</i> : —	[202]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (mannitol)	<i>In vitro</i> : 0.1% w/v	<i>In vitro</i> : 9-fold	[227]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (FD-4)	<i>In vitro</i> : 0.1% w/v	<i>In vitro</i> : 26-fold	[227]
Citric acid	Paracellular	Intracellular ATP	<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (tiludronate)	<i>In vitro</i> : 0.025% w/v	<i>In vitro</i> : 3-fold	[497]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (ranitidine)	<i>In vitro</i> : 0.1% w/v	<i>In vitro</i> : 19-fold	[498]
			<i>Ex vivo</i> (rat): perfusion	<i>Ex vivo</i> : flux (tiludronate)	<i>Ex vivo</i> : 0.025% w/v	<i>Ex vivo</i> : 7-fold	[498]
			<i>Ex vivo</i> (rat): Ussing	<i>Ex vivo</i> : flux (Phenol red)	<i>Ex vivo</i> : 20 mM	<i>Ex vivo</i> : no effect	[499]
			<i>Ex vivo</i> (rat): colon	<i>Ex vivo</i> : flux (ebiratide)	<i>Ex vivo</i> : 20 mM	<i>Ex vivo</i> : 7-fold	[245]
			<i>In situ</i> (rat): loop	<i>In situ</i> : AUC (phenol red)	<i>In situ</i> : 20 mM	<i>In situ</i> : 4-fold	[500]
			<i>In situ</i> (rat): loop	<i>In situ</i> : F (carboxyfluorescein)	<i>In situ</i> : 20 mM	<i>In situ</i> : F = 57%	[501]
			<i>In situ</i> (rat): colon	<i>In situ</i> : AUC, flux (azetirelin)	<i>In situ</i> : 2.5 mM	<i>In situ</i> : 9-fold	[86]
			<i>In situ</i> (rat): closed loop	<i>In situ</i> : PA (hCT)	<i>In situ</i> : 10 mM	<i>In situ</i> : 4-fold	[231]
			<i>In vivo</i> (dog): oral	<i>In vivo</i> : F (azetirelin)	<i>In vivo</i> : 2.5 mM	<i>In vivo</i> : F = 43.5%	[241]
Dodecyl-β-D-maltopyranoside (DDM)	Multimodal	Membrane fluidity	<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (FD-4)	<i>In vitro</i> : 1 mM	<i>In vitro</i> : 6-fold	[504]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (FD-4)	<i>In vitro</i> : 1 mM	<i>In vitro</i> : 6-fold	[503]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (FD-4)	<i>In vitro</i> : 0.25%	<i>In vitro</i> : 53-fold	[136]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (PEG 4000)	<i>In vitro</i> : 0.25%	<i>In vitro</i> : 29-fold	[88]
			<i>Ex vivo</i> (rat): colon	<i>Ex vivo</i> : flux (inulin)	<i>Ex vivo</i> : 50 mM	<i>Ex vivo</i> : —	[504]
			<i>In situ</i> (rat): loop method	<i>In situ</i> : AUC (phenol red)	<i>In situ</i> : 20 mM	<i>In situ</i> : 2-fold	[500]
			<i>In situ</i> (rat): ligated loop, colon	<i>In situ</i> : AUC (insulin)	<i>In situ</i> : 1% w/v	<i>In situ</i> : 55-fold	[20]
			<i>In vivo</i> (rabbit): oral	<i>In vivo</i> : AUC, (norfloxacin)	<i>In vivo</i> : 1:1	<i>In vivo</i> : no effect	[505]
			<i>In vivo</i> (rat): rectal, microenema	<i>In vivo</i> : flux (trypan blue)	<i>In vivo</i> : —	<i>In vivo</i> : —	[265]
			<i>In vivo</i> (rat): jejunum	<i>In vivo</i> : flux (fosfomycin)	<i>In vivo</i> : 1% w/v	<i>In vivo</i> : —	[209]
EDTA	Paracellular	Ca ²⁺ chelation PKC activation	<i>In situ</i> (rat): rectal	<i>In situ</i> : PK/PD (insulin)	<i>In situ</i> : —	<i>In situ</i> : 3-fold	[208]
			<i>In situ</i> (rat): rectal	<i>In situ</i> : F (cefmetazole)	<i>In situ</i> : 0.25 mL/kg	<i>In situ</i> : F = 18.2%	[266]
			<i>In situ</i> (rat): rectal loop	<i>In situ</i> : AUC (insulin)	<i>In situ</i> : 50 mM	<i>In situ</i> : 15-fold	[185]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : P _{app} (mannitol)	<i>In vitro</i> : 1 mM	<i>In vitro</i> : —	[257]
Glyceryl monocaprate	Transcellular	Membrane fluidity	<i>Ex vivo</i> (rat): BBM	<i>Ex vivo</i> : S (DPH)	<i>Ex vivo</i> : —	<i>Ex vivo</i> : —	[506]
Laurylocarnitine	Multimodal	Membrane fluidity —	<i>In vivo</i> (rat): rectal	<i>In vivo</i> : F (cefotaxitin)	<i>In vivo</i> : —	<i>In vivo</i> : 26-fold	[506]

			Ex vivo (rats): S-G diffusion cell	Ex vivo: flux (LY)	Ex vivo: 10 mM	Ex vivo: 20-fold	[73]
			Ex vivo (rats): S-G diffusion cell	Ex vivo: TEER	Ex vivo: 2 mM	Ex vivo: —	[73]
		Decrease Claudin level	In vivo (rat): oral (microcapsule)	In vivo: F (DMP 728)	In vivo: 8 mg/kg	In vivo: F = 6.9%	[188]
		Increase Ca ²⁺ levels	In vivo (dog): oral (EC capsule)	In vivo: F (DMP 728)	In vivo: 2 mg/kg	In vivo: F = 17%	[188]
		Decrease ATP levels	In vitro: Caco-2	In vitro: TEER, flux (FD-40)	In vitro: 100 µM	In vitro: —	[509]
			Ex vivo (rat) Ussing	Ex vivo: I _{sc} (Cl ⁻)	Ex vivo: 0.5%	Ex vivo: —	[202]
			Ex vivo (rat) Ussing	Ex vivo: flux (FD-4)	Ex vivo: 0.5%	Ex vivo: —	[202]
			In vitro: Caco-2	In vitro: TEER, flux (mannitol)	In vitro: 0.1% w/v	In vitro: 143-fold	[227]
n-Tetradecyl β-D-maltopyranoside (TDM)	Transcellular	Membrane fluidity	In vitro: Caco-2	In vitro: TEER, flux (FD-4)	In vitro: 0.1% w/v	In vitro: 153-fold	[227]
			Ex vivo (rat): Ussing (jejunal)	Ex vivo: flux (mannitol)	In vitro: 0.1% w/v	Ex vivo: 9-fold	[510]
			Ex vivo (rat): Ussing (jejunal)	Ex vivo: flux (FD-4)	In vitro: 0.1% w/v	Ex vivo: 20-fold	[510]
			In vitro: Caco-2	In vitro: flux (FD-4)	In vitro: 2.5% w/v	In vitro: 363-fold	[509]
N-Trimethylated chitosan	Multimodal	Membrane fluidity	In situ (rat): instillation	In situ: F (buserelin)	In situ: 1% w/v	In situ: 16-fold	[510]
		TJ alteration via PKC	In situ (rat): intubation	In situ: F (octreotide)	In situ: 10% w/v	In vivo: 15-fold	[77]
			In vivo (pig): oral (EC capsule)	In vivo: F (octreotide)	In vivo: 40% (70 mg)	In vivo: F = 0.5%	[427]
			Ex vivo (rat): Ussing	Ex vivo: I _{sc} (Cl ⁻)	Ex vivo: 0.5%	Ex vivo: —	[202]
			Ex vivo (rat): Ussing	Ex vivo: flux (FD-4)	Ex vivo: 0.5%	Ex vivo: —	[202]
			Ex vivo (rat): BBM	Ex vivo: fluorescence polarization	Ex vivo: —	Ex vivo: —	[506]
			In vivo (rat): rectal	In vivo: F (cefoxitin)	In vivo: —	In vivo: 34-fold	[506]
			Ex vivo (rats): S-G diffusion cell	Ex vivo: flux (LY)	Ex vivo: 5 mM	Ex vivo: 18-fold	[73]
			Ex vivo (rats): S-G diffusion cell	Ex vivo: TEER	Ex vivo: 1 mM	Ex vivo: —	[73]
		Ca ²⁺ level	In vitro: Caco-2	In vitro: TEER, flux (ruthenium red)	In vitro: 0.2 mM	In vitro: 20-fold	[511]
		ATP levels	In vitro: Caco-2	In vitro: flux (PEG 4000)	In vitro: 0.2 mM	In vitro: no effect	[511]
Palmitoylecarnitine	Multimodal	Membrane fluidity	In vivo (rat): oral (microcapsule)	In vivo: F (DMP 728)	In vivo: 8 mg/kg	In vivo: F = 14.6% (6-fold)	[188]
		Claudin modulation	In vivo (dog): oral (capsule)	In vivo: F (DMP 728)	In vivo: 2 mg/kg	In vivo: F = 20.5% (2-fold)	[188]
			In vitro: Caco-2	In vitro: TEER, flux (mannitol)	In vitro: 500 µM	In vitro: 10-fold	[503]
			In vitro: Caco-2	In vitro: flux (mannitol, Ca ²⁺)	In vitro: 100 µM	In vitro: —	[512]
			Ex vivo (rat): Ussing	Ex vivo: flux (FD-4)	Ex vivo: 0.5%	Ex vivo: 13-fold	[454]
			Ex vivo (human): Ussing	Ex vivo: flux (FD-4)	Ex vivo: 0.5%	Ex vivo: 8-fold	[454]
			In vitro: Caco-2	In vitro: TEER, flux (mannitol)	In vitro: 0.75 mM	In vitro: TEER <10%	[515]
			In vitro: Caco-2	In vitro: TEER, flux (FD-40)	In vitro: 100 µM	In vitro: —	[509]
			In vitro: Caco-2	In vitro: TEER, flux (fluorescein)	In vitro: 200 µM	In vitro: —	[514]
			In vivo (rat): oral	In vivo: PA (insulin)	In vivo: 5 mM	In vivo: 79-fold	[515]
			In vitro: Caco-2	In vitro: flux (insulin)	In vitro: 60 µM	In vitro: —	[516]
Penetratin (D- penetratin)	Transcellular	Carrier	In situ (rat): perfusion	In situ: AUC (GLP-1)	In situ: 0.5 mM	In situ: no effect	[517]
			In situ (rat): perfusion	In situ: AUC (extendin-4)	In situ: 0.5 mM	In situ: 2-fold	[517]
			In vitro: Caco-2	In vitro: flux (insulin)	In vitro: —	In vitro: 3-fold	[517]
			In vivo (rat): instillation	In vivo: F (insulin)	In vivo: —	In vivo: F = 3.1%	[519]
			In vivo (human): oral	In vivo: AUC (GLP-1)	In vivo: 2 mg	In vivo: 2-fold	[372]
			In vivo (human): oral	In vivo: AUC (PYY ^{3–36})	In vivo: 1 mg	In vivo: 2-fold	[372]
			In vitro: Caco-2	In vitro: TEER, flux (mannitol)	In vitro: 50 mg/mL	In vitro: 27-fold	[360]
SNAC	Transcellular	Carrier	In situ (rat): oral	In situ: PK/PD (heparin)	In situ: 300 mg/kg	In situ: —	[361]
			In vitro: Caco-2	In vitro: TEER, flux (insulin)	In vitro: 55 mM	In vitro: 10-fold	[363]
			In vivo (rabbit): oral	In vivo: flux (heparin)	In vivo: 120 mg/kg	In vitro: no effect	[520]
			In vitro: Caco-2	In vitro: TEER, flux (mannitol)	In vitro: 50 mg/mL	In vitro: 2-fold	[360]
			In vivo (rat): suppository	In vivo: AUC (sodium ampicillin)	In vivo: 20 µmol/kg	In vivo: 6-fold	[166]
			In vitro: Caco-2	In vitro: flux (mannitol)	In vitro: 10 mM	In vitro: 8-fold	[453]
sodium caprate (C ₁₀)	Multimodal	PLC	In situ (rat): instillation	In situ: flux (FD-4)	In situ: 100 mM	In situ: 14-fold	[70]
		MLCK	In situ (rat): colon	In situ: AUC (cefmetazole)	In situ: 0.25% w/v	In situ: 10-fold	[177]
		Membrane fluidity	In vivo (rat): oral	In vivo: AUC, F (DMP 728)	In vivo: 40.2%	In vivo: F = 6% (3-fold)	[188]
			In vivo (dog) oral	In vivo: F (DMP 728)	In vivo: 40.2%	In vivo: F = 17.7%	[188]
			In vivo (rat): suppository	In vivo: AUC (sodium ampicillin)	In vivo: 20 µmol/kg	In vivo: 5-fold	[166]
			In vitro: Caco-2	In vitro: flux (mannitol)	In vitro: 120 mM	In vitro: 2-fold	[453]
			In vivo (human): suppository	In vivo: F (cefoxitin sodium)	In vivo: 0.5 g	In vivo: 2.5-fold	[334]
sodium caprylate (C ₈)	Multimodal	Membrane fluidity	In situ (rat): suppository	In situ: AUC (acyclovir)	In situ: 4%	In situ: 2-fold	[519]
		Affinity for Ca ²⁺	In vivo: (rabbit) suppository	In vivo: flux (gentamicin)	In vivo: 0.18 mM	In vivo: —	[273]
			Ex vivo (rat): inverted sac	Ex vivo: flux (CsA)	Ex vivo: 1% w/v	Ex vivo: 9-fold	[158]
			Ex vivo (rat): everted colon	Ex vivo: flux (inulin)	Ex vivo: 0.25%	Ex vivo: —	[520]
			In situ (rat): colon	In situ: flux (inulin)	In situ: 0.25%	In situ: —	[520]

(continued on next page)

Table 1 (continued)

Enhancer	Mode	Actions	Model	Representative peptide/metric	Concentration & dose	Enhancement	Ref
sodium cholate	Multimodal	Membrane Fluidity Tight junction	<i>In situ</i> (rat): rectal loop	<i>In situ</i> : AUC (insulin)	<i>In situ</i> : 50 mM	<i>In situ</i> : no effect	[185]
			<i>In situ</i> (rat): colon	<i>In situ</i> : AUC (cefmetazole)	<i>In situ</i> : 0.25% w/v	<i>In situ</i> : 2-fold	[177]
			<i>In vivo</i> (rat): jejunal	<i>In vivo</i> : flux (fosfomycin)	<i>In vivo</i> : 1% w/v	<i>In vivo</i> : —	[209]
			<i>Ex vivo</i> (rat): colon	<i>Ex vivo</i> : flux (urea)	<i>Ex vivo</i> : 0.25%	<i>Ex vivo</i> : no effect	[179]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (mannitol)	<i>In vitro</i> : —	<i>In vitro</i> : —	[150]
			<i>In situ</i> (rat): duodenum	<i>In situ</i> : AUC (LMWH)	<i>In situ</i> : 30 mg/kg	<i>In situ</i> : —	[161]
			<i>In situ</i> (rat); perfusion	<i>In situ</i> : flux (cefazolin)	<i>In situ</i> : 10 mM	<i>In situ</i> : no effect	[521]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (cefazolin)	<i>In situ</i> : 10 mM	<i>In situ</i> : no effect	[282]
			<i>In vivo</i> (rat): suppository	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 2%	<i>In vivo</i> : —	[491]
			<i>In vivo</i> (human): microenema	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 20 mg/mL	<i>In vivo</i> : —	[251]
			<i>Ex vivo</i> (rat): everted sac	<i>Ex vivo</i> : P _{app} (cefotaxime)	<i>Ex vivo</i> : 1%w/v	<i>Ex vivo</i> : 3-fold	[158]
			<i>In vivo</i> (rabbit): (EC capsule)	<i>In vivo</i> : RH (insulin)	<i>In vivo</i> : 50 mg	<i>In vivo</i> : no effect	[524]
			<i>In vivo</i> (dog): rectal suppository	<i>In vivo</i> : RH (insulin)	<i>In vivo</i> : 50 mg	<i>In vivo</i> : RH = 40%	[525]
			<i>In situ</i> (dog): EC microtablet	<i>In situ</i> : flux, PK/PD (insulin)	<i>In situ</i> : 12% w/w	<i>In situ</i> : —	[524]
			<i>In situ</i> (dog): non-EC capsule	<i>In situ</i> : flux, PK/PD (insulin)	<i>In situ</i> : 100 mg	<i>In situ</i> : —	[524]
			<i>In vivo</i> (dog): oral EC microtablet	<i>In vivo</i> : flux, PK/PD (insulin)	<i>In vivo</i> : 12% w/w	<i>In vivo</i> : —	[524]
			<i>In situ</i> (rat): intestinal loop	<i>In situ</i> : flux (insulin, RNase)	<i>In situ</i> : 10 mg/mL	<i>In situ</i> : 4-fold, 9-fold	[37]
			<i>In vivo</i> (rat): instillation	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 10 mg/mL	<i>In vivo</i> : —	[37]
			<i>In vivo</i> (rat): enema	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 2 mg/mL	<i>In vivo</i> : —	[240]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (FD-4)	<i>In vitro</i> : 0.05%	<i>In vitro</i> : 33-fold	[527]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (Rh 123)	<i>In vitro</i> : 0.05%	<i>In vitro</i> : 12-fold	[527]
			<i>In situ</i> (rat): Instillation	<i>In situ</i> : PK/PD (insulin)	<i>In situ</i> : 2 mg	<i>In situ</i> : —	[526]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (epirubicin)	<i>In vitro</i> : 1.2 mM	<i>In vitro</i> : —	[527]
			<i>In situ</i> (rat): everted sac	<i>In situ</i> : flux (epirubicin)	<i>In situ</i> : 12 mM	<i>In situ</i> : —	[527]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : Ca ²⁺ (Fura-2-AM)	<i>In vitro</i> : 0.05%w/v	<i>In vitro</i> : no effect	[528]
sodium deoxycholate	Multimodal	Membrane Fluidity Tight junction	<i>In situ</i> (rat): instillation	<i>In situ</i> : AUC (phenol red)	<i>In situ</i> : 20 mM	<i>In situ</i> : 9-fold	[501]
			<i>Ex vivo</i> (rat): everted sac	<i>Ex vivo</i> : P _{app} (cefotaxime)	<i>Ex vivo</i> : 1%w/v	<i>Ex vivo</i> : 15-fold	[158]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (cefotaxime)	<i>In situ</i> : 1% w/v	<i>In situ</i> : 4-fold	[531]
			<i>In vivo</i> (rat): oral	<i>In vivo</i> : F	<i>In vivo</i> : 1%w/v	<i>In vivo</i> : F = 33%	[531]
			<i>Ex vivo</i> (rabbit): Ussing	<i>Ex vivo</i> : flux (insulin)	<i>Ex vivo</i> : 1%	<i>Ex vivo</i> : —	[489]
			<i>In vivo</i> (rabbit): rectal	<i>In vivo</i> : AUC (insulin)	<i>In vivo</i> : 1%	<i>In vivo</i> : —	[489]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (oxalate, urea)	<i>In situ</i> : 5 mM	<i>In situ</i> : 10-fold, 7-fold	[530]
			<i>In vivo</i> (rat): rectal Perfusion	<i>In vivo</i> : AUC (sulfanilic acid)	<i>In vivo</i> : 5 mM	<i>In vivo</i> : —	[470]
			<i>In vivo</i> (rat): gavage	<i>In vivo</i> : flux, PK/PD (heparin)	<i>In vivo</i> : 500 mg/kg	<i>In vivo</i> : —	[203]
			<i>In situ</i> (rat): intestinal loop	<i>In situ</i> : PK/PD (calcitonin)	<i>In situ</i> : 10 mM	<i>In situ</i> : 7-fold	[241]

sodium dodecyl sulphate	Transcellular	Membrane fluidity Oxidative phosphorylation ATP depletion	<i>In vivo</i> (human): oral (capsule)	<i>In vivo</i> : PK/PD (PSP)	<i>In vivo</i> : 300 mg	<i>In vivo</i> : no effect	[531]
			<i>In vivo</i> (dog): suppository	<i>In vivo</i> : RH (insulin)	<i>In vivo</i> : 150 mg	<i>In vivo</i> : 35% RH	[525]
			<i>In vivo</i> (hamster): perfusion	<i>In vivo</i> : PK/PD (inulin)	<i>In vivo</i> : 5 mM	<i>In vivo</i> : 50-fold	[534]
			<i>In vivo</i> (hamster): perfusion	<i>In vivo</i> : PK/PD (albumin)	<i>In vivo</i> : 5 mM	<i>In vivo</i> : 9-fold	[534]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (VLPVP)	<i>In vitro</i> : 100 µmol/L	<i>In vitro</i> : —	[535]
			<i>Ex vivo</i> (rat): everted gut	<i>Ex vivo</i> : flux (phenol red)	<i>Ex vivo</i> : 100 mM	<i>Ex vivo</i> : 8-fold	[534]
			<i>In situ</i> (rat): instillation	<i>In situ</i> : flux (phenol red)	<i>In situ</i> : 100 mM	<i>In situ</i> : 12-fold	[534]
			<i>In vivo</i> (rat): gastric	<i>In vivo</i> : PK/PD (phenol red)	<i>In vivo</i> : 150 µmol	<i>In vivo</i> : 3-fold	[534]
			<i>In vivo</i> (rat): enema	<i>In vivo</i> : PK/PD (insulin)	<i>In vivo</i> : 5 mg/mL	<i>In vivo</i> : —	[240]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (mannitol)	<i>In vitro</i> : 2 mM	<i>In vitro</i> : >139-fold	[216]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux	<i>In vitro</i> : 2 mM	<i>In vitro</i> : 142-fold	[216]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (mannitol)	<i>In vitro</i> : 0.4 mM	<i>In vitro</i> : 20-fold	[255]
			<i>Ex vivo</i> (rat): jejunum	<i>Ex vivo</i> : flux (amoxicillin)	<i>Ex vivo</i> : 0.2 mg/mL	<i>Ex vivo</i> : 2-fold	[537]
			<i>In vivo</i> (rat): oral gavage	<i>In vivo</i> : AUC (phenol red)	<i>In vivo</i> : 2% w/v	<i>In vivo</i> : 6-fold	[75]
			<i>In situ</i> (rat):	<i>In situ</i> : flux (cefazolin)	<i>In situ</i> : 10 mM	<i>In situ</i> : 5-fold	[282]
			<i>In situ</i> (rat): colon	<i>In situ</i> : TEER, flux (LY)	<i>In situ</i> : 5 mM	<i>In situ</i> : 24-fold	[73]
			<i>In situ</i> (rat): rectal perfusion	<i>In situ</i> : AUC (sulfanilic acid)	<i>In situ</i> : 5 mM	<i>In situ</i> : —	[470]
			<i>In situ</i> (rat): intestine	<i>In situ</i> : flux (BTDS)	<i>In situ</i> : 0.1% w/v	<i>In situ</i> : 2-fold	[536]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (cefazolin)	<i>In situ</i> : 10 mM	<i>In situ</i> : no effect	[521]
			<i>In situ</i> (rat): loop	— AUC (cefmetazole)	— 0.25%	— 2-fold	[177]
sodium taurocholate	Multimodal	—	<i>Ex vivo</i> (rat): everted sac	<i>Ex vivo</i> : flux (inulin)	<i>Ex vivo</i> : 0.25%	<i>Ex vivo</i> : —	[520]
			<i>In situ</i> (rat): colon loop	<i>In situ</i> : flux (inulin)	<i>In situ</i> : 0.25%	<i>In situ</i> : —	[520]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (FD-4, EMD)	<i>In vitro</i> : 15 mM	<i>In vitro</i> : —	[242]
			<i>In situ</i> (rat): instillation	<i>In situ</i> : AUC (Phenol red)	<i>In situ</i> : 20 mM	<i>In situ</i> : 3-fold	[501]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (mannitol)	<i>In vitro</i> : 50 mM	<i>In vitro</i> : >403-fold	[216]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux	<i>In vitro</i> : 50 mM	<i>In vitro</i> : >663-fold	[216]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : TEER, flux (heparin)	<i>In vitro</i> : 0.1%	<i>In vitro</i> : 4 -fold	[539]
			<i>In vitro</i> : Caco-2	<i>In vitro</i> : flux (CRC 220, SR 101)	<i>In vitro</i> : 40 mM	<i>In vitro</i> : 280-fold, 530-fold	[538]
			<i>In vivo</i> (dogs): suppository	<i>In vivo</i> : RH (insulin)	<i>In vivo</i> : 100 mg	<i>In vivo</i> : RH = 50%	[525]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (sulfaguanidine)	<i>In situ</i> : 20 mM	<i>In situ</i> : 2-fold	[539]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (cefazolin)	<i>In situ</i> : 10 mM	<i>In situ</i> : —	[282]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (ACDB, phenol red)	<i>In situ</i> : 40 mM	<i>In situ</i> : —	[540]
Sucrose monolaurate	Paracellular	Increase pore radius	<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (sulfanilamide)	<i>In situ</i> : 40 mM	<i>In situ</i> : no effect	[540]
			<i>Ex vivo</i> (rabbit): Ussing	<i>Ex vivo</i> : flux (insulin)	<i>Ex vivo</i> : 1%	<i>Ex vivo</i> : —	[489]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (oxalate)	<i>In situ</i> : 5 mM	<i>In situ</i> : no effect	[530]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (sulfaguanidine)	<i>In situ</i> : 20 mM	<i>In situ</i> : 3-fold	[541]
			<i>In situ</i> (rat): perfusion	<i>In situ</i> : flux (sulfanilic acid)	<i>In situ</i> : 20 mM	<i>In situ</i> : 4-fold	[541]
			<i>In situ</i> (rat): rectal loop	<i>In situ</i> : AUC (insulin)	<i>In situ</i> : 3%	<i>In situ</i> : 17-fold	[185]
			<i>Ex vivo</i> (pig): intestinal	<i>Ex vivo</i> : flux (cyclosporin A)	<i>Ex vivo</i> : 10%	<i>Ex vivo</i> : 6-fold	[542]

Table 2

Marketed therapeutic peptides (arbitrary MW cut-off: 9 kDa).

Generic name	Trade name®	Manufacturer	Delivery	MW	Classification/application
Eptifibatide	Integrilin	GSK	IV	832	Anti-platelet drug
Octreotide	Sandostatin	Novartis	SC, IV	1019	Somatostatin analogue
Desmopressin	DDAVP	Ferring	Oral	1069	Synthetic vasopressin analogue
Vasopressin	Pitressin	Goldshield	SC, IM	1084	Antidiuretic peptide
Lanreotide	Somatuline LA	Ipsen	IM	1096	Somatostatin analogue
GnRH	HRF	Intrapharm	SC	1182	Peptide hormone
Cyclosporin	Neoral	Novartis	Oral	1203	Immunosuppressant peptide
Leuporelin/leuprolide acetate	Prostap	Takeda	SC, IM	1209	GnRH agonist
Terlipressin	Glypressin	Ferring	IV	1227	Synthetic vasopressin analogue
Mifamurtide	Mepact	Takeda	IV	1238	Osteosarcoma
Buserelin	Suprefact	Sanofi-Aventis	SC, Nasal	1239	GnRH agonist
Goserelin	Zoladex	AstraZeneca	Implant	1269	GnRH super agonist
Icatibant	Firazyr	Shire HGT	SC	1305	Hereditary angioedema
Triptorelin	Decapeptyl SR	Ipsen	IM	1312	GnRH agonist
Nafarelin	Synarel	Pharmacia	Nasal	1322	GnRH agonist
Histrelin	Vantas	Orion	Implant	1324	GnRH agonist
Abarelix	Plenaxis	Speciality European Pharma Ltd		1416	Prostate cancer
Cetrorelix	Cetrotide	Merck Sorono	SC	1431	GnRH antagonist
Vancomycin	Vancomycin matrigel	Flynn Pharma	Oral (local),	1486	Antibiotic peptide
Linacotide	Linzess	Ironwood Pharma	Oral (local)	1527	IBS
Degarelix	Firmagon	Ferring	SC	1631	GnRH antagonist
Bivalirudin	Angiox	The Medicines Company	IV	2180	Anticoagulant
Tetracoactide	Synacthen	Alliance	SC	2933	ACTH analogue
Tetracosactide	Synacthen	Alliance	IM, IV	2933	Corticotrophin analogue
Salmon calcitonin	Miacalcic	Novartis	SC, IV, Nasal	3432	Anti-osteoporotic peptide
Nesiritide	Natrecor	Scios Inc	IV	3464	human B-type natriuretic peptide
Glucagon	Glucagen	Novo Nordisk	SC, IM, IV	3483	Antidiabetic peptide
Liraglutide	Victoza	Novo Nordisk	SC	3751	GLP-1 analogue agonist peptide
Teduglutide	Gattex/Nycomed	NPS Pharma	SC	3752	GLP-2 analogue agonist peptide
Pramlintide	Symlin	AstraZeneca	SC	3951	Analogue of Amylin
teriparatide	Forsteo	Lilly	SC	4118	rh parathyroid hormone (analogue)
Exenatide	Byetta	Lilly/Amylin	SC	4187	Exendin-4
Enfuvirtide	Fuzeon	Roche	SC	4492	Antiviral peptide
rh Insulin	Actrapid	Novo Nordisk	SC	5808	Antidiabetic peptide
rh Insulin	Insuman rapid	Sanofi Aventis	SC	5808	Antidiabetic peptide
rh Insulin	Humulin S	Lilly	SC	5808	Antidiabetic peptide
Insulin lispro	Humalog	Lilly	SC	5808	Analogue of rh insulin
Insulin glulisine	Apidra	Sanofi Aventis	SC	5823	Analogue of rh insulin
Insulin aspart	NovoRapid	Novo Nordisk	SC	5826	Analogue of rh insulin
Insulin detemir	Levemir	Novo Nordisk	SC	591	Analogue of rh insulin
Insulin glargine	Lantus	Sanofi Aventis	SC	6063	Analogue of rh insulin
Glatiramer acetate	Copaxone	Teva Pharma	SC	6400	Immunomodulator peptide
Ecaltantide	Kalbitor	Dyax	SC	7054	Hereditary angioedema
Mecasermin	Increlax	Ipsen	SC	7649	rh insulin like growth factor-I
rh PTH	Preotact	Nycomed	SC	9000	Anti-osteoporotic peptide

MW = molecular weight; SC = subcutaneous injection; IM = intramuscular injection; IV = Intravenous injection or infusion; rh = recombinant human.

cationic peptides could be attenuated by electrostatic interaction between anionic mucins and cationic mucoadhesive oligomers/polymers (e.g. chitosans), which could reduce loss of peptides that are positively charged at the small intestinal pH. Malhaire et al. (in this Issue [46]) address the impact of mucous on oral absorption of peptides in nano-carriers in more detail.

The primary barrier to oral peptide bioavailability however, is poor intestinal transepithelial flux, which is due to unfavourable physico-chemical properties. Intestinal epithelial permeability of peptides is predicted to be low based on typical metrics derived from Lipinski “Rule of 5” [47], the BCS [48], the Biopharmaceutics Drug Disposition Classification System (BDDCS) [49], the Developability Classification System (DCS) [50] and the Nutraceutical Classification System (NCS) [51]. The “Rule of 5” predicts that a drug will exhibit poor oral bioavailability if it does not exhibit two or more of the following parameters <5 hydrogen bond donors, <10 hydrogen bond acceptors, <500 Da MW, Log P < 5 [47], criteria that most peptides do not adhere to. In general, a Log P_{OCTANOL:WATER} value for a peptide is likely to be below −1, unsuitable for passive transcellular diffusion [22,52]. TJs at the paracellular route prevent movement of high MW solutes (>500 Da), and so this transport route is also not available for most peptides. Of note is that a peptide that does not comply with the “Rule of 5” does not necessarily preclude oral administration, as seen for desmopressin (1–4 µg dose,

0.2 mg/tablet, 15 H-bond acceptors, 14 H-bond donors, MW 1069 Da, and a Log P: −4.2). Peptides with lower potency (e.g. octreotide: 50 µg dose, Log P: −1.4, H-bond donors: 13, H-bond acceptors: 12, MW: 1019), or greater complexity (e.g. exenatide: Log P: −2, H-bond donors: 58, H-bond acceptors: 67, MW 4187 Da) need formulation assistance to be developed for oral delivery. The amphipathic nature of peptides also gives rise to pH-dependent changes in partitioning behaviour as measured by the LogD_{pH} metric. This has implications for the use of acidifiers that reduce proteolytic degradation [53], as peptides typically have high solubility and low permeability at pH values below their pI due to the protonation of basic amino acid side chains.

Beyond the intestinal epithelium, it is assumed that peptides freely diffuse through the basement membrane and lamina propria, and also passively permeate capillary endothelia endothelium owing to leaky TJs and fenestrations (20–100 nm). For insulin and GLP-1, entry into the portal vein mimics physiological secretion, yet more refined studies are required to understand the extent to which intestinal brush border and hepatic metabolism influence oral peptide bioavailability. While the BCS classifies drugs as low permeability, distinction has been made between poor permeation and high metabolism (BDDCS) [49,54]. This is an important distinction in oral peptide delivery as there may be requirement to combine intestinal PEs with delivery approaches that limit metabolic degradation, not just by peptidases. For example,

cyclosporin (Sandimmune®, Novartis, Switzerland) can permeate the intestinal epithelium up to 86%, but the fraction absorbed (F_A) is 35% due to coordinated brush border cytochrome P450 metabolism and P-glycoprotein efflux, and a further 8% is lost through hepatic metabolism to yield an oral F of 27% [55]. Given that many conventional therapeutic peptides (e.g. exenatide, octreotide, liraglutide) have been designated BDDCS Class I (high solubility/extensive metabolism) through *in silico* modelling, it is more difficult to develop peptides within this category. The most promising strategies to limit metabolism include combining PEs with hydrophobisation, structural modification (amino acid substitution and N-terminal amidation) and nanoencapsulation. Other strategies to evade metabolism have been tested pre-clinically include inhibition of cytochrome P450 3A4 (e.g. cyclosporin [56]) or by targeting regions of the GI with lower metabolic activity [57].

Enteric coating also ensures that the peptide, PE, and other additives within the formulation are contemporaneously released in the small intestine. In rat studies, enteric coated tablets increased the efficacy of the PE, palmitoyl carnitine (PC), as measured by an increase in oral F of the transport marker, cefoxitin [58]. Innovative formulation strategies can optimise release and spreading in an epithelial region in order to (i) create a diffusion gradient that improves the rate and extent of peptide flux (ii) decrease luminal residence of peptide to limit exposure to pancreatic peptidases (iii) allow formulation additives to reach a threshold concentration required to improve flux (Fig. 1). Confining the PE and peptide to a small intestinal region will in most cases improve intestinal permeability. For example, the F_{ABS} of cefoxitin was 77% when co-instilled with PC in ligated rat jejunum, but was only 18% in the un-ligated tissue [58]. There are however some exceptions, such as with the use of paracellular PEs, which are required to spread over a wider area to ensure that sufficient number of TJs are opened [59].

Intestinal transit and the requirement for liberation from a dosage form make it difficult to achieve optimal presentation of the peptide and PE at the intestinal wall. Slow infusion/high fluid volume instillation into the small intestine mimics slow dissolution of solid dosage forms, and so it was not surprising that fast infusion of sCT (25 mg/5 mL/25 s) led to increased absorption in dogs compared to slow infusion (25 mg/20 mL/600 s) [58]. The duodenum and proximal jejunum have strong absorptive capacity, but targeting release in this region is hampered by greater brush border enzyme activity [60] and fast transit [18], which invariably disperses the peptide and additives in the dosage form over a larger area. Significant dilution and spreading of a radiopaque dye and CA occurred in the duodenum following initial disintegration from a capsule dosage form. In the lower GI (ileum) however, the dye was exhibited more optimal plug flow [53]. The average small intestinal transit time for a monolithic dosage form is just over 200 min,

irrespective of its density and shape [61]. Given the approximate length of the small intestine (7m), transit can be estimated to ~3.5 cm/min, wherein lies significant difficulty for efficient use of PE. The peptide will quickly pass the optimum region of the small intestine where the peptide and PE are designed to be co-released. PEs that require several hours to increase intestinal permeability will therefore not have that opportunity in the small intestine owing to fast transit. For example C_{10} was a 5-fold more effective PE at the lower flow rate of 0.1 mL/min compared with 0.2 mL/min in a rat intestinal single pass perfusion model [62]. The most promising PEs are therefore those that work quickly, allowing local co-presentation of the peptide and PE at the intestinal epithelium in high concentrations. Accelerated transit of a formulation (e.g. secretory diarrhoea, thyrotoxicosis) requiring intimate contact with the epithelium could therefore lead to a slower barrier modification that would result in erratic absorption.

Several strategies have been attempted to improve peptide and PE localisation at the intestinal wall. In contrast to the potential benefits of mucolytics to facilitate better diffusion to the small intestinal wall, others advocate use of mucoadhesive polymeric coatings to prevent spreading and improve co-presentation. The small intestine, however, has a high rate of mucous sloughing that has seen mucoadhesive polymer coating strategies become less popular [63,64]. Thiolated polymers are mucoadhesives that form covalent disulphide bonds with mucin glycoproteins (reviewed in [65]). Mucoadhesion of polymers can be improved by functionalising with thiol (e.g. 4-fold for alginate with cysteine and 250-fold for chitosan with iminothiolane [66]). The potential of mucoadhesive polymers was highlighted in data showing relative bioavailability of insulin of 15% with respect to the S.C. route following oral administration of mini-tablets containing chitosan-6-mercaptopropionic acid in rats [67]. Thiomatrix GmbH (Austria) has researched formats of thiolated chitosan and polyacrylic acid that combine mucoadhesion with permeation enhancement and enzyme inhibition. In addition, dosage forms containing superporous hydrogels (SPH™) swell and take up 100–200 times their original volume and can bring a co-delivered PE and peptide into direct contact with the epithelium. For example, oral F of octreotide in pigs was 16% with trimethylated chitosan [68], and enteric-coated SPH™ improved intestinal residence time by 45–60 min in man [69].

Several investigations yielded insight into the importance of liberation and co-presentation of PEs and poorly permeable molecules in the small intestine. Co-delivery of C_{10} (100 mM) and FD4 in intestinal instillation in rats yielded an oral F of 33%, but staggering delivery of FD4 by 10 min after C_{10} exposure had F_{ABS} of only 9% [62]. Both *in vitro* [70,71] and *in vivo* [72] testing proved that the epithelial barrier can recover from PE-induced membrane perturbation, but it depends on the molecule, its concentration and the time of exposure [70,71,73]. Recovery of

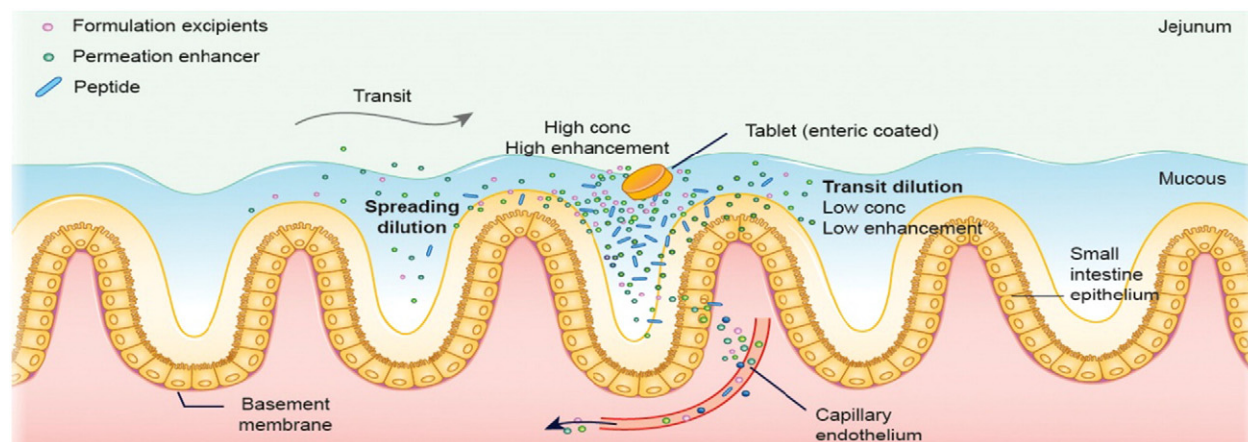


Fig. 1. Optimal liberation of peptide and PE from an oral solid dosage is required to maximise enhancement of oral bioavailability. High regional concentration of peptide and PE improves intestinal flux through the creation of a diffusion gradient and by enabling the PE to reach a threshold concentration for flux enhancement.

barrier integrity was slow in rats following intestinal exposure to both sodium dodecyl sulphate (SDS) and EDTA for 60 min, but rapid recovery was observed with C₁₀ [74]. Similarly, gavage of phenol red and SDS together to rats increased absorption of phenol red, but staggering the delivery of phenol red by 15 min post-SDS exposure had only a modest effect on dye absorption [75]. In a kinetic analysis, co-delivery of C₁₀ (100 mM) with cefoxitin for 15 min in rat perfusion was less effective in achieving bioavailability than co-administration of a lower concentration (50 mM) for a longer time period (30 min) [74]. In cases where dissolution of PE and payload are not closely matched, enhancement is weaker than when release is synchronised. For example, comparable dissolution of drug and PE was observed from matrix tablets containing sulphiride, C₁₀ and HPMC, and this formulation had high F_{ABS} of 46% for sulphiride [76]. On the other hand, for multilayer tablets where dissolution of C₁₀ (>95% at 30 min) was faster than sulphiride (<95% at 30 min), there was a consequential lowering of F_{ABS} (11%). A number of formulation strategies have been attempted to promote co-presentation including formulating peptide and PE with excipients that promote rapid disintegration and dissolution to facilitate optimal presentation at the intestinal epithelium [77]. Others attempt to reproduce pre-clinical effectiveness of fluid instillations by formulating fluidic solutions, lipoidal vehicles, or suspensions in liquid/semi-solid dosage forms that limit the role of dissolution in co-presentation strategies at the epithelium [78].

The majority of PEs in proprietary oral delivery systems are relatively mild surfactant-based detergents requiring high mM concentrations to alter permeability of mucosae *in vitro* and milligram to gram quantities to perform *in vivo*. Depending on the physicochemical properties of the PE and peptide, such high concentrations present difficulty for formulation and process optimisation. For example, salts of medium chain fatty acids have good flowability when granulated, but require inclusion of disintegrants to ensure rapid release from solid dosage forms. On the other hand, it is difficult to formulate tablets containing high quantities of non-ionic surfactants due to low melting points. For example, low temperature trituration and filtration of the PE, sucrose laurate (MP: 37–43 °C) through a wire mesh (180 µm) yielded a sticky cohesive powder with poor flowability (Maher S, and Brayden DJ, unpublished). At the high pressures required for tableting (1.5 kN, 3 kN) sucrose laurate underwent pressure-dependent solid-to-liquid phase transition which immediately reversed upon removal of the compression force, yielding waxy defective tablets with excessive hardness and poor disintegration. Such behaviour can be prevented by inclusion of other excipients, but this may lower the overall concentration of PE available for the formulation. While most peptides are more stable in the relatively quiescent solid form, there are stability considerations that can impact formulation. Inclusion of CA in oral peptide formulations can be problematic in cases where the peptide does not exhibit solid state stability at low pH values. For example, amorphous (lyophilised) insulin at low pH (3–5) decomposed to [desamidoA21] insulin or the covalent amide linked dimer, [Asp_{A21}-Phe_{B1}], which is accelerated by the presence of water [79]. Alternatively, acidifiers and alkalizing agents can attenuate chemical degradation [80]. This is not to say that aqueous solutions are preferable to solid peptide formulations, but one must take into consideration solid state stability of the final dosage form [81].

Low and erratic intestinal flux of peptides from oral formulations is likely to be influenced by intestinal fluid volume. Fasted state fluid volume ranges from 45 mL to 319 mL (83 mL median) that is distributed heterogeneously along the intestine, with an increase in fluid filled pockets (12 mL median volume) in the distal small intestine [18]. The most clinically advanced PEs obey concentration-dependence, so if the dosage form dissolves in a larger intestinal fluid volume, the concentration range required for permeation enhancement may not be reached. Hence high PE doses are required in oral dosage forms in order to overcome potential dilution effects [82]. On the other hand, if excessive concentrations of PE are required to compensate for dilution, there is the possibility of superficial mucosal damage in low volume compartments [83].

The average residence time in the colon (20–30 h) is far higher than that of the small intestine, and the colonic fluid volume is lower (mean fasted state volume: 13 mL; mean fed state volume 11 mL [18]); these features could facilitate optimal co-presentation of peptide with PE. Efforts to target this region focus on regional differences in enzyme digestion, pH, time, and pressure [84]. Success in colonic targeting is largely restricted to delayed release formulations of small molecules to locally treat inflammatory bowel disease (e.g. mesalazine, Pentasa®, Ferring, Switzerland) and also to gastro-resistant prodrug formulations that are activated by bacterial azoreductases (e.g. sulfasalazine, Salazopyrin®, Pfizer, USA). Specific advantages to colonic targeting for systemically-delivering therapeutic peptides are greater sensitivity of the colonic mucosa to PEs [85,86] and lower levels of luminal and brush border peptidases [21]. However, delayed absorption and low and inconsistent luminal volumes [18] will lead to variable dissolution and reduce access to the colonic mucosal surface. In sum, oral peptide formulation targeting to the colon in association with PEs ultimately has a rather weak rationale for systemic delivery.

4. Intestinal PEs

Intestinal PEs are a candidate excipient class that transiently increase permeability of co-administered payloads across the small intestinal epithelium (Fig. 2). The use of PEs has long been met with suspicion, which is influenced by the diversity of substances that alter the intestinal barrier and induce toxicity. Tables 1 and S1 together list over 250 substances that have demonstrated permeability enhancement action in pre-clinical studies. These include a range of natural, semi-synthetic- and synthetic substances: solvents (e.g. ethanol [87]) to chelating agents (EDTA [88]), surfactants (sodium caprate (C₁₀) [89]), endogenous secretions (bile salts [90]), drugs (acetylsalicylic acid [91]), and high MW polymers (e.g. polysaccharides [92]) and bacterial toxins [93]. In 1961, sodium EDTA (50 mg/kg) enabled absorption of heparin following oral administration in gelatin capsules to dogs [94]; three years later it was tested by rectal and sublingual delivery routes in man [95]. Since then, C₁₀ was approved in Scandinavia and Asia to improve rectal bioavailability of ampicillin from a suppository (Doktacillin™, Meda Pharmaceuticals, Sweden), although the efficacy of this approach was correlated to non-specific damage to the rectal mucosae rather than paracellular permeability modification [83].

Some PEs are marketed for delivery of poorly absorbed drugs by other administration sites including topical (ethyl oleate/glyceryl monooleate in Minitrans® patch; Meda Pharmaceuticals) and buccal (bile salts in Oralyn®; Generex, Canada). Although there are currently no marketed oral peptide formulations containing PEs, several formulations are in clinical development and Chiasma Pharma (USA/Israel) recently filed an NDA for oral octreotide (Mycapssa™). The Eligen® carrier (Sodium N-[8-(2-hydroxybenzoyl) Amino] caprylate (SNAC) was approved for oral delivery of vitamin B₁₂ (Eligen® B12, Emisphere, USA) and is in Phase III for oral delivery of the GLP-1 analogue, semaglutide (Novo Nordisk, Denmark).

PEs act either paracellularly via the opening of TJs or transcellularly through an increase in plasma membrane permeability, or a combination of both (Fig. 2). The number of substances that have been reported to increase transcellular intestinal epithelial permeability outweighs those that increase paracellular permeability by over 10 to 1 (Tables 1, S1). This is not surprising as it is far easier to non-specifically perturb the intestinal mucosae than to selectively open junctional complexes, but it is important to note that <5% of transcellular PEs have progressed to clinical assessment for oral peptide delivery. This indicates high attrition of substances that do not address (i) significant but temporary enhancement, (ii) epithelial recovery and (iii) established safety profile in man. New generation TJ modulators specifically target the paracellular route, and promising results from pre-clinical studies could translate into technologies suitable for clinical trials.

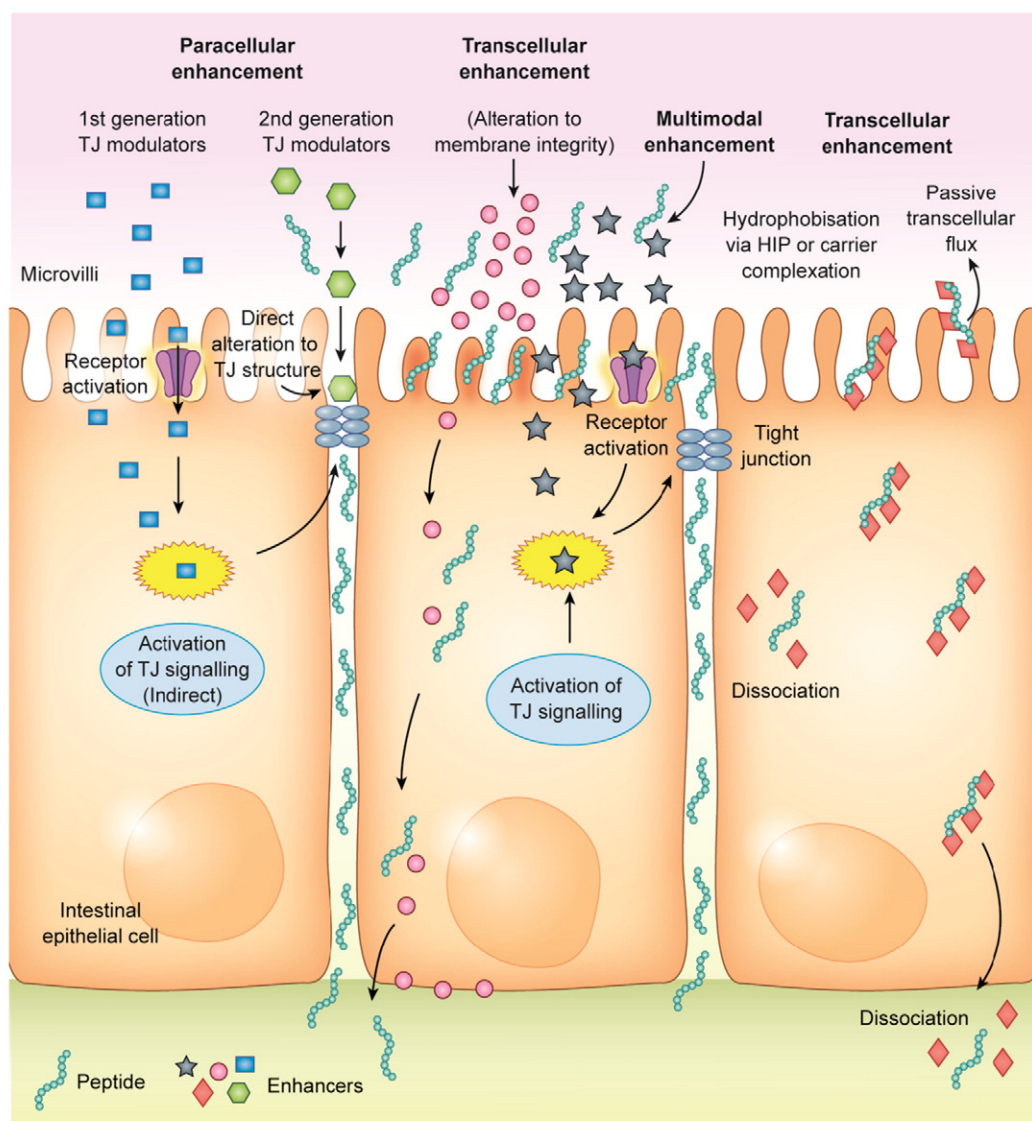


Fig. 2. Modes of PE action. Paracellular PEs are divided into two classes. 1st Generation paracellular PEs increase intestinal permeability by targeting cell signalling pathways involved in disbandment of TJs. 2nd generation directly target the physical disruption of TJ by interfering in intercellular homophilic interactions. Transcellular PEs act via alteration to the integrity of the cell plasma membrane or via hydrophobisation of the target therapeutic peptide. Selected permeation PEs exhibit both paracellular and transcellular enhancement action in a concentration and/or time dependent fashion, and these PEs are referred to multimodal PEs.

4.1. Paracellular PEs

A system of categorisation designates TJ openers as either 1st generation or 2nd generation. For the most part, 1st generation act through alteration of intracellular signalling mechanisms involved in modulation of TJs, while 2nd generation directly disrupt homophilic interactions at cell adhesion recognition (CAR) sequences between TJ or adherens junction (AJ) proteins in adjacent epithelial cells (Fig. 2). The elucidation of proteins that regulate TJ function, in particular actomyosin, occludin, claudin, and tricellulin, has provided molecular understandings of actions induced by both 1st and 2nd generation TJ openers. In the 1980–90s, 1st generation modulators included cytochalasins [96,97], C₁₀ [89], ZoT [98], nitric oxide donors [99], and EDTA [88]. These PEs aided the development of 2nd generation modulators, in particular, sequences targeting the extracellular loop of E-cadherin [100], occludin [101], and claudin (C-CPE [102]) (Table S1). Many patent filings disclose 2nd generation PEs that disrupt AJ and TJ complexes [103]. Selected 1st generation TJ modulators are more clinically advanced, and continuing efforts to discover novel PEs in this category

are ongoing. Omitted from discussion in this Section are paracellular PEs that have multimodal action, including C₁₀ and chitosan derivatives. In addition, PEs that increase epithelial permeability in MDCK monolayers were omitted if they have not also been reported to increase permeability in GI-related models.

4.1.1. Paracellular PEs emerging from the study of toxins

A number of toxins have been shown to alter paracellular permeability ([104] and Table S1); knowledge that has assisted in our understanding of TJ structure/function. For example, cytochalasins are a group of small molecule fungal metabolites that cause actin-mediated disbandment of TJ [105]. Likewise, although a large number of patents have been filed for ZoT and its capacity to generate oral insulin delivery in rats [106], the more important output was the discovery of zonulin, an endogenous mediator of epithelial permeability involved in intrinsic control of barrier integrity [107]. An associated discovery was that of larazotide acetate (Alba Therapeutics, USA), an 8-mer peptide that promotes TJ assembly [108] and is currently in clinical development for management of coeliac disease [109,110].

In their native form, toxins are not realistic candidate PEs due to safety considerations. The initial discovery of protein-based toxins like ZoT, melittin, and CPE was followed by structure activity studies to identify modified analogues that retained enhancement action but not the elements pertaining to toxicity. This approach was effective for CPE (C-CPE) and ZoT (AT1002 [111]), but not melittin [112]. Native ZoT (~45 kDa) increased small intestinal permeability via PKC-dependent cytoskeletal contraction [113], and SAR studies localised the enhancement action to the 12 kDa carboxyl terminal region (termed Delta G) and subsequently to its first six amino acids [114]. This short peptide sequence was designated AT1002 (H-FCIGRL-OH) (Alba Therapeutics (USA)) [111]. While the enhancement action of AT1002 was reversible, high mg concentrations were required to induce a 40-fold enhancement of Lucifer Yellow in Caco-2 monolayers [111] and in addition, the peptide did not increase duodenal absorption of cyclosporin A unless co-administered with a peptidase inhibitor (bestatin) and a surfactant (benzalkonium chloride) [115]. Melittin is a small cell penetrating and antimicrobial peptide that also modulates the function of a number of mammalian receptors and signalling proteins (e.g. calmodulin [116]). While melittin was an efficacious PE in Caco-2 monolayers [117], there was a reduction in its efficacy in isolated GI tissue mucosae [118] and in rat models [119]. Substitution of selected amino acid sequences in melittin was effective in eliminating the peptide's ability to form pores in mammalian membranes, but the capacity for permeation enhancement was lost [112]. Viral protein 8 (VP8) is a PE derived from rotavirus that improved enteral delivery of insulin in diabetics rats [120]. A number of patents have been filed for VP8 relating to oral insulin delivery [121]. VP8 altered the distribution pattern of TJ-associated claudin, ZO-1, and occludin, but did not decrease overall expression. Sequence analysis of VP8 showed that regions of this protein have sequence homology with extracellular loops of claudin and occludin. If prospective studies demonstrate that VP8 acts through direct disruption of homophilic interactions between claudins or occludin in adjacent enterocytes, analogues of VP8 could become promising 2nd generation PEs.

4.1.2. Paracellular PEs that bind claudins

CPE is one of at least 14 toxins produced by *C. perfringens* which triggers food borne illness through the formation of ion selective channels in enterocyte plasma membranes, resulting in fluid secretion and Ca^{2+} -dependent enterocyte apoptosis [122]. SAR analysis found that the N-terminal was responsible for pore formation, and the C-terminal sequence (C-CPE_{184–319}) was the first protein reported to modulate the function of claudin [102]. While C-CPE is a toxin, its capacity to bind claudins and the design of novel claudin modulating sequences warrants distinction from conventional toxins.

Claudins are a family of TJ proteins (~23 kDa) having at least 25 members; they are expressed at different sites within the body in distinct combinations typically containing 3–5 family members. Claudin proteins are positioned at the apical most point of TJ structures organized at the apical neck of polarized intestinal epithelial cells. All claudin family members contain four transmembrane domains that form two extracellular loops and can establish cis- and trans-interactions that are both homophilic and heterophilic in nature to establish the permselective paracellular barrier properties of a specific epithelium. These extracellular adhesion points are therefore the most targetable TJ component for modulation of the paracellular barrier. PEs that bind either the first (Claudin-1_{53–80} peptide [123], C₁C₂ [124] and C-CPE [125]) or second (C-CPE [102]) extracellular CAR motifs interfere in TJ formation leading to an increase in paracellular permeability (Table S1).

Comparison of the enhancement action of C-CPE with C₁₀ revealed comparable efficacy in rats, but at a 400-fold lower dose for C-CPE (0.1 mg/mL versus 40 mg/mL) [93]. However, these experiments were performed in closed intestinal rat loops, and C-CPE had a lag time of ~60 min before jejunal enhancement of FD4 flux could be detected, whereas C₁₀ maximally enhanced FD4 flux in <60 min. Slow onset of

enhancement action is not uncommon for peptides that directly modulate the function of claudin as both claudin-derived peptide [123] and the claudin-1 peptidomimetic (C1C2) [124] required long incubation times before inducing a significant alteration in epithelial permeability. Pre-treatment with C-CPE for 4 h prior to administration of PTH improved F_{ABS} of the peptide from 0.6% (co-administration) to 2.7% (pre-treatment) when delivered in a rat jejunal loop, which suggests slow recovery of barrier integrity with this PE [126]. However, formulators cannot use pre-treatment in development of an oral peptide dosage form. Removal of the first 10 amino acids from native C-CPE_{184–319} yielded a truncated structural variant (C-CPE_{194–319}) that had a fast onset of action in rat jejunal loops. However, the concentration required to elicit such a response was high at 4 mg/mL (F_{ABS} : 1.3%) and the potency relative to C₁₀ was reduced to only 10-fold [126]. Recombinant engineering of C-CPE through random mutation at positions 304 (S-to-A), 305 (S-to-P), 307 (S-to-R), 309 (N-to-H), and 313 (S-to-H) yielded a variant termed m19, which exhibited a 2.1-fold increase in enhancement action compared to native C-CPE, as measured in rat jejunal loops [127].

4.1.3. Paracellular PEs that target E-cadherin and Ca^{2+}

An increase in intracellular Ca^{2+} can increase epithelial permeability through the activation of calmodulin, which activates myosin light chain kinase (MLCK) to phosphorylate actomyosin II, culminating in disbandment of TJs [128]; a mode of action proposed for C₁₀ [129,130]. In contrast, bilateral or basolateral depletion of extracellular Ca^{2+} from cultured intestinal epithelial monolayers increased paracellular permeability by disrupting Ca^{2+} dependent E-cadherin at AJs [131]. Depletion can be achieved by incubating cells or tissues in low Ca^{2+} buffers or using chelating agents (ethyleneglycoltetraacetic acid (EGTA) [131] and EDTA [132]). In addition, barrier alteration is reversed by restoring Ca^{2+} levels [133]. While the AJ does not solely control the gate function of the paracellular route, it plays an important role in formation and modulation of TJs through indirect binding to TJ-associated proteins (ZO-1) and cytoskeletal reorganisation [134]. For example, treatment with EGTA can activate mg^{2+} myosin ATPase that is followed by cytoskeletal contraction and disbanding of the TJ [135].

Chelation of extracellular Ca^{2+} remains one of the most widely studied approaches to alter paracellular permeability that has been pursued with excipients including EDTA, EGTA, and CA; frequently as sodium salts (Table 1). EDTA forms water soluble complexes with several metals of physiological importance. Although the Ca^{2+} -EDTA chelate is weak compared to lead, zinc, copper, and iron, this interaction is sufficient to alter intestinal paracellular permeability through sequestering the Ca^{2+} required by E-cadherin [133] and by activating PKC [136] (reviewed in [137]). It is not clear how PKC activation results in decreased TJ permeability, since activation by phorbol myristate acetate phosphorylates and inactivates MLCK, leading to a decrease in MLC phosphatase (MLCP) and an increase in TJ barrier integrity [138]. EDTA also improved intestinal permeability in a number of pre-clinical delivery models (Table 1).

The ability of EDTA to chelate and deplete extracellular Ca^{2+} can also protect the peptide, as Ca^{2+} is a co-factor for many proteases. The permitted use concentration of EDTA typically ranges between 0.01 and 0.1% w/v, and the FDA Inactive Ingredients List permits levels of up to 5mg in oral tablets. In theory, if a tablet containing 5mg of EDTA was to dissolve to completion in 5–50 mL of intestinal fluid, the expected regional concentration would range from 0.1 to 0.01% w/v. The permeation enhancement concentration of EDTA tested in animal models ranges between 0.03 and 0.6% w/v (Table 1). In one study, its concentration was 50 mg/mL (5% w/v), which far exceeds the regulatory limits [94]. *In vitro*, the enhancement action of EDTA can often be erratic when solely added to the mucosal surface as opposed to bilateral addition. Apical addition of EDTA (2.5 mM) in Caco-2 monolayers had only a 2-fold increase on the permeability of FD4, whereas basolateral EDTA

application produced a 10-fold increase and a bilateral application of the chelator resulted in a 322-fold increase in the permeation of this solute [132].

Oramed Pharma (Israel) has developed Peptide Oral Delivery (POD™), which improved oral delivery of antidiabetic peptides in dog and pig studies, a technology that is now in Phase 2 trials. Formulations of exenatide used in a canine duodenal cannulation model contained 150 mg of sodium EDTA [26], a quantity that could result in an estimated luminal concentration of ~0.3–3% (w/v); which is higher than the maximum recommended levels in the FDA Inactive Ingredients List. Oramed have also combined EDTA with insulin (8 mg, SBTI [125 mg], aprotinin [150,000 IU] and omega-3 fish oil [1 mL]) [139]. A formulation containing insulin (8 mg), Carrier (≤ 150 mg), Adjuvant_A (≤ 125 mg), and Adjuvant_B (≤ 24 mg) improved glycaemic control in type 1 diabetics [140]. The mode of enhancement action of EDTA at high concentrations has not been comprehensively studied. Pre-clinical testing showed that the effects of high concentrations of EDTA (50 mg/mL; 5% w/v) can be blocked by Ca^{2+} salts, which suggests that enhancement mechanism remains coupled to a paracellular mode of action. Given the role of Ca^{2+} in cell metabolism, attempts have been made by others to avoid the use of chelators by selectively targeting E-cadherin using peptides that scramble intercellular contact interactions [141].

4.1.4. Paracellular PEs that target occludin

Elucidation of the occludin structure, along with reports showing the effectiveness of C-CPE in modulating claudin expression, led to the rational design of a prototype 2nd generation TJ modulator that targets occludin. Peptide sequences that correspond to the first extracellular loop motif of occludin (OP_{90–113}, OP_{90–135} and OP_{90–103}) can scramble homophilic interactions between adjacent epithelial cells and improve intestinal permeability [101]. However, even the shortest occludin peptide (OP_{90–103}) was only active when added bilaterally (68-fold benefit) or basolaterally (11-fold benefit) to epithelial monolayers on Transwells®. Efficacy (enzymatic/physical stability, targeting) of OP_{90–103} was improved by synthesis of an all D-form, inclusion of peptidase inhibitors, and conjugation to a hydrocarbon moiety (C₁₄-OP_{90–103}) but, to our knowledge, there has been no further progression of this PE or others that target occludin. A number of other occludin binding peptides have shown enhancement action in epithelial models derived from kidney but there are no equivalent reports in GI delivery models [142].

4.1.5. Paracellular PEs and cytoskeletal reorganisation

Cytoskeletal control of TJ permeability is a widely studied target to improve oral peptide delivery. The phosphorylation of MLC by MLCK results in a conformational shift in the structure of myosin II in enterocytes resulting in cytoskeletal contraction and increased TJ permeability; possibly through ZO proteins and/or cingulin [143]. This is a transient physiological process under non-pathological conditions, but activated MLCK can be associated with chronic inflammatory conditions of certain pathologies [144]. A number of PEs increase MLCK activity including EGTA [135], cytochalasin B [145], cytochalasin D [146] and C₁₀ [146]. The cellular control of cytoskeletal organisation is dynamic, with contraction occurring when MLC is phosphorylated and relaxation occurring when MLC is de-phosphorylated via an endogenous phosphatase. Given such dynamic control of the phosphorylation state of MLC, selected inhibition of MLCP prevents dephosphorylation, thereby indirectly increasing the level of MLC-P without the necessity for MLCK activation. Two rationally-designed candidates, permeant inhibitor of phosphatase (PIP) peptide 640 (RRDYKVEVRRKKR-NH₂) and PIP peptide 250 (RRFKVKTKKKR-NH₂), target the interaction between sub-units of MLCP; specifically interaction between protein phosphatase 1 (PP1) and MYPT1 (myosin phosphatase target subunit) (PIP640) and the interaction between PP1 and CPI-17 (C-kinase-activated protein phosphatase-1 (PP1) inhibitor–17kDa) [147]. Both PIP 640 and PIP 250 were all D-forms and cationic amino acids were introduced to mimic the behaviour of cell penetrating peptides (CPPs). Both

peptides increased FD4 permeability across Caco-2 monolayers and also insulin delivery in rat intestinal loop instillations (F_{REL} 4% (PIP 640) and 3% (PIP 250)) (Table S1). PIPs caused a gradual decrease in transepithelial electrical resistance (TEER) in Caco-2, but the high concentrations of PIP 250 used in instillations were effective with a shorter onset time.

4.2. Transcellular PEs

Permeability enhancement is achieved for the most part by PEs acting on the plasma membrane including fatty acids, acylcarnitines, acylated amino acids, alkyl polyethoxylates, glyceryl polyethoxylates, channel forming peptides, bile salts, glycerides, sucrose esters, polysorbates, enamines, and salicylates (Tables 1, S1). The majority of these are surfactants, and while several thousand surfactants are used in a diverse range of pharmaceutical, cosmetic and industrial applications, only a few meet the criteria for success (Section 4). Transcellular PEs can be broken down into (i) surfactants that alter the integrity of the enterocyte plasma membrane to enable better transepithelial flux of the co-administered molecule (this Section), (ii) complexing agents that bind to the molecules and transiently increase their passive transcellular diffusion (Section 4.3), and (iii) non-surfactant PEs with poorly defined modes of action (Section 4.4).

4.2.1. Soluble surfactant PEs

Surface active agents share the capacity to preferentially adsorb at the boundary between distinct phases of matter to lower interfacial tension, which is a direct consequence of structural amphiphilicity. They are broadly categorised as soluble (this section) or insoluble (Section 4.2.2) amphiphiles, which depends on the relative contribution of the hydrophobic moiety (hydrocarbon chain length, degree of branching, aromaticity) and the hydrophilic moiety (anionic, non-ionic, amphoteric or cationic). While both soluble and insoluble amphiphiles are surface-active and are surfactants *per se*, the term is more commonly applied to soluble amphiphiles.

The most common surfactant-based PEs are soluble in an aqueous continuous phase, and the most effective enhancers are those that exist in high concentrations in the monomeric form, but not so high as to impede their ability to insert into enterocyte plasma membranes. Soluble surfactants can be further sub-divided based on their physical structure at high concentration, with sub-group 1 exhibiting lyotropic mesomorphism, a capacity to form supramolecular cubic, hexagonal or lamellar liquid crystal structures. Most PEs lie within the soluble sub-group 1, including fatty acid salts, non-ionic surfactants (polysorbate 20) and acylcarnitines. Surfactants in soluble sub-group 2 do not exhibit lyotropic mesomorphism [148], although this group also contains PEs that have progressed to clinical evaluation (*e.g.* bile salts). The hydrophobic moieties of sub-group 1 are usually aliphatic, while sub-group 2 are aromatic. This distinction is important because the behaviour of bile salts differs from aliphatic PEs, making it difficult to correlate physicochemical properties and permeation enhancement. A wide range of metrics relate surfactant structure and physical behaviour to transcellular permeation enhancement, including the critical micelle concentration (CMC), HLB, as well as the magnitude of change to surface tension ($\Delta\gamma$), and aggregation number (*n*).

4.2.1.1. CMC in permeation enhancement. In dilute form, the monomeric surfactant exists as a molecular dispersion ($<1\text{nm}$) and above the CMC they self-associate into micelles, which are thermodynamically stable, isotropic dispersions that lie within colloidal dimensions ($1\text{--}500\text{nm}$). Although there is no linear relationship between CMC and permeation enhancement, it represents the concentration of surfactant that exists in solution (monomeric form), which is ultimately responsible for detergent action and transcellular permeation enhancement. CMC is an indication of the solubility of the monomeric form, as any further addition leads to formation of micelles. Therefore, the CMC

value indicates the amount of free surfactant available to diffuse from the bulk vehicle to the plasma membrane to initiate enhancement. It is a common misconception that detergent effects only begin above the CMC; however, it is unclear if micelles can alter membrane permeability directly. For example, C₁₀ exhibits detergent-based perturbation of Caco-2 monolayers at 10 mM [149], but its CMC in physiological buffers is ~26 mM [150]. While the intestinal epithelial enhancement capacity of MCFAs clearly increases above their CMC, this is because micelles are a reservoir for efficient replenishment of free monomer lost from solution upon interaction with enterocytes, in addition to solubilising fragments of cell membrane following detergent action.

In general, the higher the CMC the greater the detergent effect, which is why anionic surfactants are more effective PEs compared to non-ionic surfactants, although these actions also correlate with a higher level of membrane damage. The typical CMC range for non-ionic and amphoteric surfactants is 0.1–1 mM, which means that the free surfactant monomer concentration is far lower than for anionic surfactants [151–153]. In addition to a lower free monomeric concentration, non-ionic surfactants commonly form larger micelles, which make them effective drug solubilisers. Non-ionic surfactants can however, still exhibit strong enhancement. Despite their higher CMC values, the efficiency by which surfactants penetrate the plasma membrane often decreases at much higher CMC values [153]. From a molecular perspective, surfactants that have very high aqueous solubility have a significant imbalance in their HLB. For example, ionic surfactants (e.g. carboxylates) with very high CMC values normally have very short hydrocarbon tails that are less efficient at perturbing phospholipid bilayers (e.g. hydrocarbon chains <C₆). Likewise, non-ionic surfactants that have very high HLB values have either very short alkyl chains (<C₄) or very long ethoxylate moieties (E₂₀ +), both of which influence their interaction with the plasma membrane.

The relationship between CMC and enhancement action is confounded when using the surfactant at concentrations below their CMC value. Consider the CMC values in water of an homologous series of fatty acid carboxylates, C₆ (CMC = 1000 mM [154]), C₈ (CMC = 351 mM [155]), C₁₀ (CMC = 95.5 mM [155]), C₁₂ (CMC = 23 mM [155]), C₁₄ (CMC = 6.9 mM [155]), C₁₆ (CMC = 3.9 mM [156]), C₁₈ (CMC = 1.8 mM [156]). When comparisons are made between C₈–C₁₂ at relatively low concentrations (e.g. 10 mM, the common concentration used in Caco-2) the detergent effect is strongest for C₁₂ because its greater hydrophobicity will permit more efficient insertion into the plasma membrane. While C₁₄ to C₁₈ are even better able to insert into the plasma membrane, the free monomer is only present at 6.9–1.8 mM concentrations, indicating that there is less free monomer available for longer chain fatty acids. If the concentration of C₁₀ is increased to 100 mM (the level of C₁₀ used in animal models), the monomer concentration of C₁₀ (95 mM) that is available to alter permeability is far in excess of that of C₁₂ (23 mM), indicating that C₁₀ will demonstrate more efficacious enhancement compared to C₁₂. This point is important in candidate development, because a PE that is the most effective in Caco-2 monolayers and isolated intestinal mucosae (where effective concentrations lie below the CMC), might not be the most effective in animal models. Furthermore, C₆ and C₈ have significantly higher CMC values than MCFAs owing to very high aqueous solubility, but their short alkyl chain length does not favour membrane penetration. It is not always the case that surfactants with low CMC values are more efficient at inserting into the plasma membrane. Due to unusual stereochemistry a lower molar ratio of cholate (0.5) was required to completely solubilise PC bilayers compared with sodium deoxycholate (0.7) despite a higher CMC (16 mM versus 6 mM) [157]. Nevertheless, sub-CMC concentrations of deoxycholate had significantly greater enhancement action compared with cholate in Caco-2 monolayers [158].

Any molecule or physical change within the dispersion that lowers the free monomer concentration can alter the rate and extent of enhancement. Sequestration of free surfactant into mixed surfactant micelles with endogenous or dietary surfactants (e.g. bile salts, phospholipids,

mono- and di-acyl glycerol), emulsification of dietary lipids and/or adsorption to solid particulates all have the potential to alter enhancement action. Although the formation of mixed micelles does not reduce the CMC of a surfactant PE, the sequestration of free monomers into mixed micelles with endogenous surfactants effectively removes the monomeric surfactant and limits enhancement action. Enhancement action of alkyl maltopyranosides (C₁₀–C₁₄) across Caco-2 monolayers were therefore attenuated in the presence of bile salts and lecithin due to removal of monomers into mixed micelles [159]. PEs cannot avoid endogenous bile salts within the upper GI, where levels are 2–6 mM in the fasted state and 7–16 mM in the fed state [160], although one would expect such loss of efficacy to be reduced at *in vivo* PE dose levels above their CMC, as free monomer concentration would be efficiently restored and can offset initial removal into mixed micelles by the presence of bile salts. Low concentrations of surfactant-based PEs often cause a rapid decrease in TEER in Caco-2 monolayers, but then no further decrease is detected over time [150,161]. Initially, the predominant monomeric form is likely to have stripped phospholipids from the plasma membrane to reduce TEER and increase paracellular permeability, but then the free monomer concentration is reduced due to incorporation into the membrane and into mixed micelles. It is clear therefore that the free monomer concentration of surfactant significantly varies depending on the environment.

The capacity of ionic surfactants to form micelles can be impacted by increasing electrolyte concentration or by altering pH to reduce surfactant ionisation. Given that micelle formation is an index between water/hydrocarbon repulsion and ionic repulsion between head groups, an increase in ionic strength of the counter-ion can neutralise repulsion between head groups and ultimately reduce the CMC (e.g. bile salts [157]). On the other hand, decreasing the pH of C₁₀ below its pK_a leads to the formation of insoluble capric acid, which does not form micelles, and effectively converts C₁₀ into an insoluble less efficient PE surfactant that interacts poorly with the intestinal epithelium.

4.2.1.2. HLB in permeation enhancement. High HLB values (10–20 +) are assigned to surfactants with high aqueous solubility and low values (1–10) to those with low solubility. In general, the most effective PEs in advanced pre-clinical or clinical testing are low MW, anionic/non-ionic surfactants that have HLB values of ~15–25 (e.g. C₁₀, lauroyl carnitine). It is however, not possible to predict enhancement potential for a candidate PE from HLB alone, but prediction accuracy can be improved when values are aligned with other common metrics including CMC, change in surface tension ($\Delta\gamma$), and aggregation number as well as structural information (MW, alkyl chain length (C_x), ethoxylate chain length (E_x) and pK_a). Detergent effects are evident at HLB values of 13–18, yet many anionic detergents and PEs have HLB values above this range, such as the most effective carboxylates (C₁₀, HLB 21) and sulphates (SDS, HLB 40). A high HLB value therefore does not necessarily correlate with a high CMC, nor will it predict a strong detergent efficiency or capacity for permeation enhancement. While both the CMC and HLB give an indication of surfactant dispersibility, the key difference is the inability of HLB to distinguish between solubility in the monomer and micellar forms. For example, Poloxamer 188 (P188) ((polyoxyethylene-polyoxypropylene-polyoxyethylene (E₈₀P₃₀E₈₀)) has a HLB value of 29, which might suggest a high CMC (high free monomer solubility), yet monomer solubility for P188 is low (CMC 0.8 mM), indicating good solubilisation but poor detergent and permeation enhancement effects.

4.2.1.3. Surfactant structure in permeation enhancement. Surfactant structure plays a significant role in enhancement action. Ionisable functional groups in charged surfactants prevent close packing of hydrophilic head groups and the efficiency with which new surfactant monomers are inserted into micelles. This ultimately leads to higher CMC values, improved detergency, and enhancement action compared with non-ionic surfactants. The CMC for a comparable series of dodecyl hydrocarbon chains varies by three orders of magnitude depending on the hydrophilic head group: dodecyltrimethylammonium chloride (CMC 20 mM),

SDS (CMC 8 mM), disodium 1,2-dodecylsulphate (CMC 40 mM) and $C_{12}E_6$ (CMC 0.09 mM). Divalent counter ions can also reduce repulsion between ionised hydrophilic moieties leading to significant lowering of the CMC. In some cases, divalent ions are known to complex and neutralise surfactant charge leading to precipitation of fatty acid (for example as Ca^{2+} salts) and decreased enhancement action [162]. Therefore dietary Ca^{2+} could have an impact on enhancement action; for example, milk altered the interaction of both fatty acids and bile salts with colonic epithelial monolayers [163].

Cationic and anionic surfactants are more commonly used to improve penetration through skin because the *stratum corneum* is more resilient than the intestinal epithelium. When these surfactants are used to improve flux across the skin they are referred to as penetration enhancers. The behaviour of zwitterionic surfactants is similar to mild non-ionic surfactants and some examples show promise as PE candidates in intestinal tissue (e.g. dimethyl palmitoyl ammonio propanesulfonate (PPS) [164]). Ethoxylates are commonly used non-ionic surfactants and ethoxylate chain lengths between E_{10} and E_{20} are more effective PEs than E_{20} – E_{40} , furthermore, alkyl chain lengths between C_{12} and C_{16} had the strongest enhancement of poorly permeable marker molecules [165]. In rat stomach, $C_{12:0}$ and $C_{18:1}$ from an alkyl series of C_{12} – C_{18} exhibited the strongest enhancement action with ethoxylate chains between E_{10} and E_{20} [165]. The hydrophobic moiety of the surfactant also influences enhancement action. Traube's rule states that for a homologous surfactant series, the concentration required for an equal lowering of surface tension in dilute solution decreases 3-fold for each additional methyl functional group. Such a rule indicates that the surface saturation for surfactants will occur at lower concentrations, which is why CMC decreases so significantly with increasing alkyl chain length, and goes some way to explaining why there is such a significant increase in intestinal epithelial permeation enhancement in MCFA moving from C_8 to C_{12} . When the CMC and efficiency of membrane insertion are both considered, the most effective alkyl chain length for MCFA and alkyl sulphates is between C_{10} and C_{12} [150,166]. Fatty acids of uneven chain length are less widely studied than conventional fatty acids, but they also demonstrate efficacy (e.g. C_9 , C_{11} , $C_{11:1}$ [70]). Enhancement action of $C_{11:1}$ (FD4, F_{ABS} 21.8%) was comparable with C_{10} (FD4, F_{ABS} 22.4%) in rat jejunal intestinal instillations. However, performance $C_{11:1}$ (FD4, F_{ABS} 40.2%) in solid dosage forms was improved in the same model compared to C_{10} (FD4, F_{ABS} 26.3%). The difference in CMC between C_{10} (23 mM) and saturated $C_{11:0}$ (5 mM) would suggest that a higher level of C_{10} will be available to enhance permeability in instillations at a concentration of 100mM, yet unsaturation of the $C_{11:1}$ (17 mM) at carbon 10 significantly raises the CMC. Here, the increase in CMC for a longer alkyl chain with improved membrane perturbation appears to strengthen the permeation enhancement potential of $C_{11:1}$ compared with both $C_{10:0}$ and saturated $C_{11:0}$ [70].

4.2.1.4. Mechanism of action of soluble surfactants. The capacity of surfactants to alter intestinal permeability relates to detergent-like action, where the monomeric form adsorbs and penetrates the mesophasic plasma membrane leading to removal of membrane constituents. The concentration of surfactant that can be accommodated in the plasma membrane influences permeability, and so enhancement is a function of both the physicochemical properties of the PE and the composition of the enterocyte plasma membrane. For example, the phospholipid profile of enterocyte plasma membrane is dependent on the species, the levels of maturation along the villi, the region of the intestine, and even within specific micro-domains of individual enterocytes [167]. Diet can also alter composition of the plasma membrane, which can influence fluidity and permeability [168]. The plasma membrane itself can therefore contribute to intestinal region-specific enhancement and variable efficacy of surfactant-based PEs. For example, it is more difficult for the surfactant Triton® X-100 to penetrate membrane domains rich in sphingomyelin (SM) compared with phosphatidylcholine (PC)

[169]. Although both phospholipids are insoluble swellable amphiphiles, SM forms more tightly packed bilayers due to higher hydrocarbon saturation [170] and strong association with cholesterol [171] (reviewed in [172]).

Surfactant-based PEs increase transcellular flux by altering the integrity of the plasma membrane of intestinal epithelial cells. There is initial adsorption of the surfactant, followed by penetration of the monomeric form into the outer layer of the phospholipid bilayer, and then a flip-flop into the inner leaflet [173]. Studies in model membranes indicate that bilayers can accommodate some surfactant without loss of integrity [152]. At a certain concentration, the penetration of surfactant may fluidise the plasma membrane. For example, insertion of octylglucoside into unilamellar phosphatidyl choline (PC) vesicles led to an increase in diphenylhexatriene fluorescence anisotropy [174]. An increase in membrane fluidity is facilitated by an initial swelling of the bilayer in to a more expanded state (e.g. swelling upon insertion of octylglucoside into unilamellar PC vesicles [152]). In this state, the surfactant may lead to an increase in membrane permeability without causing lysis and micellar solubilisation. For example, increased permeability of phosphatidylcholine (PC) bilayers to a marker solute (Mn^{2+}) was observed following treatment with sodium deoxycholate [175]. The insertion of a non-native surfactant into bilayers alters the natural packing integrity of cholesterol, phospholipids and proteins, which alters membrane fluidity. C_{10} appears to disrupt intermolecular forces and packing organisation of brush border membrane vesicles (BBMV), as indicated by a decrease in fluorescence polarisation of two fluorophores that specifically label the interior and exterior of the bilayer [176,177]. Similar behaviour was observed with other fatty acids and bile salts [158].

A threshold of surfactant is eventually reached where no further expansion of the bilayer can occur without destabilisation. In membrane models, the continuous addition of surfactant results in linear increase in surface pressure to a plateau, above which surface pressure decreases and membrane integrity is reduced (e.g. Brij® 70 [175]). Because mixed micelles are not detected immediately following a drop in surface pressure [165], it is possible that phospholipids (PC and SM) and proteins are dislodged and exuded into the luminal fluid, an event that could be associated with increased permeability without initial lysis [152]. The driving factors for removal of membrane constituents are the high surface pressure, the dynamic adsorption/desorption of soluble surfactant, and their ability to bind proteins and lipids. Above this concentration, the destabilising effect of surfactant on the membrane causes buckling, lysis and fragmentation in to laminar membrane segments that are exuded into the luminal media. These fragments form laminar cell membrane fragment-surfactant mixed micelles, as illustrated in the Small and Mixed-Disc models [148]. Further addition of surfactant decreases the ratio of phospholipid, protein and cholesterol in the mixed micelles and reduces their size [178].

4.2.1.5. Lead soluble surfactants as intestinal PEs. Soluble surfactants represent the largest category of PEs, with some being used in oral peptide clinical trials (Aguirre *et al* in this Issue [4]). In many cases the extent of permeation enhancement is still relatively low because either the CMC is too low to permit efficient alteration in membrane integrity or because their alkyl chain length does not permit insertion into the plasma membrane. At the other extreme, strong detergency may cause effects beyond that of endogenous detergents and might not be safe. The leading candidates are discussed in following sections.

Case 1. C_{10}

The capacity of C_{10} to clinically improve intestinal drug absorption was demonstrated in rectal suppositories of ampicillin, for which it gained approval in Sweden (1992) and Japan (1985). Doktacillin™ suppositories were marketed by Astra Pharma (Sweden) and contained ampicillin (250 mg), C_{10} (25 mg) and hard fat (950 mg, Pharnasol™

B-105, NOF Corp. Japan). The inclusion of C₁₀ in this formulation improved rectal F of ampicillin in man from 13% to 23% [83]. Doktacillin™ is now part of the portfolio of Meda (Sweden), but it is no longer marketed. In the early 1990s studies demonstrating enhancement action and mode of action were carried out in Caco-2 monolayers [162], rat intestinal mucosae mounted in Ussing chambers [179], and rat *in situ* intestinal closed loops [20] (Table 1).

In the last 30 years, C₁₀ became a candidate PE in a number of proprietary formulations of biomolecules such as proprietary oligonucleotides [180], desmopressin [181], acyline [182], low MW heparin (LMWH) [72] as well as poorly permeable small molecules (e.g. alendronate [181], pemetrexed [181], and zoledronic acid [89])). C₁₀ is the key MCFA in the Gastrointestinal Permeation Enhancement Technology (GIPET™, Merriam Pharmaceuticals, Ireland [183]) and enteric-coated oral solid dose formulations containing C₁₀ have been in clinical trials for oral delivery of insulin and GLP-1 analogues [181]. C₁₀ is also listed in patent filings by Biocon (India) as an additive to improve oral delivery of a PEGylated alkylated insulin (reviewed in [2]). C₁₀ is efficacious in both small and large intestine, although it was more effective in the colon for FD4 and insulin in rat intestinal instillation [20,70,118,184,185]. Disparity is also noted between the effective enhancement concentrations of C₁₀ *in vitro* and *in vivo*, which relates to differences in GI transit, release kinetics, dilution, interaction with bile salts and dietary lipids.

Typical effective concentrations of C₁₀ vary between those for Caco-2 monolayers (10 mM; 2.5 mg/mL buffer [119]), isolated tissue mucosae (10 mM; 2 mg/mL buffer [119]), loop gut instillations (50–100 mM; 10–20 mg/mL saline [118]), and mini-tablet insertion to rat loops (500 mM; 100 mg/mL saline [70]). Efficacious concentrations in larger species include those for intra-duodenal intubation in dogs (185–500 mM; 550–1650 mg/15mL [180]), intra-duodenal administration to pigs (500–1000 mM; 2.5 g [186]) as well as oral tablet delivery in dogs (150–990 mg) and man (660 mg) [187]. In general, the formulations tested in pre-clinical models have to be refined for clinical application for practical reasons. When mini-tablets (50 mg) containing FD4 (10 mg), C₁₀ (30 mg) and microcrystalline cellulose (Avicel® PH102, FMC Biopolymer, UK) were inserted into the rat small or large intestine with a small volume of PBS, the F_{ABS} of FD4 was improved from 7% to 26% in jejunum, and from 2% to 30% in colon. In rats, the oral F of a short cyclic peptide that functions as a GP IIB/IIIA receptor antagonist, DMP 728, was increased from 2.4 to 6% when delivered in size 9 microcapsules containing ratios of C₁₀ (40%), PEG (13.4%), PEG1450 (13.4%), water (26.8%) and the DMP 728 salt (6.3%) [188]. Formulation of DMP 728 in hard gelatin capsules containing 150 mg of C₁₀ improved absorption in dogs by approximately 3 fold with an increase in oral F from 5% to 19%, again with considerable inter-subject variability. Relating pre-clinical data in rodents to man is difficult because normalised doses of active and excipient used in rats are far higher than the doses that can be administered in clinical trials. For example, administration of 10 mg of FD4 to rodents (40 mg/kg) would normalise to a dose of nearly 2.8 g in man.

The ability of C₁₀ to improve oral absorption of oligonucleotides was tested in pigs and dogs, and in clinical testing by Ionis Pharma (USA, formerly Isis Pharmaceuticals). Oral delivery of enteric-coated tablets with an oligonucleotide (240 mg) and C₁₀ (990 mg) yielded an oral F of 1.7% in beagles, although a therapeutic threshold was achieved [180]. Disintegration of this oligonucleotide formulation occurred in less than 60 min, where the rate of C₁₀ release was linear from 5% at 15 min to 85% after 45 min, suggesting that ~25% of C₁₀ was released in 30 min at an approximate rate of 9 mg/min. Although no data is known about release *in vivo* or on the local concentration that is present at the intestinal mucosa, it is probable that slow release along with relatively fast transit and variable fluid volume results in jejunal apical membrane exposure to low mM concentrations of C₁₀ for a short duration, consistent with *in vitro* studies. As the concentration of C₁₀ that is effective *in vitro* is less effective in intestinal instillations, slow release from the Ionis

formulation may have been responsible for the poor data in dogs. Delivery of C₁₀ (2.5 g; 500–1000 mM) with 250 mg oligonucleotides to pigs via jejunal intubation was equally disappointing (F for Ionis-2503: 1%, F_{ABS} for Ionis-1004838: 3%). In man however, delivery of Ionis-104838 (100 mg) in enteric-coated capsules loaded with immediate release mini-tablets containing C₁₀ (660 mg) improved oral F to 10% compared to s.c., with the authors concluding that delivery of 4 × 100 mg capsules delivered weekly may lead to the required systemic absorption [187]. Nevertheless, both inter- (F_{REL}: 0.02–28.1%) and intra- (F_{REL}: 0.03–28.1%) patient variability was high, and Ionis discontinued oral development of oligonucleotides.

Case 2. acyl carnitines in CA-based formulations

Acyl carnitines are soluble surfactants that improve intestinal permeability in animal models (Tables 1, S1). The presence of a quaternary ammonium imparts a positive charge at physiological pH values and, as the pK_a of a carboxylate is 3.8, acyl carnitines exist as zwitterions in the small intestine and their CMC values are lower than anionic surfactants (e.g. lauroyl carnitine chloride (LCC): 1.2 mM) [189]. Several carnitine esters, in particular palmitoyl carnitine chloride (PCC), improved rectal absorption of cefoxitin and a somatostatin analogue in dogs [190]. PCC also improves absorption when formulated in enteric coated tablets and delivered to dogs (200 mg cefoxitin, 600 mg PCC) [58]. Formulation of DMP728 (18–20 mg) with LCC (100 mg, enteric coated) or PCC (100mg, non-enteric coated) in gelatin capsules improved oral bioavailability in dogs almost 2-fold to 21% and 17%, respectively [188]. Both LCC and PCC are the main PEs in Peptelligence™ [191], the oral peptide delivery system of Enteris BioPharm (USA). Peptelligence™ is an enteric coated solid dosage form containing the API, acyl carnitine, and CA. The CA-containing particles are coated to prevent interaction with the enteric coating. In 2009, Peptelligence™ was licensed to Tarsa Therapeutics (USA) for clinical development of an oral formulation of recombinant sCT, which successfully achieved its primary endpoint in a phase III trial of Oral Calcitonin in Postmenopausal Osteoporosis (ORACAL) [192] (discussed by Aguirre *et al* in this Issue [4]). Importantly, the dose of sCT (0.2 mg, 1200 IU) in Peptelligence™ tablets containing 500 mg CA was 6-times higher than the nasal preparation (200 IU Miacalcin®, Novartis, Switzerland) and 12 times higher than the s.c. dose (100 IU Miacalcin®, Novartis).

The iteration of Peptelligence™ used in the ORACAL trial did not contain an acyl carnitine [192], despite their inclusion in clinical trials by Enteris with other peptides [191,193]. In preclinical testing, oral sCT absorption was increased in rat by 10-fold when LCC was included, increasing F_{ABS} to 8% [194]. The Peptelligence™ formulation also contains a disintegrant to accelerate dissolution and hence facilitate more rapid absorption following gastric emptying. However, the residence time of enteric coated monolithic formulations within the stomach is erratic (4–120 min). This is because ingestion of dosage forms is not synchronised with muscular contractions (migrating myoelectric complex) that bring about gastric emptying. Within the study protocol for oral PTH 1–34, patients were directed to self-administer the Enteris formulation 2 h after their evening meal; which depending on the nature of the diet could impact on gastric emptying time [195]. In optimisation of Peptelligence™ formulations containing CA and LCC, the polymer used for enteric coating of capsules influenced both T_{MAX} and oral F of an LHRH analogue in dogs in the rank order of Eudragit® L30 D55 (10% w/w, duodenal release T_{MAX}: 111 min, F: 3%) < Eudragit® L30 D55 (15% w/w, duodenal release T_{MAX}: 116 min, F: 4.6%) < Eudragit® L30 D55, and FS 30 D (12% w/w, duodenal/colonic release T_{max}: 152 min, F: 7.2%) [191]. These data suggest that delayed release can impact oral peptide delivery.

The dose of acyl carnitine used in the clinical studies has not been reported, although an iteration tested in a proof of concept study contained CA (473 mg), taurodeoxycholic acid (75 mg), LCC (75mg) and sCT (0.82 mg) in Eudragit® L30-D55-coated capsules. The mean

C_{MAX} of calcitonin for 5 patients was reported to be 5–10 times above the targeted therapeutic plasma level [194]. It is not clear if the interaction between ingredients has an impact on enhancement, especially the dissociated form of CA and the quaternary ammonium ion of LCC, but testing in rodent instillations [194,196] and oral delivery in dogs [197] clearly indicate the benefit of combining CA and LCC. For example, duodenal delivery of sCT (0.1 mg) with just CA (77 mg) had a F_{ABS} of 0.7% compared with a F_{ABS} of 5.4% in a formulation containing sCT (0.1 mg), CA (48 mg), and LCC (5 mg) [194]. Likewise, LCC (55 mg) in the absence of CA had minimal effect on absorption of sCT from enteric coated capsules, yet the combination of LCC (56 mg) with CA (565 mg) improved oral absorption [198]. Effort has been afforded to assess the effect of regional pH on absorption of sCT, as well as several other biopharmaceutical considerations [53,82,197,199]. These studies show the relationship between intestinal pH and absorption of sCT, and while the magnitude of CA's effect on absorption of sCT is less than the combination of CA with LCC, the current NDA reflected a desire to omit PEs for regulatory reasons, whereas CA is an accepted excipient [192,200].

The action of CA has been attributed to stabilisation of sCT by reduction of intestinal pH [53,198]. Following gastric emptying enteric coated capsules containing CA (656 mg) and radiopaque dye took 40 min to disintegrate, after which the pH remained below pH 5 for 90 min [53]. A correlation was found between reduction in luminal pH and improved absorption of sCT in beagle dogs [198]. However, there is a lack of clarity on the effect of CA on barrier integrity. While relatively low concentrations (4% w/v) of CA had no impact on TEER in isolated rat jejunal mucosae mounted in Ussing chambers [53], such a concentration is lower than that present in rat intestinal instillations (192 mg/mL; 19% w/v) and potentially lower than the regional concentration following release from dosage forms. The luminal pH measurements from pH 1 to pH 1.5 [198] tend to correlate with more concentrated solutions of CA (10–40 %w/v). CA (0.57 % w/v) and the basic citrate form increased the permeability of mannitol, but not FD4 in isolated rat colonic mucosae [25]. CA had stronger enhancement action than salicylic acid, taurocholic acid, and polysorbate 80. In the same study, an enteric-coated capsule formulation containing sCT (0.5 mg) and CA (10mg) induced a F_{ABS} of 1.8% in rats [201]. The existence of CA in the non-dissociated form at low pH impedes its ability to chelate extracellular Ca^{2+} and hence alter paracellular permeability. At the same time, increased intestinal permeability in isolated rat colon by CA and tartaric acid has led to the theory that absorption of acidic forms of organic acids results in intracellular acidosis (leading to an increase in intracellular Ca^{2+} and subsequent contraction of actomyosin) or depletion of ATP (resulting in Ca^{2+} -ATPase inhibition, phospholipase activation and cytoskeletal destabilisation) [202].

Case 3. ethoxylates

The enhancement action of a diverse group of non-ionic surfactants has been reported in pre-clinical delivery models. These comprise medium and long chain sucrose esters, ethoxylated sorbitan esters, medium chain ethoxylates (C_xE_y , where C_x is the hydrocarbon number and E_y is the ethylene oxide number) and ethoxylated glycerides (Tables 1, S1). The naming convention for the hydrophilic moiety of non-ionic surfactants makes it difficult to identify suitable candidates. For instance, polyethylene glycol (PEG) has the International Non-proprietary Name (INN) macrogol, but it is frequently referred to as polyoxyethylene (POE), polyethylene oxide (PEO) and ethoxylate (E_y), depending on the regulatory jurisdiction, supplier, and end user. Proprietary names, identification codes and even rebranding can also impact tracking of non-ionic surfactants, for example polyethoxylated-30 castor oil is marketed under the trade name Cremophor® EL (new trade name Kolliphor® EL).

Fatty acid- and fatty alcohol-ethoxylates enhanced oral and rectal bioavailability of heparin and small molecules in animal models, in particular those within the series C_{10} – C_{16} and E_8 to E_{25} [203,204], although

strong enhancement is sometimes reported outside this range (e.g. $C_{16}E_2$ [203]). A wide range of surfactants in this series are classified as excipients in several regulatory jurisdictions, primarily in topical delivery as emulsifiers, wetting agents and solubilisers. As might be expected, surfactants in this series that are permitted for use in enteral delivery have longer hydrocarbon chain length and ethoxylate chain length that lie outside of the optimal region for detergent- and enhancement actions (e.g. $C_{18}E_{40}$ (Myrj® 52; Croda, UK) [205]). The most effective PEs in this category are however, potent detergents (examples including $C_{12}E_9$ and $C_{12}E_{23}$ [206]) that were originally categorised as strong enhancers with slow recovery (Class III, see Section 6).

Fatty ethoxylates typically form viscous liquids, unctuous semi-solids, and/or malleable soft solids that are difficult to incorporate into oral solid dosage forms compared to salts of fatty acids. For example, $C_{12}E_8$ (50mg) increased the P_{app} of [^{14}C]-mannitol by 6-fold across isolated rat colonic mucosae compared with a 9-fold increase for C_{10} , but $C_{12}E_8$ is a liquid at room temperature. Formulation development strategies can however, occasionally permit transformation of liquids into solids for solid dose formulation (e.g. through the adsorption of liquid on to granulated magnesium aluminometasilicate [207]). The requirement for a relatively high quantity of adsorbent, and other excipients, reduces the quantity of the non-ionic surfactant per unit dose, which could compromise the overall enhancement action of the dosage form. On the other hand, non-ionic surfactants can be incorporated more easily into microemulsions and oily suspensions and this is an important research area.

$C_{12}E_9$ is one of the most efficacious PEs in the C_xE_y ethoxylate series with comparable enhancement action to EDTA in rectal administration [208] and to C_{10} in jejunal and colonic instillation in rats [209] (Table 1). The suspension of insulin in a mixture of corn oil and $C_{12}E_9$ (BL-9-EX, Nikko, Japan) improved rectal absorption of insulin from suppositories [210]. In a similar study, rectal suppositories (2 g) of insulin dispersed in a hard fat base (Witepsol® W35, Cremer Oleo, Germany) with $C_{12}E_9$ (1%, 20 mg) reduced plasma glucose level in streptozotocin-induced diabetic dogs, but when higher concentrations of $C_{12}E_9$ were used, enhancement action was reduced, which led the authors to suggest that micellar solubilisation of insulin reduced the free concentration of insulin [211]. Formulation of non-ionic surfactants in gel-based preparations for rectal administration gives further insight into how non-ionic surfactants might be presented to the intestinal epithelium. Inclusion of $C_{12}E_9$ improved absorption of eel calcitonin following rectal delivery in a polyacrylic acid-based hydrocolloid gel (Carbopol® 941, Lubrizol, Japan) [212].

Alkyl phenol ethoxylates are a series of detergents sold under the trade names, Triton® (Dow, USA), Ipegal® (Rhodia, France) and Nonident® (Shell Co, Netherlands). Several of these surfactants have been tested in oral delivery models (Table S1). In particular, octyl phenol ethoxylate (average $E_{9.5}$) (Triton® X-100) is a detergent that, unlike sodium lauryl sulphate, does not denature membrane proteins [213]. A combination of Triton® X-100 (5% w/v) and the mucolytic, N-acetylcysteine (1% w/v) improved F_{ABS} of sCT from 0.1% to 1.6% in rodent gut loop instillations [44]. Comparison of several octyl phenol ethoxylates within the range of $E_{7.5}$ –to- E_{100} , showed a centred symmetrical pattern of enhancement versus HLB that peaked at an ethoxylate chain length of 15 (HLB: 15) [43]. Likewise, in the case of nonyl phenol ethoxylates, significant enhancement of phenol red absorption was observed in the rat gut perfusion model for E_9 (HLB: 13), $E_{10.5}$ (HLB: 14) and E_{20} (HLB: 16), but not E_{30} (HLB: 17), E_{50} (HLB: 18) or E_{100} (HLB: 19) [214]. Furthermore, there was a good correlation between membrane perturbation and enhancement.

Case 4. fatty acid- and ethoxylated sugar esters

Fatty acid esters of monosaccharides (Sorbitan, glucose) and disaccharides (sucrose, maltose), and their ethoxylated variants give rise to a diverse range of surfactants with application in oral delivery. Ethoxylated

sorbitan esters (polysorbates) are common excipients used in oral delivery of poorly soluble drugs (e.g. polysorbate 20 and polysorbate 80 [215]). Several polysorbates exhibit modest to poor intestinal permeation enhancement capacity that is lower than that of anionic surfactants (Table S1) [216]. While polysorbates have relatively high HLB values, the quantity of free surfactant available to elicit membrane detergency is low, which is reflected in modest enhancement (e.g. polysorbate 20, CMC: 0.006%; polysorbate 85, CMC: 0.002% [217]). Polysorbate 80 (PEG-20 sorbitan monooleate) did not improve rectal absorption of insulin (up to 5%) from an acrylic gel delivered to rodents, unlike $C_{12}E_9$ [212]. However, when dispersed in oily vehicles (e.g. glyceride base (Miglyol®; Sasol, South Africa)), enhancement effects of polysorbates were improved compared to that seen in aqueous dispersions [218]. For example, polysorbate 20 (PEG-20 sorbitan monolaurate) improved rectal absorption of marker sulfanilic acid by 4-fold but when dispersed in Miglyol® it was improved by 33-fold. Inclusion of polysorbate 60 in suppositories [containing a 1:1 mixture of hard fat (Witepsol® H15) and PEG (400:1000:6000 in a part ratio of 1:8:1)] improved rectal absorption of insulin in rabbits [219].

Sucrose esters are a group of non-ionic surfactants formed by esterification of sucrose with up to eight fatty acids, giving rise to a variety of functionally and structurally diverse non-ionic surfactants. This group of surfactants are permitted in oral formulations in major regulatory jurisdictions with functions including solubility enhancement, lubrication and emulsification [220]. Excipient grades of sucrose esters are not purified forms of monoesters rather they are supplied as heterogeneous mixtures containing percentages of sucrose mono-, di- and poly- esters. For example, sucrose laurate (D-1216, Mitsubishi Science, Japan; Table 1) contains 80% sucrose monolaurate (HLB of 15) but also contains 20% of the di-, tri- and poly-ester. Given that sucrose di-laurate, with a HLB value of 5 is practically insoluble in water, the dispersions formed by commercial sucrose esters are not simple micellar systems; rather they are complex dispersions that range from mixed micellar systems to microemulsion pre-concentrates composed of monoester (surfactant) and diester/polyester (co-surfactant/oil). This is similar to other oil-in-water (o/w) microemulsion pre-concentrates including Labrasol® (Gattefosse, France), which are composed of 90% surfactant (PEG-8 C8/C₁₀ glycerides) and 10% co-surfactant/oil (medium chain glycerides).

The effect of five food-grade sucrose esters (0.25–1% w/v) was compared in a rat intestinal loop study, with sucrose stearate (75 % monoester) demonstrating higher efficacy than sucrose laurate (80% monoester) using a marker molecule, carboxyfluorescein (CF) [221]. The AUC_{0–240min} of CF was improved by 5-fold with sucrose laurate (1% w/w) and to 6-fold with sucrose stearate (0.5% w/w). Rectal insulin absorption was improved with sucrose caprate and sucrose laurate, but not with sucrose palmitate or oleate (all at 80% monoester) [185]. Interestingly, both sucrose caprate and sucrose laurate induced comparable CF intestinal permeation enhancement to MCFAs. Sucrose laurate also demonstrated comparable enhancement action to C₁₀ and Triton® X-100 in a rat duodenal instillation of FD4 [222]. Importantly, enhancement of sucrose laurate was MW-dependent, with the greatest effect seen on absorption of FD4 (14-fold), and FD10 (8-fold), but no effect on FD70 absorption was observed [222].

These data and the regulatory status present a strong case for development of sucrose esters for oral peptide delivery. However, in a comparison of enhancement action between C₁₀ and the pure form of sucrose monolaurate, a far higher quantity of the non-ionic surfactant was required to deliver comparable enhancement of ¹⁴C mannitol across isolated rat intestinal mucosae (Maher and Brayden, unpublished). Given that 500 mg doses of C₁₀ have been tested in oral dosage forms designed for man, the level of sucrose laurate that would be required to improve oral delivery in dosage forms *in vivo* requires further evaluation. Furthermore, sucrose esters are malleable/waxy solids that are difficult to manipulate in the solid form especially at high temperature. Of concern were the low quality tablets that formed when high

quantities of sucrose laurate (42 mg and 8 mg fluorescein) were made into standard mini-tablets. These tablets exhibited sticking, tackiness, and poor disintegration (>60 min) [223]. While sub-optimal formulation and process properties may be addressed, the relatively low level of sucrose laurate capable of being encapsulated within the formulation is problematic.

Case 5. alkyl maltosides and glucosides

Alkyl maltosides and glucosides have a similar structure and function to sucrose esters, the distinction being that their lipophilic moiety is typically linked to the carbohydrate via glycosidic or thioglycosidic bonds. In general, the most effective disaccharide maltosides have better enhancement capacity than the most effective monosaccharide glucosides (Tables 1, S1). The order of enhancement for boosting CF absorption in rat rectal instillations was dodecyl maltoside (DDM) (10 mM, F_{ABS}: 73%), decyl maltoside (20 mM, F_{ABS}: 62%), decyl glucoside (20 mM, F_{ABS}: 61%), octyl glucoside (30 mM, F_{ABS}: 48%), dodecyl glucoside (20 mM, F_{ABS}: 26%), and it is noteworthy that the highest DDM concentration was more effective than C₁₀ (50 mM, F_{ABS}: 35%) and C₁₂ (50 mM, F_{ABS}: 68%) [224]. Direct comparisons do not, however, consistently indicate that maltosides are more effect than glucosides, as variation in both CMC values and efficiency of membrane perturbation impacts permeation enhancement. For example, the CMC of hexyl glucoside is 250 mM, but despite such a high free monomer concentration, its lower capacity as a detergent reduces its efficacy for membrane perturbation and enhancement (F_{REL} insulin: 8%, buccal insulin absorption in rats). Glucosides with more hydrophobic side chains have lower CMC values but generate higher levels of membrane perturbation; giving rise to better permeation enhancement (e.g. dodecylmaltoside: CMC 0.6 mM, F_{REL} insulin: 30%) [225]. A greater improvement in epithelial permeability was observed with decyl glucoside (F_{REL}: 18%, CMC: 3mM) compared to decyl maltoside (F_{REL}: 6%, CMC: 1.6 mM), the data suggesting that the balance between free surfactant and membrane perturbation facilitated better enhancement for the glucoside [225]. On the other hand, the more hydrophobic dodecyl glucoside adversely changes the balance between CMC (lowered CMC to 0.13 mM) and membrane perturbation, which reduced enhancement action (F_{REL}: 7%). However, the balance between CMC (0.6 mM) and membrane perturbation is clearly better with dodecyl maltoside, which induced good intestinal permeation enhancement (F_{REL}: 30%) [225].

The most prominent members of this group of PEs are DDM and tetradecyl maltoside (TDM), which form part of the Intravail® platform (Aegis Therapeutics, USA) that is currently focused on nasal peptide delivery [226] but is also being examined in preclinical testing for oral delivery of octreotide and buccal delivery of desmopressin. Both DDM and TDM have been tested in oral delivery models (Table 1). DDM (0.1% w/v, 26-fold) had a greater effect on intestinal FD4 permeability compared to TDM (0.1% w/v, 6-fold) in isolated rat colonic mucosae [227]. In a follow on study, TDM improved colonic absorption of sCT from rat intestinal instillations [228]. However, like many surfactants, the actions of TDM were region-specific, in the rank order of colon>ileum>jejunum. This effect was also observed with decyl maltoside in rat intestinal instillation with CF [224]. Colon specific delivery of CF (1 mg) with DDM (1 mg) improved oral F_{ABS} from 17% in oral solution to 68% in enteric coated colon specific chitosan capsules in rats [86]. Nevertheless, convincing colonic delivery data for systemic absorption of peptides has not yet been presented for these PEs [229]. There are also reports of enhancement efficacy in the upper GI; DDM increased insulin absorption following duodenal instillation in rats by 13% [230]. Oral formulation of the thyrotropin-releasing hormone (TRH) analogue, azetirelin, in enteric coated capsules containing lactose (120 mg), DDM (15 mg) and CA (100 mg) improved oral F_{ABS} from 15% to 44% in beagle dogs [231]. It is noteworthy that CA was first loaded into gelatin capsules that were subsequently inserted into the enteric coated dosage form, and that over 90% dissolution occurred in <30 min. Furthermore, the F_{REL} of octreotide

was 4% when delivered as an oral solution to rats with 0.5% DDM, EDTA (0.1%) and sodium acetate (pH 4.5). At higher concentrations, DDM was less effective e.g. 1.5% (F_{REL} : 0.7%) and 3% (F_{REL} : 0.5%) [232,233]. In the same study, absorption was biphasic with T_{MAX} at both 10 min and 30 min, which has not been observed in other pre-clinical delivery models. Intravail® (0.3% DDM in PBS) also improved oral absorption (F_{REL} s.c.) of the 7-mer peptide, [D-Leu-4]OB3 from 1% to 4% in mice [234]. Both studies are impressive, especially as both [D-Leu-4]OB3 and octreotide were administered via oral gavage.

Case 6. bile salts

Bile salts are another of the most widely tested intestinal PEs in oral peptide delivery (Tables 1, S1). In initial PE classifications, common bile salts were defined as Class II (moderate efficacy, fast recovery) [206]. Many studies have focused on their role in intestinal physiology and pathophysiological conditions (e.g. [235]). The most widely tested bile salts in oral peptide delivery are sodium deoxycholate, cholate, taurodeoxycholate, and taurocholate, all of which exhibit detergent characteristics, but have varying extents of free surfactant monomer and capacity to interact with plasma membranes [236]. Each of these bile salts can reduce the insulin degrading activity of intestinal brush border homogenates [237]. More specifically, sodium taurodeoxycholate (2 mM) inhibited enzymatic activity of endopeptidases-2 (47%) and DPPIV (32%) [237]. Unlike fatty acids, no clear order of intestinal permeation enhancement can be delineated between bile salts analogues (Table 1), but improved intestinal permeation has been observed for a range of poorly permeable payloads: calcitonin [82], heparin [238], HRP [239], insulin [240], acylated calcitonin [241], a proprietary cyclopeptide [242], rhEGF [243], interferon [244], ebitaride [245], and EPO [246].

A number of proprietary oral peptide delivery systems list specific bile salts in patents (e.g. Genexer [247], Biocon [248], Oramed [139], Aegis [249], Unigene [194], and Nod Pharma [250]). Sandoz Pharma (Basel, Switzerland) evaluated an oral formulation of octreotide (4 mg) containing either ursodeoxycholate (100 mg) or chenodeoxycholate (100 mg) administered to healthy volunteers in hard gelatin capsules [90]. While only modest absorption was noted for the dosage form containing ursodeoxycholate (F_{ABS} : 0.26%), the formulation containing chenodeoxycholate improved F_{ABS} to 1.26 % and AUC ($ng \cdot h \cdot mL^{-1}$) to 56% of the i.v. dose (100 μg). On the other hand, delivery of insulin (150 IU) and sodium cholate (50 mg) in an enema had only modest effect on rectal absorption of insulin in healthy volunteers [251].

Biocon have a portfolio of patents on a proprietary PEGylated and alkylated form of insulin, IN-105. Although this analogue reduced post-prandial glucose excursion in phase II trials [252], it failed to adequately reduce glycated haemoglobin (HbA1c) in phase III. Additional patents refer to application of PEs in delivery of the conjugate. Several bile salts significantly enhanced glycaemic control of Biocon anti-diabetic peptides in rat, including sodium cholate, glycodeoxycholate, ursodeoxycholate, taurocholate, taurodeoxycholate [248]. An oral liquid formulation of IN-105, in which patients were dosed with IN-105 (0.06–0.25 mg/kg), contained sodium cholate (30 mg/mL), capric acid (5 mg/mL), lauric acid (5 mg/mL) and CA (62.5 mg/mL). Effort to improve absorption of IN-105 from a solid dosage form has focused more on other PEs including C_{10} , SLS and cyclodextrins [253]. In one example, IN-105 was formulated with granulated C_{10} and a super-disintegrant (crospovidone (Kollidon® CL, BASF)), a formulation shown to improve oral IN-105 delivery in beagle dogs [253].

Bile salts are also amenable to oral solid dose formulation as their free flowing solid form has similar physical characteristics to that of MCFA salts. This is emphasised by inclusion of sodium taurodeoxycholate in Peptelligence™ formulations developed by Enteris Pharma [194], where sodium taurodeoxycholate had comparable effect on oral bioavailability of sCT (F_{ABS} : 1.3–1.4%) compared with lauroyl carnitine when tested in enteric coated (hypromellose phthalate) gelatin capsules. In another

iteration of this dosage form, F_{ABS} of sCT (10mg) was 1.5% following oral administration to dogs with CA (600 mg) and taurodeoxycholate (80 mg) [194]. In an alternative presentation, oral insulin absorption was improved upon delivery in enteric coated particulates, which contained insulin (625 IU), soybean trypsin inhibitor (200 mg) and sodium cholate (615 mg).

Case 7. alkyl sulphates

Comparison of a range of alkyl sulphates of C_6 -to- C_{14} showed that maximal membrane perturbation (as quantified by protein release) is observed in the order decyl>lauryl>myristyl>octyl>hexyl [254] (Tables 1, S1). SDS is the most prominent alkyl sulphate tested in oral peptide delivery and one of the most widely used anionic surfactants in formulation science (Table 1). While the terms SDS and sodium lauryl sulphate (SLS) are interchangeable names for the C_{12} hydrocarbon tail, monographs state that SLS must contain $\geq 85\%$ SDS (with other alkyl sulphates making up the remaining 15%); indicating a subtle difference between SDS and SLS. The interaction of SDS with biological membranes differs in comparison to other surfactant-based PEs, as it is strongly denaturing [153]. Nevertheless, SDS is an excipient and a quantity of up to 95 mg (SLS-USP) is listed in the FDA Inactive Ingredients List. SDS was originally categorised as a Class III agent (strong enhancement and slow recovery) and relatively low concentrations are required to enhance permeability in Caco-2 monolayers (e.g. 0.4 mM SDS [255] versus 13 mM C_{10} [162]). Although rats showed recovery of barrier integrity following oral gavage of SDS, the relatively long time for recovery from even small quantities (20 mg/mL) suggest that application of this excipient in routine oral peptide delivery could lead to progressive deterioration. Biocon also evaluated SDS (as SLS) as a PE to improve oral IN-105 in beagle dogs [253]. Each tablet contained IN-105 (6 mg), SLS (50 mg), crospovidone (15 mg, Kollidon® CL, BASF) as well as mannitol (78 Pearlitol® SD 200, Roquette Pharma, France), silica (0.75 mg, Aerosil® 200, Evonik Corp) and magnesium stearate (0.75 mg).

4.2.2. Insoluble surfactants

Insoluble surfactants reduce surface tension by forming monolayer films at the surface of water, although they are practically insoluble and do not form stable micelles. This group can be sub-divided into non-swelling amphiphiles that do not exist in the aqueous phase (e.g. di- and tri-glycerides, cholesterol, long chain fatty acids) and swelling amphiphiles which can exist as lamellar liquid crystals (e.g. phospholipids, monoglycerides) [148]. The dispersion behaviour of insoluble surfactants can be explained according to HLB values. Surfactants of HLB 1–4 do not disperse and readily separate, HLB 3–6 disperse with difficulty, HLB 6–10 form coarse emulsions, HLB 10–13 form cloudy to clear dispersions and HLB 13–20+ form micellar solutions [256].

This PE category includes medium chain mono- and di-glycerides (e.g. monocaprin [257], MGK™ [258] (Table S1)). While insoluble surfactants have significantly lower intestinal permeation enhancement capacity compared with the soluble surfactants, this group is important because (i) membrane phospholipids and cholesterol are insoluble surfactants and their interaction with soluble surfactants forms the basis for enhancement action of the soluble class, (ii) some soluble anionic surfactants can undergo a shift to an insoluble form depending on their pKa and the pH of the vehicle, and (iii) because they contribute enhancement action in lipid-based delivery systems in clinical development in oral peptide delivery. Examples include those present in oily suspensions (e.g. Transient Permeation Enhancement, TPE™, Chiasma, Israel), reverse micelles (Soligenix Inc., USA) and microemulsions (Macrulin™, Provalis, UK). In most cases, insoluble surfactants are lipophilic oils, malleable semi-solids or waxes that function as both lipids and hydrophobic surfactants within the Lipid Formulation Classification System (LFCs) [259]. For example, dutasteride is formulated as a solution in mono- and di-glycerides of C_8 and C_{10} in soft gelatin capsules (Avodart®, GlaxoSmithKline, UK). The physicochemical attributes of

insoluble surfactants are problematic for formulation of hydrophilic peptides because the latter do not dissolve in oils; therefore formulated peptides require coarse dispersion in oil (in suspensions, reverse micelles, or w/o (micro) emulsions), which could present physical stability issues. If not properly stabilised, suspended drug forms are susceptible to aggregation and caking during storage, which can impede dissolution.

The physical behaviour of fluidic, insoluble surfactants in the presence of low concentrations of water presents the possibility for the formation of reverse micelles, where an aqueous micellar core is formed around an oil continuous phase. In simulations, hydrogen bonding between glycerol backbones of glyceryl monolaurate and glyceryl dilaurate formed reverse micelles [260]. If the size of the aqueous reverse micelle core can be optimised for solubilisation of a hydrophilic peptide, this type of micelle can protect it because it would partition within the nano-sized aqueous core and would be inaccessible to peptidases. A problem with application of reverse micelles and w/o dispersions is deterioration when diluted in aqueous environments. The addition of a large volume of water to a balanced system containing droplets of water-in-oil can result in phase inversion, cracking, or diffusion of the peptide to the diluent aqueous phase [261]. An oily dispersion of anhydrous reverse micelles of insulin was formed by freeze-drying an aqueous dispersion of insulin-ursodeoxycholate-phospholipid, followed by dispersion in an oil phase (polysorbate 20/caprylocaproyl glycerides) [262]. Dilution of this dispersion in water led to the formation of a coarse emulsion, but it destabilised the reverse micelle leading to release of 60% insulin into the aqueous environment. The authors concluded that the modest absorption of insulin in cannulated rabbit duodenum related to the enhancement of surfactant-based PEs rather than flux of the peptide in the reverse micelle form. In other examples where phase inversion was not observed, release of insulin from soybean phosphatidylcholine reverse micelles in medium chain triglycerides was low at 7–12% over 24 h, yet bioavailability was still observed from oral administration to diabetic rats [263]. Physical and chemical stability of reverse micelles has been demonstrated for 12 months [262]. Soligenix (USA) has developed a number of proprietary dispersions to assist oral delivery of peptides including liposomes (Orasome™, now discontinued) and reverse micelles (Lipid Polymer Micelle, (LPM™)) [264]. The LPM™ system is a reverse micelle comprising medium chain mono- and di-glycerides and glyceride ethoxylates, where patent disclosures include reverse micelles prepared from medium chain polyoxyglycerides (both PEG-6 (Softigen® 767, Cremer Germany) and PEG-8 (Labrasol®, Gattefosse)). Included are stabilisers (e.g. gelatin, poly-lactide and/or poly-glycolide) that slow the rate of phase inversion upon dispersion in luminal fluid. In rat duodenal instillations, LPM™ increased F_{REL} of leuprolide to 28% compared with 0.23% in solution form. When insoluble PE surfactants exist in semi-solid or solid forms, they can still be formulated in hard gelatin capsules. Overall, insoluble surfactants induce only modest intestinal permeation enhancement, slow release, and poor dispersion. This surfactant group is better suited for rectal formulations, as maintenance of close and prolonged contact with the rectal mucosae is possible from suppositories [265].

Case 8. acyl glycerols

Medium and long chain acyl glycerols or glycerides are excipients used as vehicles, emulsifiers and solubilisers in the delivery of poorly soluble drugs (e.g. Capmul®; Abitec, USA and Imwitor®; Sasol, South Africa). Medium chain monoacyl glycerols such as monacaprins and monacapyrlins (Capmul® 708G; Abitec, US) are generally the most effective intestinal PEs within this category [266], although efficacy as oral PEs is still moderate compared with soluble surfactants (Tables 1, S1). The non-ionic glycerol moiety is a weak hydrophilic moiety compared with anionic, cationic or medium-to-long ethoxylate groups. Thus, monoglycerides have poor aqueous solubility and low HLB values (e.g. glyceryl monooctanoate, (HLB 6.6), glyceryl monolaurate (HLB 5.2)). The rank order of enhancement of acyl glycerols in rectal delivery of a

small molecule, cefmetazole, from an oily suspension was monacapyrlin (F_{ABS} : 37%) > monacaprins (F_{ABS} : 18%) > monolaurin (F_{ABS} : 14%) [266]. In contrast, monacaprins were more effective than monacapyrlins for rectal delivery of cefoxitin in rats [267]. Acyl glycerols are useful in rectal delivery because they can be formulated as suppositories that intimately contact the rectal mucosa, whereas when orally-delivered in gelatin capsules, they have poor dispersibility in small intestinal fluid. Non-dispersible lipoidal vehicles are an important class within the LFCS [268], and are used for delivery of molecules with low aqueous solubility. In addition, the NDA for oral octreotide in an oily suspension (Chiasma's Mycapssa™) contains glyceryl monacapyrlate and glyceryl tricapyrylate, illustrating the relevance of these agents for oral peptides. In many instances, acyl glycerols comprise the core oil phase and are also co-surfactants in complex dispersions containing mixtures of soluble and insoluble surfactants. In such complex dispersions, there is only a moderate intestinal permeation enhancement action of acyl glycerols compared to those of soluble surfactants (C_8 , or glyceride ethoxylates). The intestinal epithelial surface area exposed to acyl glycerols is increased by formulation in emulsified forms with soluble surfactants such as Cremophor®, demonstrating the positive interaction between the two types of surfactants [269].

Nikkol MGK™ is a mixture of glycerides that has demonstrated intestinal epithelial permeation enhancement in rat [266] and dog models [270]. MGK™ is composed of glyceryl monacapyrylate (54%), glyceryl dicaprylate (36%), glyceryl tricapyrylate (6%) and caprylic acid (3%) [266]. For example, rectal absorption of encephalin in rats was improved from F_{ABS} 4% to 20% in the presence of these glycerides [271]. Rectal delivery of insulin in an MGK™ oily suspension had only a modest effect on rectal absorption in dogs (15% reduction in plasma glucose), but when insulin was first dissolved in water and emulsified in MGK™ to form a w/o emulsion, plasma glucose level decreased to 40% for 1 h [272]. Monacapyrlin, dicaprylin and tricapyrylin improved rectal F_{ABS} of cefmetazole from 0% to 37%, 7% and 3.8%, respectively. Glyceryl monacapyrylate (Capmul® 708G) also improved rectal absorption of gentamycin in rabbits, to a level that was half that of C_8 [273]. While monoglycerides are typically the most effective glyceride PEs, intestinal enhancement also occurs with medium chain triglycerides. The extent of enhancement by triglycerides is largely attributable to the free fatty acids and mono-/di-glycerides liberated by intestinal lipases. For example, trilaurin had no effect on absorption of cefoxitin in rat intestinal instillations *per se*, but addition of lipases enabled it [274].

4.2.2.1. Permeation enhancement from complex lipoidal dispersions. The combination of insoluble surfactant with soluble surfactant, water, co-solvent and/or oil gives rise to a wide range of pharmaceutical dispersions from mixed micelles to liposomes, solid lipid nanoparticles (SLNs) to coarse emulsions and microemulsions. These delivery systems are principally formulated in the o/w orientation, and are typically used for oral delivery of drugs with low aqueous solubility [275]. A growing number of delivery systems based on lipoidal dispersions improve oral peptide delivery. In many cases, the surfactants used in the stabilisation of lipoidal dispersions also function as transcellular PEs [161]. Some of the surfactants used in the formation of lipoidal dispersions (e.g. w/o microemulsions or w/o/w multiple emulsions) are less efficient PEs, so improved absorption more likely relates to improved dispersion and subsequent interaction of lipids with the epithelium.

4.2.2.2. Permeation enhancement from oil-in-water systems. Coarse or simple emulsions are thermodynamically unstable, anisotropic dispersions with a droplet size of approximately 0.2–10 μm [276]. Fixed oils are typically emulsified with soluble surfactants to form o/w dispersions that can be delivered as liquids or in gelatin capsules. The first formulation of oral cyclosporin (CsA) was a pre-concentrate coarse o/w emulsion (Sandimmune®, Novartis) containing corn oil, linoleoyl polyoxyl-6 glycerides (Labrafil® M2125CS), ethanol (12.7%) and glycerol in soft gelatin capsules. Oral F of CsA in Sandimmune® was erratic with values

ranging from 10%–89%. In contrast, microemulsions are optically isotropic, thermodynamically stable dispersions with a typical oil droplet size of $<0.2\ \mu\text{m}$; their smaller droplet size relative to coarse emulsions provides a larger interface for digestion and ultimately permits more efficient uptake via endogenous lipid absorption pathways. The second generation CsA formulation (Neoral®, Novartis) is an o/w microemulsion pre-concentrate that contains corn oil mono- di and tri-glycerides, polyoxyl 40 hydrogenated castor oil (Cremophor® RH40), dl- α -tocopherol, ethanol, glycerol and propylene glycol. The oral F of CsA from Neoral ranges between 20 and 50% [277]. As most therapeutic peptides have high aqueous solubility, they do not partition in oil droplets, and hence o/w systems do not protect the peptide from peptidases, although the soluble surfactants may improve permeability. Several excipient suppliers provide pre-concentrated fluidic or semi-solid vehicles that have a high proportion of oil/co-surfactant and soluble surfactants, making it possible for hydrophilic peptides to be dispersed as either oily suspensions or molecular dispersions that form by solubilisation of the peptide in hydrophilic micellar structures.

Excipients used in formulation of emulsions and microemulsions are often complex mixtures of non-ionic surfactants, co-surfactants, co-solvents and/or oils that are supplied as pre-concentrated vehicles with differing degrees of dispersibility in aqueous environments. In some instances mixtures of co-surfactants alone (e.g. glycerides, fatty acid esters of propylene glycol) provide an environment for dissolution of lipophilic drugs, but these are poorly dispersed in water and have poor intestinal epithelial permeation enhancement. Such mixtures can be formulated with soluble surfactants to form vehicles that when mixed under high shear form simple emulsions or spontaneously emulsify in an aqueous media. The Bancroft rule indicates the phase (oil or water) in which the surfactant is soluble, dictates whether oil (o/w) or water (w/o) droplets are formed, and so most oil in water emulsions form due to the presence of surfactant with solubility in the external aqueous phase (i.e. soluble surfactants). Although the soluble surfactant is in dynamic equilibrium with the emulsion interface and micellar structures [278], the free soluble surfactant monomer is still capable of eliciting transcellular enhancement action. For example, macrogolglycerides (50 mg/kg) improved absorption of heparin following jejunal delivery in rats *in vivo* [279]. These data are not surprising as the most effective non-ionic surfactants tested in oral peptide delivery have similar hydrocarbon (C_{8-12}) and ethoxylate ($_{8-20}$) chain length [280]. Not all ethoxylated lipoidal vehicles alter intestinal permeability, for example, Kolliphor® EL (BASF, Germany) [281] and Cremophor® RH60 [282] are poor PEs. Complex mixtures of macrogolglycerides and free glycerides form vehicles that are capable of perturbing the intestinal epithelium, reducing TEER, and increasing transcellular flux *in vitro* [283]. These include Labrasol®, Gelucire® (Gattefosse, France), Capmul® MCM (Abitec, USA), and Labrafil® (Gattefosse). However the concentration of surfactant in these dispersions that is available to elicit transcellular enhancement is difficult to ascertain compared with pure surfactant.

Case 10. Labrasol® and Gelucire® 44/14

Labrasol® is composed of 90% free PEG and medium chain polyoxylglycerides (termed macrogolglycerides in the EU) and 10% medium chain glycerides. The major PE component in Labrasol® is the mixture of the non-ionic surfactants, C_8 and C_{10} mono- and di- glyceride ethoxylates (E_8) (caprylocaproyl macrogol-8 glycerides). Labrasol® improved permeation of poorly permeable macromolecules (e.g. LMWH [161], erythropoietin [284], and FD4 [285]) in pre-clinical models (Table 1), and this feature has been outlined further in patents (e.g. [286]). Labrasol® can also improve jejunal and colonic absorption of insulin, although F_{ABS} was low at 0.2%, and less than the effects observed with other non-ionic surfactants [287]. Comparison of Labrasol® with C_{10} in rat intestinal loop instillations showed a 6-fold lower capacity of the former to boost absorption of LMWH [161]. Nevertheless, dispersions of Labrasol® have demonstrated strong enhancement action in

particulate dispersions: adsorption of Labrasol® to carbon nanotubes formed liquid filled nanoparticles that when mixed with sodium starch glycolate (Explotab®, JRS Pharma, Germany), casein, and EPO (100 IU/kg) formed a solid dispersion that improved F_{ABS} of EPO from 0.6% to 11.5% in rats [288]. Labrasol® has also been transformed into a solid for delivery of macromolecules. For example, heparin (25mg) was dispersed in Labrasol® (0.5mL) followed by solidification using Fluorite RE (Eisai Co. Ltd., Japan), Neusilin® US2 (Fuji Chemical Ind. Co. Ltd., Japan) or Sylysia™ 320 (Fuji Sylysia Co. Ltd., Japan), and loaded into enteric coated capsules for oral delivery in dogs [289]. Dissolution of heparin occurred in the order Sylysia™ 320 > Fluorite RE > Neusilin® US2, and enhancement of heparin bioavailability occurred in the order Fluorite RE (19%) > Sylysia (13%) > Neusilin US2 (5%). Solidified forms of Labrasol® can be prepared with relatively low quantities of carrier (Neusilin®), but the preparation of powders that exhibit the necessary characteristics for solid dose formulation (flowability, tableting, tablet disintegration, dissolution, hardness, uniformity) reduces the effective quantity of Labrasol® in the tablets (Maher S, and Brayden DJ, unpublished). Formulation of rhPTH 1–34 was achieved in a w/o microemulsion consisting of water (15%) and oil (85%), with the oil phase consisting of 6:2:1:1 of Labrasol®, medium chain triglycerides (Crodamol® GTCC, Croda, UK), macrogol-15 hydroxystearate (Kolliphor® HS15, BASF, Germany) and tocopherol acetate [290]. This dispersion delayed enzymatic degradation of rhPTH 1–34, increased permeability across Caco-2 monolayers, and improved bioavailability to 5% and 12% in rats by oral gavage and intestinal instillation, respectively. It is however noteworthy, that the daily dose of PTH in that study was in excess of the recommended daily injectable dose in man. Gelucire® 44/14 is a semi-solid excipient mixture containing 20% C_{12} glycerides and 80% C_{12} mono- and di-glyceride ethoxylates (E_{32}) (lauroyl macrogol-32 glycerides). Although Gelucire® 44/14 (HLB: 14) has a comparable HLB to Labrasol® (HLB: 12), and both belong to Class III of the LFCS, the ethoxylate chain for Gelucire® 44/14 is above the most effective length for permeation enhancement. In rodent studies there was less absorption of EPO [284] and LMWH [161] for Gelucire® 44/14-containing formulations compared to Labrasol®. Gelucire® 44/14 is also a constituent in an experimental multiple emulsion (water-in-oil-in-water) that improved bioavailability of calcein from 1.8% to 8% [291].

Case 11. innovative lipid blends

Oral delivery of antidiabetic peptides in fluidic dispersions has been disclosed in patent filings (Table S2). In one embodiment, oral delivery of a high dose of insulin in a non-aqueous dispersion of propylene glycol, Capmul® MCM, Pluronic® F127, and PEG 3350 in enteric-coated capsules lowered blood sugar in dogs [292]. In another, self-emulsified lipoidal vehicles showed efficacious permeation enhancement of hydrophobic forms of insulin (30–60 nmol/kg) in rat and canine intestinal instillations [293]. The formulation was based on five key ingredients, propylene glycol, Tween® 20, Labrasol®, and diglycerol caprylate blended in different proportions. There was no obvious concentration dependency observed with the PEs (Labrasol® and Tween® 20), rather the most efficacious absorption enhancement was observed with dispersions forming the smallest particle size. In general, self-microemulsified delivery systems demonstrated increased intestinal permeation enhancement relative to self-emulsified dispersions. Delivery of hydrophobised forms of insulin were also improved in solutions of propylene glycol injected into canine small intestine [294]. Other iterations evaluated in oral delivery of modified insulins include combinations with propylene glycol, medium chain monoglycerides (Capmul® MCM C_8 , MCM C_8/C_{10}), propylene glycol monocaprylate (Capmul® PG8), and Labrasol® [294]. In these patent filings, enhancement was demonstrated from enteric-coated formulations containing physiological insulin doses (0.17 mg/kg), which suggests that lipoidal dispersions may have promise.

Sigmoid Pharma (Ireland) has performed preclinical testing for oral delivery of sCT by formulating it in a lipoidal dispersion composed of an

oily mixture (medium and long chain triglycerides (Miglyol® 818, Sasol), diethylene glycol monoethyl ether (Transcutol® HP, Gattefosse) and PEG-35 castor oil (Kolliphor® EL, BASF)), which is mixed with an external aqueous phase containing gelatin and a PE (C_{10} , sodium taurodeoxycholate or coco-glucoside) to which sCT was added [295]. This dispersion was then extruded into cold oil to form semi-solid minispheres (1–2 mm). The physical properties of these lipoidal minispheres have not been reported, although the presence of Kolliphor® EL and a gelatin emulsifier suggest the formation of an o/w emulsion. While this dispersion had only a modest effect on intestinal absorption of sCT in rat instillations, the application of particulate forms of semi-solid lipoidal dispersions formulated in an enteric-coated hard gelatin capsules offers potential.

4.2.2.3. Permeation enhancement from water-in-oil systems. A number of peptides have been dispersed within the aqueous core of w/o microemulsions including arginyl-glycyl-aspartic acid (RGD) peptide, vasopressin, calcitonin and insulin [296]. Provalis (UK) developed a w/o microemulsion (Macrulin™) to improve oral delivery of insulin. Macrulin™ is composed of an external oil phase (Labrafil® M1944CS) with an aqueous disperse phase stabilised with lecithin and alcohol [297]. The optimised dispersion exhibited physical stability for at least six months at 4°C, 25°C and 40°C, although pseudo-ternary phase diagrams prepared during the preparation of the microemulsion clearly indicate the destabilising effect of water [297]. Intra-gastric delivery of 200 IU insulin in this microemulsion lowered blood sugar in diabetic rats to the same level as the s.c. dose (0.3 IU), but dose correction revealed a pharmacological activity (PA) of less than 0.2% [298]. Delivering Macrulin™ to healthy patients via intra-duodenal intubation increased plasma insulin levels and lowered blood sugar [299]. Development of Macrulin™ has since discontinued, but the potential for optimised enteric-coated formulations that regionally release fluidic w/o microemulsions in the small intestine warrants future study.

The aqueous environment of w/o dispersions can be tailored to suit the optimal solubility and stability for peptides leading to improved bioavailability [296]. For example, the PA% of insulin was 5% in dogs when dispersed in a lecithin-based w/o microemulsion and delivered in capsules coated for colonic release [300]. Significant improvement in oral insulin absorption in rats was reported from an experimental w/o microemulsion formulated with the cationic surfactant, didodecyldimethylammonium bromide (DMAB), propylene glycol (co-solvent) and tracetin (oil) [301]. In that study, pseudo-ternary phase diagrams suggest that the microemulsion retained its isotropic properties with significant aqueous dilution [302]. In general however, w/o microemulsions that are stabilised with soluble surfactants (such as DMAB), are sensitive to destabilisation and phase inversion following dilution. Oral delivery of w/o systems to average sized rats, where the total fasted volume is less than 0.2 mL [303], could mask potential phase inversion in man, where the fasted fluid volume is significantly higher [18]. Analysis of phase inversion of a w/o microemulsion (Miglyol® 812, Capmul® MCM, Tween® 80, and water) that improved intestinal uptake of a marker peptide showed that the dispersion was susceptible to phase inversion when diluted 5-fold with water, and partial inversion following 2-fold dilution [304]. Not all w/o microemulsions invert at such low ratios of water, as inversion has been reported in the presence of 100-fold to 1000-fold excess water [261], indicating that careful selection of dispersion additives can influence stability. However, improved stability will compromise enhancement action as the presence of soluble PE surfactants is one of the driving forces for phase inversion. Therefore, w/o microemulsions require a delicate balance between transcellular permeation enhancement and physical stability. The most effective strategy to deliver w/o systems is therefore within enteric-coated systems where the maximal volume of fluid that is likely to be encountered in the proximal small intestine is relatively low. A panel of w/o microemulsions were

evaluated in delivery of the RGD peptide following intra-duodenal delivery in rats [305]. There was no clear correlation between particle size and enhancement, but dispersions with higher concentrations of Capmul® MCM and Cremophor® EL had a greater effect on bioavailability. The most effective w/o microemulsion (RGD; F_{ABS} : 29% versus saline F_{ABS} : 0.5%) consisted of saline dispersed in propylene glycol dicaprylocaprate (Captex® 200; Abitec, USA), Capmul® MCM, lecithin (Centrophase™ 31, Central Soya Company Inc., USA) and Cremophor® EL. The inclusion of established PEs (e.g. C_8 and C_{10}) in w/o emulsions added a boost to bioavailability of calcinein administered via intra-duodenal instillation in rats [306]. A microemulsion containing these PEs improved bioavailability by 27-fold from 1.3% in solution to 36% in the w/o dispersion.

4.2.2.4. Permeation enhancement from multiple emulsions. Water-in-oil-in-water (w/o/w) emulsions are thermodynamically unstable multiple emulsions that are difficult to stabilise, which is a factor that restricts their application in delivery of proteins (reviewed in [307]). Intestinal delivery of peptides in w/o/w emulsions was first tested over 45 years ago [308] and several prototypes have been shown to improve peptide permeability in animal models, including insulin [309] and sCT [310]. The excipients used in the stabilisation of multiple emulsions can also alter intestinal permeability (e.g. SLS [309]). PEs can be added to either the oil phase (e.g. long chain fatty acids [311]) or either the external [312] or internal [313] aqueous phases. Furthermore, many of the insoluble surfactants used in the stabilisation of w/o emulsions have associated intestinal permeability enhancement action (e.g. fatty acids [314,315]). A multiple emulsion containing either eicosapentaenoic acid (EPA) or docosahexaenoic acid (DHA) in the oil phase increased absorption of insulin in rat colon and rectal instillations, although significant variability was reported [316]. Dilution of an insulin multiple emulsion with sodium taurocholate accelerated proteolytic degradation of insulin by pancreatin, suggesting that the dispersion could be destabilised in the small intestine [313]. The inclusion of peptidase inhibitors and PEs in such emulsions raises questions about their overall capacity to protect and improve intestinal permeation. This is not surprising, as lipases destabilise multiple emulsions leading to release of constituents in the internal water phase [317]. Other factors such as oil droplet diameter, composition of the oil phase, and constituents of the external phase also impact performance [318]. While physical stability of multiple emulsions have been reported for up to 2 years at 25°C, those loaded with insulin and stored at 15°C had stability for only 1–3 months [319], and the lack of stability in the presence of the peptides seems to be a widespread problem [320].

4.2.2.5. Particulates in PE-based lipoidal systems. The dispersion of nanoparticles or microparticles within lipid based drug delivery systems is a research area of growing interest. Nanoparticles for oral peptides were developed on the premise that they can protect the peptide from pre-systemic digestion and shuttle the cargo across the intestinal epithelium. However, the inability of most prototype particles to be sufficiently internalised by the epithelium has hampered development. A somewhat weaker rationale is that dispersion of peptide-loaded particulates within fluidised lipoidal delivery vehicles could facilitate better presentation of the nanoparticle at the intestinal epithelium and improve absorption of the released peptide. Furthermore, encapsulation of the peptide within a solid matrix could improve stability of the peptide within the lipoidal vehicle. The term solid-in-oil-in-water (s/o/w) generally refers to microparticles or nanoparticles that are dispersed in an oily vehicle that is then emulsified in an external aqueous phase [321]. The term has also been applied to reverse micelles within o/w emulsions, despite micelles existing in a dynamic fluidic state of matter, as opposed to more rigid colloids and coarse suspensions [322]. Insulin loaded reverse micelles were prepared in the lipophilic surfactant, sucrose erurate (HLB 2, ER-290, Mitsubishi Kagawa, Japan), and dispersed in soybean oil containing sodium cholate and sucrose laurate. When

mixed with water, the micelles formed an o/w emulsion with kinetic stability for 30 days. Oral delivery of this dispersion to rats reduced blood sugar over 3–6 h [322]. Further iteration of this dispersion led to stabilisation of the o/w emulsion droplets into a more rigid multi-molecular film with hydroxypropylmethylcellulose phthalate (HPMCP), which also functions as an enteric coating for controlled release [323]. Polyelectrolyte complexes of insulin and chitosan formed 100 nm diameter nanoparticles when dispersed in an oily vehicle (oleic acid, glyceryl-6 dioleate (Plurol® Oleic, CG, Gattefosse, France) and Labrasol®). When administered orally to fasted streptozocin rats, the PA of insulin was 2.6% over 24 h when diet was unrestricted and 7% when diet was controlled [324].

Case 13. TPE™

The most clinically advanced oral peptide delivery system is TPE™, developed by Chiasma (Israel) for the delivery of octreotide [325–329]. In 2015, an NDA for oral octreotide was submitted to the FDA under the brand name, Mycapssa™. Chiasma has also demonstrated enhancement with iterations of TPE™ in rectal (GLP-1 [330]) and nasal (IFN [330]) delivery as well as oral delivery of several other peptides (including insulin, hGH, teriparatide, exenatide [329]). TPE™ is an oily suspension of hydrophilic peptide that forms a coarse dispersion when mixed with other additives (soluble surfactant and suspending agent) and dispersed within an oil phase. TPE™ consists of enteric-coated capsules/tablets (acryl-EZE®, Colorcon, UK) containing a lyophilised mixture of octreotide, C₈ or C₁₂ and Povidone, dispersed in an oil phase composed of glyceryl monocaprylate, glyceryl tricaprylate, and polysorbate 80 [331]. The formulation improved absorption of octreotide in both rat and monkey [331] and in clinical trials [3]. Kinetics of the absorption process indicated that the actions of the oily suspension were partially reversed after 10 min and fully reversed between 30 and 60 min [331]. Reversibility is not unique to these oily suspensions, as similar barrier recovery was observed with other permeation enhancers [62]. Chiasma also reported an upper MW limit to enhancement by TPE™ of 10 kDa, while safety has been demonstrated by daily administration to primates over a 9 month period [331]. While the overall effectiveness of TPE™ has been determined in phase III trials conducted over 13 months in acromegaly patients, 58% of patients required up-titration to 40 mg, 60 mg or 80 mg (40 mg bid) to maintain response, the later representing an 800-fold increase in dose relative to the s.c., route although a therapeutic threshold was achieved and there was low variability in plasma octreotide level [332]. Nevertheless, the very high dose required for such a low MW peptide challenges the potential wider application of this delivery system in its current format.

C₈ is the best known PE in TPE™ compositions but glyceryl monocaprylate also demonstrates enhancement action [331] (Table 1). While C₈ is structurally similar to C₁₀ and C₁₂, it is less effective in head-to-head testing in Caco-2 monolayers [150], rat intestinal loop models [20,161], and rat rectal infusions [166,333]. However, alone it improved rectal bioavailability of cefoxitin by over 3 fold from 5% to 17 % in healthy volunteers; highlighting its ability to improve intestinal permeation [334]. In rat intestinal instillations, the reported level of C₈ in TPE is 5.5 mg/mL (~33 mM), which lies within the range where enhancement action has been reported in other studies (33–50 mM, Table 1). Delivery of FD4 in an admixed solution with C₈ (1.65 mg) improved absorption in rodent intestinal instillations, but an equivalent concentration of the surfactant within TPE™ further increased AUC by 5-fold [331]. The presentation of peptide and PE in an oily suspension therefore has an advantage over presentation in solution, hence the novelty of TPE™. This vehicle could also be responsible for why TPE™ was more effective in the small intestine compared with colonic delivery; which differs from several reports for MCFA (including C₈) when delivered in aqueous solution or mini-tablets [70].

4.3. Peptide hydrophobisation

The hydrophobisation of a peptide aims to improve passive transcellular permeation, and can be achieved by either covalent (e.g. alkylation or bile acid conjugates [335]) or physical complexation (e.g. hydrophobic ion pairing (HIP) [336] or non-ionic interactions). A diverse group of complexing agents induce peptide hydrophobisation including polyelectrolytes (e.g. chitosan, oligo-L-arginine PEG₂₀₀₀), monoprotic complexing agents (e.g. fatty acids, bile salts) and small molecule carriers (e.g. SNAC). HIP is the ionic complexation of an ionisable peptide group with a counter ion of comparable charge to conventional hydrophilic counter ions in salt formation, but lower capacity for solvation due to the presence of a hydrophobic moiety. The neutralised complex lowers the aqueous solubility of the ionisable drug to an extent dictated by the nature of the hydrophobic moiety of the counter-ion. Ad-mixture typically leads to precipitation in water and an increase in lipophilicity, which should improve passive transcellular flux across epithelia [337]. Surfactants are effective complexing agents because they have strong ionisable functional groups and distinct hydrophobic regions.

Hydrophobisation is effective for low and high MW species, with improved fluxes across model membranes observed over two log orders of MW from oxytocin-derived tetra peptide (MW 448 Da) to bovine serum albumen (66 kDa) [338]. Similar to pharmaceutical salt formation, different pH and counter-ions are used for physical complexation of acidic and basic drugs. Detergents like SDS have long been known to undergo HIP with peptides and proteins when mixed in stoichiometric proportions to oppositely-charged amino acid side chains [339]. As most therapeutic peptides are amphoteric, complexation can be performed at pH values above their pI where the peptide displays an anionic charge and where a cationic complexing agent is used (e.g. N α -deoxychoyl-L-lysine-methyl ester (DCK)), or below their pI where the peptide holds a cationic charge and therefore an anionic complexing agent is used (e.g. SDS). Insulin with a pI of 5.5 has six acidic and six basic functional groups, and when the pH is adjusted to 2.5, each of the acidic and basic functional groups are protonated to yield an overall +6 charge. In an acidic solution, complexation, for example with SDS (1:6 ratio of peptide:SDS) leads to a 3.4-fold increase in log P. In cases where the pH is above the pI, the majority of acidic and basic functional groups are deprotonated, resulting in overall anionic charge (-6). The pH at which complexes form can impact dissociation kinetics, because complexes formed at low pH dissociate at the pH in the small intestine, while those formed at high pH dissociate in the stomach. Although it is noteworthy that complex dissociation is not always evident at pH values that are predictive of dissociation [340]. It seems logical to complex the peptide in the anionic form at physiological pH in order to avoid pH-dependent complex dissociation prior to drug absorption. However, there are safety concerns regarding the internal use of cationic detergents as complexing agents. More importantly, while the proportion of anionic functional groups available for complexation can be high at physiological pH, the individual pK_a values for basic amino acids means that not all cationic amino acid side chains are deprotonated at physiological pH, and so positive charges in the molecules could impede passive transcellular diffusion.

In cases where surfactants are weak acids or weak bases, consideration must be given to their pK_a relative to the isoelectric point of the peptide, and ultimately the pH of the environment, because the low pH environment required to protonate weakly basic amino acid side chains in the peptide will often result in protonation of the acidic counter ions in the complexing agent leading to precipitation of the complexing agent. As such, it is often necessary to consider strongly acidic or strongly basic surfactants where pH does not influence their ionisation in physiological conditions (e.g. benzalkonium chloride, SDS). Other factors that influence complexation efficiency include the type and number of ionisable functional groups in the peptide as well as the ionic strength of the environment [341]. Ionic complexation does not always lead to precipitation from an aqueous solution, as

different carriers impart different levels of hydrophobicity. Testing a series of acyl sulphonates of different chain lengths with insulin showed that the dodecyl and decyl chains led to complete peptide precipitation, octyl led to partial precipitation, while hexyl and butyl did not precipitate the peptide [338]. At the same time, the order of flux through a methylene chloride layer followed the order $C_{12} > C_{10} > C_8 > C_6 > C_4$ with flux ranging from $0.1 \times 10^{13} \text{ mol} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}$ to $3.2 \times 10^{13} \text{ mol} \cdot \text{cm}^{-2} \cdot \text{s}^{-1}$. This is not surprising, as shorter chain surfactants have lower affinity for both self-association and association with proteins [342].

Several surfactants used in HIP are also established PEs that can alter barrier integrity e.g. SDS, sodium deoxycholate and fatty acids. The quantity of PE used in complexation is low relative to those used to alter the barrier, but this quantity is a function of potency of the peptide. For example, each 100 IU of insulin (3.5 mg) requires only 1 mg of SDS to achieve a saturated complex (1:6 molar ratio), which is likely to be below the quantity of SDS needed to alter intestinal permeability. The required concentration of SDS can be still lower for more potent peptides like exenatide. However, in cases where the complexing agent is added in excess of stoichiometry proportions and above its CMC, the complex can be solubilised into micelles [340]. There has not yet been significant research evaluating strategies to formulate hydrophobised peptides. The loss of aqueous solubility is often accompanied by improved dispersibility in solvents of lower polarity. For example, the insulin-SDS complex is soluble at a concentration of 3 mg/mL in octanol without loss of secondary structure. This compares with solubility in octanol at a concentration $<0.03 \text{ mg/mL}$ for native insulin [343]. Good solubility of insulin-SDS complexes has also been observed in propylene glycol ($>0.9 \text{ mg/mL}$), PEG 400 ($>0.14 \text{ mg/mL}$) and ethanol ($>0.9 \text{ mg/mL}$) [343]. Furthermore, dispersion of hydrophobised insulin in lipid-based delivery systems has been proposed for insulin:distearyldimethylammonium bromide (DSAB) or insulin-PC complexes [344]. Therefore, hydrophobisation offers the prospect of formulating peptides in a wide range of established non-aqueous delivery vehicles already developed for poorly soluble drugs (e.g. CsA).

The data from HIP complexation has only been reported in pre-clinical models. Hydrophobisation of insulin using the semi-synthetic bile salt, deoxycholate-based synthetic NR-deoxycholy-L-lysyl-methylester (DCK, Mediplex, South Korea [345]) in a 1:10 molar ratio increased the $\log P_{\text{METHYLENE CHLORIDE:WATER}}$ by 146-fold from 0.08 to 11.64 and increased transcellular flux across Caco-2 monolayers by 15-fold versus the native peptide [337]. Oral delivery of the insulin-DCK complex improved oral insulin absorption in rats by 6-fold [346]. In dogs, absorption of the insulin-DCK complex (42 IU/kg) was comparable to the i.v. route albeit at a higher dose [337]. DCK has also been shown to hydrophobise other drugs including ceftriaxone [347], heparin [348], and risedronate [349].

HIP complexation of acidified insulin with sodium deoxycholate increased $\log P$ by two log orders from 0.004 mg/mL to 0.4 mg/mL, which was accompanied by a 23-fold increase in relative bioavailability in rats to 12% when formulated in poly(lactic-co-glycolic) acid (PLGA) nanocapsules [350]. Other examples of HIP complexation include heparin with deoxycholyethylamine [348] and insulin with PC [351]. HIP was also used as a strategy to prevent acylation of octreotide during release from microparticles composed of lactide and glycolide [352].

Case 13. Eligen®

Hydrophobisation can also be performed by exploiting weak dipole-dipole interactions. Eligen® (Emisphere, USA) is a family of several hundred proprietary carriers that physically interact with a wide range of drugs to improve passive permeation across the intestinal epithelium. The most widely tested of these carriers are SNAC (sodium N-[8-(2-hydroxybenzoyl)amino]caprylate), 5-CNAC (N-(5-chlorosalicyloyl)-8-aminocaprylic acid), 4-CNAB (4-[(4-chloro-2-hydroxy-benzoyl)amino]butanoic acid), SNAD (N-(10-[2-hydroxybenzoyl]-amino)decanoic acid), 5-CNAB (monosodium

N-(4-chlorosalicyloyl)-4-aminobutyrate) and 4-MOAC (N-[8-(2-hydroxy-4-methoxy)benzoyl]amino caprylic acid). Eligen® carriers have been evaluated for oral delivery of insulin, sCT, Peptide YY₃₋₃₆, PTH, hGH, and several GLP-1 analogues including semaglutide (Novo-Nordisk). There has been more clinical testing performed on Eligen® carriers than any other PE-based delivery system, yet there is considerable debate on how they alter intestinal permeability.

Initial research suggested that these acylated amino acids self-assemble to form microspheres [353] and that constituents thereof could improve oral delivery of sCT [353]. SAR testing indicated that lipophilicity [354] and hydrogen bonding [355] may play a role in intestinal permeation enhancement induced by Eligen®, although a consistent effect was not observed. A preliminary screen of 11 carriers showed that an optimal bell-shaped window of lipophilicity was required to improve oral heparin absorption [356]. Analysis of the interaction of Eligen® carriers with rhGH found a correlation between drug absorption and stabilisation of protein structure [356,357]. When hGH was mixed with 4-[4-[(2-hydroxybenzoyl)amino]phenyl] butyric acid (E414) [358] there was a peak shift in electrophoretic migration suggesting physical interaction. Effective carriers also bound to specific residues within the protein structure, but not with specific amino acids *per se* (e.g. to His²¹ in helix 1, and to Tyr¹⁶⁴ Arg¹⁶⁷ Lys¹⁶⁸ Asp¹⁷¹ Thr¹⁷⁵ in helix 3, but not to all His residues) [357]. While many Eligen® carriers have acidic functional groups, their interaction is not exclusively with amino acids that have basic side chains. In fact, the most effective ones interact with the anionic molecules, cromolyn [359], ampicillin, and heparin [360].

Emisphere reported the capacity of SNAC to improve permeation of heparin across Caco-2 monolayers and to induce inhibition of anti-Factor Xa in rat intestinal instillations [360] and in cynomolgous monkeys [361] (Table 1). The SNAC-heparin dispersion was prepared to a final pH of 7.5–8.5 and contained SNAC (100 mg/mL) and heparin (33 mg/mL) dissolved in 25% v/v propylene glycol [361]. The effective dose range of SNAC (88–300 mg/kg) was significantly higher than that used for conventional PEs, and this placed requirements on both dosage form capacity and the use of potent peptides and proteins. The interaction between SNAC and heparin was based on increased lipophilicity through hydrogen bonding and/or hydrophobic interactions [362]. In relation to carrier-peptide interactions, analysis by 4-4-bis-1-phenylamino-8-naphthalene sulphonate (bis-ANS) fluorescence showed that SNAC increased the lipophilic surface area of insulin through non-covalent bonding and/or conformational changes to the peptide, leading to exposure of hydrophobic peptide regions amenable to transcellular permeation [363].

While improved permeation of insulin can be uncoupled from an effect on barrier integrity, there are contradicting reports on the nature of the interaction between Eligen® carriers and the intestinal epithelium at concentrations required for oral delivery. For example, low concentrations of SNAC (17 mg/mL) in Caco-2 monolayers improved transepithelial permeation of insulin but not mannitol, suggesting that neither alteration to barrier integrity nor TJ opening was involved in the mechanism [363]. In another Caco-2 study however, SNAC caused complete loss of TEER and a 36-fold increase in mannitol permeability at the concentration (50 mg/mL) required to improve heparin permeation in an instillation model [360]. In isolated rat colonic mucosae, SNAC concentrations $>50 \text{ mg/mL}$ increased P_{APP} of mannitol and reduced TEER [364]. These two studies used such high concentrations that loss of TEER was inevitable and is likely associated with epithelial damage, so no conclusions on mechanism can be made. There are also reports that challenge the theory that SNAC acts through hydrophobisation. For example, SNAC did not cause an increase in the partition of cromolyn in either octanol ($\log D_{\text{pH}7.4}$) or chloroform, but increased epithelial membrane fluidity as measured by fluorescence anisotropy [365]. Although SNAC has demonstrated surface action (CMC: 56 mM in PBS (pH 7.4) [365]), the distribution of hydrophilic functional groups in the more hydrophobic salicylamide region of the carrier do not give rise to efficient detergent action, and this

suggests a basis for the high concentrations required to induce transcellular permeability *in vivo*. It therefore remains unclear whether the high concentrations of SNAC required for oral peptide delivery simply relate to a weak detergent action or whether there is true carrier-based delivery. It bears noting that SNAC is a weak acid (pK_a 5.08 [366]) and so could undergo HIP complexation with peptides at physiological pH, although HIP cannot explain the hydrophobisation of anionic drugs like heparin.

Whatever the mode of action, SNAC was granted GRAS status in 2009 and Emisphere recently marketed Eligen-B₁₂TM, an oral vitamin formulation containing SNAC [367]. This carrier had a no observed adverse effect level (NOAEL) of 1 g/kg/day in rats [368], well above the doses used in oral peptide formulations. SNAC was tested in a number of clinical studies especially with heparin in phase III (PROTECT), where an oral liquid dose of heparin-SNAC failed to meet its primary endpoint; moreover compliance in this trial was low due to the bitterness of SNAC in solution, which had previously been noted in proof of principle clinical testing [369]. The dose of SNAC that was required in preliminary clinical evaluation of oral heparin (90,000 IU) was 2.25 g delivered in a 15 mL volume every 8 h pre- or post- prandially [369]. Higher doses of SNAC (10.5 g) delivered to patients by nasogastric intubation were accompanied by emesis in healthy volunteers. Heparin-SNAC was subsequently formulated into soft gelatin capsules and delivered orally to patients where it improved absorption of heparin relative to an unenhanced formulation, but it has sub-optimal pharmacodynamics relative to the s.c. form [370]. Ultimately, delivery of heparin in an optimal oral solid dosage form was not feasible in one unit dose due to the large dose requirements of carrier and payload (1.9–2.8 g), even before the addition of formulation or process excipients [371]. A conservative estimate of 4×750 mg tablets three times daily indicated that such an oral heparin dosage form might only be suitable for short term use. Given the challenges facing oral heparin, development of heparin-SNAC was discontinued. Lower quantities of SNAC and other Eligen® carriers have been tested in oral peptide delivery due in part to increased potency of such peptides relative to heparin. For example, 150 mg of SNAC was tested in oral preparation of PYY and GLP-1 [372]. In a 90 day trial, patients that received up to 40 mg of insulin daily in 4 divided doses saw a reduction in HbA1c of 0.74% when their initial baseline was between 7 and 8.9% [373]. SNAC has also been disclosed in patents filed by Oramed, where an oral insulin formulation containing insulin (6 mg) and a synergistic mixture of SNAC (250 mg) and SBTI (125 mg) significantly lowered blood sugar levels in preliminary clinical testing [374]. Administration of insulin (400IU) and SNAC (2.1g) in capsules ($4 \times$ size 000 gelatin) significantly increase plasma insulin level between 20 and 50 min, and lowered plasma glucose levels between 30 and 50 min [375]. More recent developments relate to the licensing of Eligen® to Novo Nordisk for oral delivery of insulin, GLP-1 (NN9924) and selected structural analogues (e.g. semaglutide [376, 377]). In a 600 patient phase II trial, daily oral delivery of semaglutide (40 mg) formulated with SNAC lowered HbA1c by 0.7–1.9% compared with 1.9% for the s.c. group [378]. The oral dose of semaglutide was 300-fold higher than the s.c. dose (1 mg), and Novo Nordisk have proceeded to phase III(a) (PIONEER) trials with oral doses of 3, 7, and 14 mg semaglutide. In animal testing, semaglutide (10 mg) was formulated in an oral tablet containing SNAC (150, 300, or 600 mg), povidone (2, 4, or 7 mg), sodium starch glycolate (Avicel® PH102; 36, 82, 76 mg) and Mg²⁺ stearate (3, 4, or 7 mg) [379]. The semaglutide formulation containing 300 mg SNAC had F_{ABS} of 0.63 % following oral administration in Beagle dogs, furthermore a dose dependency was observed for 5 mg (F_{ABS} : 0.33%), 10 mg (F_{ABS} : 0.63 %), 15 mg (F_{ABS} : 1.2%) and 20 mg (F_{ABS} : 1.4%).

5-CNAC has also been evaluated for the oral delivery of sCT [380] and PTH [381] under licence with Nordic Bioscience (Denmark), and partnered with Novartis. Like SNAC, 5-CNAC forms a lipophilic complex with peptides to improve intestinal permeation [382]. Interaction between 5-CNAC and sCT is likely to prominently involve HIP because sCT has a high isoelectric point (pI 7) and a higher proportion of amino acid

side chains will be positively charged at pH 7 compared to insulin (pI 5.5). An insoluble complex is initially formed between sCT and 5-CNAC at low pH due to the higher proportion of cationic functional groups, but the complex is not stable at physiological pH in the small intestine.

The dose of sCT (0.8 mg) in oral formulation with 5-CNAC (200 mg) was higher than the nasal formulation (200 IU or 33 µg), and accordingly C_{MAX} was significantly higher in oral (145 pg/mL) versus nasal (11.4 pg/mL) routes [383]. Despite promising clinical performance of sCT formulations containing 5-CNAC [380,384–388], these formulations did not reach primary endpoints in two phase III trials [389]. Publication of the trial data by Nordic Biosciences has assisted development of oral peptide formulation [390]. For example, administration of sCT:5-CNAC with 50 mL of water resulted in a three-fold increase in absorption compared to that obtained with 200 mL [385], while a significant food effect was also observed [391]. Nordic also evaluated the most effective time of the day to administer oral sCT in post-menopausal women, where administration 1 h before dinner (5 pm) was more effective than after overnight fasting (8 am) or 4 h after an evening meal (10 pm) [391].

Case 14. BridgelockTM, MacrosolTM and AxcressTM

Cortecs Ltd (UK) was one of the first companies to apply oily formulations in oral delivery of peptides. BridgelockTM was an oily peptide dispersion formed by spraying a w/o emulsion onto sodium carboxymethyl cellulose followed by evaporation [78]. The dehydrated aqueous phase contained sCT, aprotinin, CA, polyoxyl 40 stearate, hydroxypropyl cellulose, NaCl, PC, and phosphatidyl glycerol, while the oil phase contained lecithin, monoolein, polysorbate 80, cholesterol and oleic acid [392]. A number of these agents can act as PEs, in particular oleic acid (Table S1). Intra-jejunal delivery of BridgelockTM improved absorption of sCT in pigs as measured by reduction in plasma Ca²⁺ [392]. An iteration led to MacrosolTM, an isotropic lipid based peptide dispersion that is formed by reconstitution of an anhydrous peptide-amphiphile mix in oil. The surfactant forms a sheath" around the peptide by interaction with its hydrophilic head group; a process facilitated by lyophilisation of an ad-mixed solution [78]. When the anhydrous mix is dispersed in oil, it forms a molecular dispersion; which distinguishes it from particulates in oil. However, unlike MacrosolTM, not all complexed peptides are soluble in lipid vehicles, rather form particulates in oil (solid/oil systems). For example, improved oral delivery of hGH in rats was measured from an oily suspension of protein complexed with sucrose erucate dispersed in soybean oil [393].

MacrosolTM is part of the portfolio of Proxima (UK) (AxcressTM) and has been licenced for oral delivery of insulin (CapsulinTM; Diabetology, UK), calcitonin (CapsitoninTM; Bone Medical, Australia) and PTH (CaPTHymoneTM, Bone Medical, Australia). In CapsulinTM, insulin is mixed with sodium ursodeoxycholate, dispersed in benzyl alcohol, and inserted to size 4 soft gelatin capsules, which lowered blood sugar following oral delivery in pigs [394]. Repeated oral administration of CapsulinTM to diabetic patients 30 min prior to breakfast and an evening meal reduced HbA1c levels by 0.2%, while the percentage of patients remaining below the recommended excursion level increased from 10% to 36 % [395]. The oral CapsitoninTM and CaPTHymoneTM formulations had comparable effects to their respective injectable forms (Miacalcin® and Forteo®) in lowering of plasma C-terminal telopeptide (CTX-1) and plasma Ca²⁺ [396]. Many of the excipients used in AxcressTM formulations are listed in national compendia, for example phenoxy ethanol, benzyl alcohol, butylated hydroxyanisole and propyl gallate [397,398]. However, the quantities used in oral peptide delivery are significantly higher than those listed in the FDA Inactive Ingredients list.

4.4. Non-surfactant PEs

Aside from the new generation of paracellular PEs, several non-surfactant PEs have been evaluated in pre-clinical and pilot clinical

testing (Tables 1, S1). A number of PEs in this category have not progressed in recent years, such as sodium taurodiethylhydrofusedate (STDHF) [399], salicylates and enamines [400]. Likewise, a cohort of this PE group have low clinical potential because of active pharmacology (e.g. salicylate [401]) and sodium nitroprusside [402]) or known systemic toxicity (p-chloromercuriphenyl sulphate [403]).

Case 15. salicylates and enamines

Studies on the action of sodium salicylate and 5-methoxy salicylate have contributed significantly to the potential application of PEs as vehicles to improve transmucosal peptide delivery, in particular across the rectal mucosa (Table S1). However, as salicylate is the active form of aspirin, its application in oral peptide delivery is limited by its anti-inflammatory and antiplatelet actions. Rectal suppositories of insulin (5–50 IU) containing triglyceride (700 mg), lecithin (70 mg) and sodium salicylate (300 mg) reduced plasma glucose in healthy dogs [404]. Moreover, absorption of hGH from a suspension of sodium salicylate in mineral oil exceeded enhancement from an aqueous solution in rat intestinal instillations [405]. In clinical testing of 10 T2D patients and 4 healthy volunteers, suppositories containing insulin (100 IU), hard fat (Witepsol H15) and sodium salicylate (200 mg) lowered plasma glucose by 28% over 2 h in T2D, after which blood sugar returned to basal level [406]. In another clinical evaluation, suppositories containing insulin in CA (0.5 M) and salicylate (300mg) improved rectal insulin absorption compared to the suppository base alone (Witepsol® H15) [407]. The enamine, D,L-phenylalanine ethyl acetoacetate, had comparable effect on rectal absorption of insulin. This enamine was formed by reacting phenylalanine and the food additive ethyl acetoacetate, but which is hydrolysed to phenylalanine and ethyl acetoacetate in the body; suggesting that safety might not be a significant consideration. A variety of enamines have been shown to improve intestinal permeability in pre-clinical models, with phenylalanine ethyl acetoacetate exhibiting stronger enhancement action than several other enamines [400], and improved rectal absorption of insulin from a suppository in diabetic dogs [408].

Case 16. chitosan and its derivatives

Chitosan is one of the most widely studied semi-synthetic polymers in the delivery field, and its capacity to improve intestinal permeability is well-known [92,409–412]. However, despite its promising action as an intestinal PE *in vitro*, it has not been assessed to date in clinical trials for oral peptide delivery. It is a polymeric PE formed by deacetylation of chitin to form a heteropolymer of N-acetylglucosamine and D-glucosamine. The primary amine of glucosamine has an approximate pKa of 6.5 and is therefore protonated in acidic conditions to yield the soluble cationic form, which is responsible for enhancement action. Permeation enhancement across Caco-2 monolayers was more evident for high MW variants and those that have a higher proportion of D-glucosamine *i.e.* a low degree of acetylation (<35%) [413]. As enteric coating is a critical component of oral peptide delivery systems, the pKa of chitosan is problematic, as variable ionisation within the jejunum (pH 6.1–7.1) creates variability in the proportion of soluble chitosan, which ultimately contributes to low bioavailability. Different chitosan salts have different efficacy as intestinal PEs [411], and a number of these salts have been shown to improve the transmucosal flux of peptides (e.g. HCl [414] and glutamate [415]). For example, rat intestinal instillation of chitosan HCl (1.5%) improved F_{ABS} of busserelin to 5%, which if translated in man would be impressive, given that the marketed nasal formulation has an F_{ABS} of 3.3% [414]. However, while these salt forms are likely to improve dissolution from enteric-coated formulations, chitosan is sensitive to pH-dependent precipitation. Another drawback to native chitosan is the gelling properties at concentrations as low as 1.5% [410], which impacts optimal release and presentation of peptide and PE at the intestinal epithelium. An assessment of

chitosan acetate as a tablet binder showed that inclusion of low concentrations (1, 2, or 3 % w/w) prolonged dissolution of theophylline from tablets to beyond 3 h, which contrasted to dissolution from tablets containing an established binder (PVP K30), where complete release was demonstrated within 1 h [416]. In assessment in compendial media, release from HCl (0.1 N) occurred in 6 h and release from phosphate buffer (pH 6.8) reached 97% and 100% within 16 and 24 h [416]. Although inclusion of excipients (e.g. Kollidon® VA64 (BASF)) can alter the disintegration of tablets that contain chitosan (5% w/w) [417], it is not clear if timely dissolution can be achieved for the higher concentrations of chitosan that are required for enhancement action. In a recent clinical evaluation, an oral dispersion of chitosan HCl (1 g/150 mL) reduced absorption of the BCS Class III drug, acyclovir (200 mg) in 12 healthy volunteers [418]. It seems that oral delivery of chitosan in fluidic dispersions in man do not mirror the enhancement data observed in pre-clinical delivery models. The safety of chitosan has been extensively reviewed with the finding that it has the potential to be a safe pharmaceutical excipient [419], although damage to intestinal epithelial cells has been recorded in cell culture models [413]. In general, the systemic toxicity of such polymeric PEs is predicted to be low as they are poorly absorbed.

The formulation and process attributes of chitosan (125 kDa, 91% deacetylation) in matrix tablets has been evaluated [420]. This chitosan variant suffered from poor particle size uniformity (44.7% ranged between 63 and 150 μ m) and poor powder flowability and compressability, which influences weight uniformity in tablet production [420]. Increasing degrees of deacetylation (80%, 85% and 90%) impact both compressability/flowability and uniformity [421]. Inclusion of a glidant and restriction of particle size limits to >212 μ m were effective in controlling weight variability [420]. Chitosan had tablet and tableting properties that were similar to microcrystalline cellulose and was considered appropriate for tableting [422]. However, the compression force used in tableting resulted in slight melting of chitosan and the formation of hard tablets (300 N) that are likely to exhibit slow release. The mean dissolution time (MDT) of isoniazid from chitosan tablets was 7.65 min, a value that increase to 30 min upon inclusion of citric acid (8%); not surprising as CA increased the solubility of chitosan leading to the formation of an interfacial gel layer that restricts interfacial migration of the drug into the bulk of the phase [420].

Several analogues have been developed to address chitosan precipitation at small intestinal pH values including trimethylation (TMC) [423] (Table 1), triethylation [424] and combinations therein [424] to form quaternary ammonium compounds that are charged at intestinal physiological pH values. Other derivation strategies have involved the formation of mono-N-carboxymethyl chitosan which gives rise to ampholytic variants that are soluble at physiological pH [425]. The most comprehensively tested chitosan derivative is TMC, which has high aqueous solubility even at low acetylation (<10%) and lower intrinsic viscosity [426]. However, higher concentrations of TMC (>10%) used to improve oral octreotide delivery in pigs formed a gel and exhibited comparable viscosity to chitosan HCl [77]. While this gel improved oral octreotide bioavailability in pigs (F_{ABS} : 1.7% versus F_{ABS} : 24.8%), the rheological behaviour of TMC remains problematic for oral peptide delivery [59]. Dissolution of tablets containing TMC was also found to be challenging in that they did not dissolve in water even when formulated with disintegrants [427]. To address this problem, DDAVP (0.05–0.1 mg) and TMC (7.5–15 mg) were wet-granulated with microcrystalline cellulose 10–15% (Avicel® PH-101, FMC Biopolymers, USA), and subsequently mixed with a super-disintegrant (Ac-Di-Sol®, FMC Biopolymers) prior to tableting in to mini-tablets (3 mm). While dissolution of DDAVP was not impeded, only 50% of the TMC was released from each mini-tablet after 2 h, lower concentrations than those needed to improve oral octreotide delivery [77]. This formulation was adapted for oral delivery of octreotide in pigs, where mini-tablets of octreotide and TMC were loaded in enteric-

coated capsules [410], but no improvement in bioavailability was observed [410].

Thiolated chitosan derivatives are a family of thiolated polymers or thiomers reported to alter intestinal permeability to improve oral peptide delivery. The enhancement action of thiomers is typically not as efficacious as the leading surfactant-based PEs, but certain analogues can combine enhancement action with mucoadhesion, peptidase inhibition, and efflux pump inhibition [428,429]. Chitosan–thioglycolic acid (TGA) improved permeability of leuprolide in isolated rat intestinal mucosae by 4-fold [430]. In oral delivery to rats, the F_{ABS} of leuprolide was increased by 3.8-fold from a gel formulation containing chitosan–TGA (8 mg/mL) [430]. Research effort has focused on the physical behaviour of thiomers in oral solid dosage forms. Several thiomers sustain drug release from tablets [431] and mini-tablets [67,432]. Given that thiomers are mucoadhesive and can inhibit peptidase activity, it is not surprising that they retain the ability to improve intestinal permeability, although whether this is directly due to permeation enhancement or whether such enhancement can effectively translate to man is unknown.

Another thiomers, chitosan 4-thiobutylamidine (TBA) improved permeability of acyclovir across Caco-2 monolayers and isolated rat intestinal mucosa [432]. In the same study, release of acyclovir from mini-tablets (30 mg) containing chitosan 4-TBA was MW-dependent in the order of 9.4 kDa > 150 kDa > 600 kDa, yielding dissolution values of 90%, 60% and 40% after 1 h. Oral delivery of the decapeptide, antide, in a matrix tablet (2 × 500 mg) containing 400 mg of chitosan 4-TBA per tablet improved oral bioavailability of the peptide in pigs from an undetectable level to F_{ABS} of 1.1% (and R_{REL} of 3.2%) [431]. In a similar study, an enteric-coated oral tablet (10mg) containing insulin (2.8 mg), chitosan TBA (5mg) and two peptide inhibitor conjugates of chitosan (chitosan–BBI (0.75 mg) and chitosan–elastatinal (0.75 mg)), as well as CA (0.1 M) was tested in rats [433]. This formulation sustained insulin release over at least 8 h, which was mirrored by lowering of blood sugar in rats; however, given the large dose of insulin, both F_{ABS} (0.7%) and F_{REL} (1.7%) were low. Similar results were observed from an oral formulation (5 mg) containing sCT (50 µg), chitosan TBA (3.75 mg) and a chitosan–peptidase inhibitor conjugate, although that formulation was not enteric coated [434].

Other thiomers also improve peptide delivery from oral solid dosage forms. Oral delivery of mini-tablets (30 mg) containing insulin and chitosan 6-MNA (6-mercaptopurine, 20 kDa, 1:4 ratio) improved oral absorption of insulin in rats (F_{ABS} : 0.73%) compared with control tablets containing chitosan alone (F_{ABS} : 0.15%) [67]. The rate of release of insulin from mini-tablets containing chitosan 6-MNA was comparable to native chitosan (60–80% after 2 h) despite a 5-fold increase in bioavailability. It is noteworthy that the dissolution media contained 30%v/v DMSO, which has a likely impact on the predicted release kinetics in man. One of the reasons for this difference was attributed to the >80-fold increase in mucoadhesion for chitosan 6-MNA, which highlights the importance of regional retention and localisation in achieving efficient intestinal permeation enhancement. On a note of caution, it is necessary to reiterate that mucoadhesion is clearly limited by the rate of mucous turnover in the gastrointestinal tract, which is species specific [435]. All of the thiomers outlined above are in the portfolio of Thiomatrix (Austria), however evidence of the clinical effectiveness of thiomers in oral peptide delivery has yet to be reported.

Case 17. CPPs

Cell penetrating peptides (CPPs) are a group of peptides that can improve intestinal peptide permeability [436]. Research in CPPs is built around three sequence types: protein-derived CPPs (e.g. HIV transactivator of transcription (*tat*) peptide and penetratin (Table 1)), chimeric peptides (e.g. transportan [437]) and designed/synthetic peptides (octa-arginine) [438] (Table S1). Exactly how CPPs increase intestinal permeability is related to their capacity to initiate endocytosis, direct translocation and to formation of channels within the cell

membrane at high concentrations [439]. In many cases the therapeutic peptide is fused with the CPP or loaded into microparticles or nanoparticles coated with CPP to improve translocation efficiency. For example, Enteris Pharma (NJ, USA) disclosed a membrane translocator fusion peptide derived from *tat* to improve oral delivery of sCT [440]. Intestinal instillation of the sCT-membrane translocator (4.5 mg / 43 mg CA) led to an improvement in F_{ABS} from <1% to 21% in rats [441]. The enhancement action of unconjugated CPPs is often observed at concentrations that are 2 log orders lower than conventional surfactants. For example, transportan (10 µM) reduced TEER in Caco-2 monolayers [442], far lower than the concentration of 10–13 mM required for C₁₀ [89]. CPPs also improve intestinal epithelial permeation of associated antidiabetic peptides. Intestinal instillation of L-penetratin (0.6 mg) in rats improved absorption of GLP-1 (0.1 mg/kg) from 0.89% to 11% [443]. The ability to engineer structural analogues of CPPs permits optimisation of both enhancement action and safety. SAR analysis of a panel of penetratin analogues led to development of a more efficacious stable analogue, PenetraMax™ [444]. Absorption of insulin in rat instillations was improved in the order of D-PentraMax (F_{REL} : 26%) > L-PentraMax (19%) > L-Penetratin (12%) > octa-arginine (4%) [445]. A number of these analogues have been disclosed in patents registered by Toray Pharmaceutical Inc. (Japan), who have performed additional pre-clinical evaluation [446]. Further analysis of the potential of CPPs in oral peptide delivery is covered by Giralt *et al* (this Issue).

4.5. Multiple modes of enhancement action

Many of the PEs that act primarily via the transcellular route have also been shown to alter paracellular permeability. In some reports, a clear concentration dependency is observed between paracellular and transcellular actions, but others report paracellular enhancement at concentrations that more closely align with transcellular perturbation. However, a drop in TEER across Caco-2 monolayers is sometimes assumed to be an increase in “paracellular permeability”, but any physical or chemical insult can non-specifically increase conductance across epithelial monolayers. More complex measurements using impedance spectroscopy have been reported [447], but such models require strict microscopic verification that the PE does not lead to intracellular uptake of a paracellular dye (i.e. transcellular perturbation), and applications that fail to do so can overestimate the contribution of the paracellular route [448]. In some cases, a paracellular mode can be uncoupled from transcellular enhancement in reductionist models, although the lower concentrations that avoid transcellular action may not be representative of the effective concentrations *in vivo*. While pharmacological inhibitors of enzymes, receptors, and signalling molecules have been shown to attenuate enhancement action of selected PEs, their actions are not comprehensively understood nor are they effective at higher concentrations that are necessary for enhancement in animal models.

There is a desire amongst investigators to research PEs that exclusively alter paracellular permeability, primarily due to perceptions of safety issues relating to transcellular perturbation. This is one of the reasons why specialist delivery companies favour delivery technologies that have a history of safe use in man or have been designated as GRAS. A wide range of tools have been used to evaluate detergent interaction with the plasma membrane of intestinal epithelial cells including cell integrity assays, BBMV and more recently, high content image analysis. A greater contribution from the paracellular route is observed at low and intermediate concentrations of membrane perturbants [449] owing to unpredictable actions ranging from (i) modulation of intracellular mediators (e.g. Ca^{2+} and ATP [202]) (ii) receptor activation (e.g. phospholipase C (PLC) [129]), (iii) selective removal of TJ proteins from fluidic regions of the membrane (e.g. claudin [450]) and (iv) cellular repair mechanisms. These diverse actions are most commonly observed with surfactants such as medium chain fatty acids, acyl carnitines and sucrose esters (Tables 1, S1).

There are a number of signalling molecules that, if depleted by transcellular perturbation, could lead to alteration in TJ structure. For example, an alteration in membrane fluidity can lead to leakage of intracellular mediators like ATP, and its depletion has been associated with Ca^{2+} -dependent alteration in TJs [451]. A group of PEs deplete intracellular ATP including C_{10} [129] and acyl carnitines [202]. These actions are lost above a threshold concentration of PE, because the perturbation caused by the surfactant deactivates the cell, and in such a case, cells that have survived the initial perturbation by the surfactant PE stimulate intracellular signalling processes that are involved in mucosal repair. This begins with disbandment of TJs, spreading and protection of exposed surface, and is concluded by resealing [452].

Ca^{2+} is another intracellular signalling molecule associated with multimodal PEs, but the nature of its role has not been fully elucidated. Electron micrographs of Caco-2 monolayers treated with 10 mM C_{10} revealed dilation of 42% of TJs in one study [453] and transcellular perturbation in another [149]. *In vitro* experiments with C_{10} on monolayers and tissue mucosae are performed in the absence of extracellular Ca^{2+} on the apical side, so increases in intracellular Ca^{2+} are caused by release from intracellular stores, but whether alterations in barrier function are due to a defined signalling mechanism or perturbation of organelles involved in intracellular storage is not clear. At lower concentrations, C_{10} (2.5 mM) increased intracellular Ca^{2+} in Caco-2 cells before other cytotoxicity metrics could be observed using high content image analysis (e.g. plasma membrane permeability changes at 8.5 mM) [150]. In cases where alteration to intracellular Ca^{2+} can be dissociated from plasma membrane perturbation, the PE has only modest effect on TEER at such concentration (40% drop in TEER after 60 min). At widely studied concentrations *in vitro*, C_{10} (8.5–13 mM) altered plasma membrane and mitochondrial membrane integrity. Most mechanism of action studies with C_{10} are however performed at higher concentrations (10–13 mM). C_{10} (13 mM) reduced localisation of ZO-1 and occludin in Caco-2, and its effect on permeability could be reversed with several pharmacological inhibitors of cellular signalling molecules including PLC, calmodulin, diacylglycerol (DAG), inositol-3-phosphate (IP3), and Ca^{2+} [129] as well as MLCK [146]. This led to the theory that C_{10} increases intracellular Ca^{2+} through the activation of PLC, which activates calmodulin-dependent phosphorylation of MLCK, in turn phosphorylating MLC, leading to cytoskeletal contraction and disbandment of TJs [89].

The elucidation of mode of action is complicated by the capacity of certain surfactants to concurrently remove TJ proteins from regions of the plasma membrane that are more sensitive to detergents. Thus, C_{10} displaced claudin 4 and 5 from lipid rafts in MDCK cells, showing the importance of protein solubilisation [450]. Likewise, the interaction of surfactants with the plasma membrane can indirectly modulate the activity of membrane proteins by exposing the receptor to ligands. The transcellular route should therefore be considered in assessment of specific intracellular signalling mechanisms. Given that C_{10} alters multiple cellular metrics at 13 mM, it is not correct to consider it an exclusive paracellular PE, as its complete spectrum of actions cannot be dissociated from those of transcellular perturbation. Superficially, the paracellular mode of action appears to be less relevant *in vivo* as the dose of C_{10} used in oral peptide animal testing and those tested in oral dosage forms in man are far in excess of those used *in vitro*. However, concentrations at the intestinal epithelium could still be relatively low *in vivo* when dissolution, spreading and dilution are taken into account. It is noteworthy that pharmacological inhibitors associated with paracellular action are also effective in tissue models and pre-clinical animal testing, and so while transcellular enhancement is the predominant mode of action in animal models, a significant paracellular contribution is also possible. For example, a calmodulin inhibitor (W7) attenuated the enhancement action (TEER and FD4 flux) of both C_{10} and LCC in isolated rat and human colonic mucosae [454]. Likewise, W7 also attenuated enhancement action of C_{10} in rat rectal delivery, but while the authors conclude that C_{10} acts through calmodulin-

dependent cytoskeletal contraction, the mechanism is far from clear since each suppository had 39 mg of C_{10} , which if dissolved in 0.3–1 mL rectal volume would reach a concentration of 200–670 mM [455].

A mechanistic study in Caco-2 cells performed on 51 PEs from 11 structural groups found that PEs can exploit both the paracellular and transcellular routes *in vitro* [449]. The screen involved calculation of K values (relative contribution of paracellular route) which measured from 0 (transcellular) to 1 (paracellular) from the formula $K = \text{EP} - \text{LP} / \text{EP}$ where EP is enhancement potential ($\text{EP} = [100\% - \text{TEER}_{\text{TEST}}] \div [100\% - \text{TEER}_{\text{TRITON X-100}}]$) and LP is LDH Potential ($\text{LDH}_{\text{TEST}} / \text{LDH}_{\text{MAX}}$). The model assumes that LDH is an acceptable marker for transcellular enhancement, although its sole use could overestimate the contribution of the paracellular route, as the absence of LDH release is not necessarily an optimal marker of cell perturbation. This point is emphasised when the IC_{50} of Triton X-100 was compared in different cytotoxicity assays: LDH (80 μM) > MTT (44 μM) > Neutral red (31 μM) > ATP (43 μM) [456]. High concentrations of PE surfactants including SLS ($K = 0$), Triton® X-100 ($K = 0.06$), sodium oleate ($K = 0.18$), PCC ($K = 0$) and sodium deoxycholate ($K = 0$) exhibit transcellular enhancement action, but at lower concentrations, the contribution of some PEs was associated with the paracellular route (e.g. sodium deoxycholate ($K = 0.71$) and sodium oleate ($K = 0.96$)). The action of established paracellular PEs like EDTA was comparable to literature observations ($K = 0.72$), but others like sodium salicylate ($K = 0.8$) had an unexpected paracellular action. The separation of paracellular from transcellular PEs allowed the discovery of a linear relationship between Log P and transcellular enhancement, and an inverse relationship between Log P and paracellular enhancement. A linear effect between Log P and enhancement was not observed for surfactant-like PEs, as specific criteria relating to HLB and CMC are more important than outright lipophilicity. Further, quantitative SAR using this dataset and an additional panel of physico-chemical properties permitted further prediction around mechanism of action [457].

5. Safety and regulation of PEs

Toxicity has long been cited as a potential drawback to the application of PEs in oral peptide delivery [458]. Each PE has specific attributes that must be considered in an overall risk benefit analysis and generalisation on toxicity is not appropriate. Enhancement action reported for the most clinically advanced PEs was often accompanied by evidence of regional toxicity in reductionist models, such as loss of cell viability in cytotoxicity assays, or focal and superficial mucosal injury in histological analysis of intestinal tissue. However, to our knowledge there have been no significant adverse events reported for any of the leading PEs tested in clinical trials to date. Given the fast rate of intestinal transit, spreading, dilution and absorption (Fig. 1), it is improbable that the intestinal epithelium will be exposed to PEs at high concentrations for prolonged periods locally, as was observed in some pre-clinical delivery models. Inferences relating to PE safety based on cytotoxicity measurements in static cell culture models are therefore not reflective of the dynamic *in vivo* environment or the capacity of the GI tract to repair from superficial mucosal injury.

5.1. Transcellular enhancers and membrane perturbation

Surfactants fulfil a wide array of applications in a variety of fields from household cleaning products, to stabilisation of cosmetics to heavy industrial applications. There are concerns relating to the application of strong detergents such as those used in industrial applications, but there are no studies advocating use of strong detergents in oral peptide delivery. Rather, the clinically-advanced PEs have established safety profiles, for instance SNAC is designated GRAS, C_{10} has Food Additive Status and sucrose laurate is an allowed excipient in the USA. Furthermore, no clinical evidence has been presented to date that suggest formulations containing PEs cause serious mucosal damage. Surfactants

are widely used in formulation of both enteral and injectable dosage forms [215]. These excipients have been shown to alter barrier integrity, and relevant examples include polysorbate 20, polyethoxylated castor oil, PEG-8 glycerides, long chain fatty acids and medium chain monoglycerides. Some alteration in barrier integrity and cell viability is anticipated when surfactant excipients are used in oral formulation. The dose of Cremophor® EL (up to 600 mg) and Cremophor® RH40 (up to 405 mg) used in oral formulations suggests that the concentration of these additives could reach levels that alter intestinal epithelial cell viability (EL: 5 mg/mL and RH40: 10 mg/mL) [459].

Surfactants are also licensed without evidence of serious side effects. Sodium docusate is a stool softener used on a daily basis in the treatment of constipation. The intestinal permeation enhancement action of docusate has been reported (Table S1). Bile salts like ursodeoxycholate (Ursofalk®, DrFalk Pharma, Germany) are also used for the dissolution of gallstones (750 mg/day) or bile replacement in primary biliary cirrhosis (up to 1.75 g/day). Jejunal instillation of ursodeoxycholate (10 mg) increased F_{ABS} of octreotide from 0.3% to 4.9% in rats [90]. Enhancement was associated with a time and concentration dependent release of LDH in Caco-2 suggesting a transcellular mechanism. Chenodeoxycholate has also been used in dissolution of gallstones, and this bile acid increased absorption of octreotide in healthy human volunteers at a dose as low as 100 mg [90]. The most widely cited side effect of these bile acids is diarrhoea. Therefore, administration of selected molecules that alter barrier integrity can be administered in man without manifestation of toxicity.

It is tempting to justify the use of PEs by citing precedence in the routine use of substances capable of altering barrier integrity. However, the capacity of aspirin, alcohol or other substance to reversibly perturb the intestinal mucosae does not set precedent for approval of medicines containing PEs. In the US, the regulatory agency does not provide an approval path for excipients, but as part of the submission of a product formulation. Once an excipient is used in an approved oral drug formulation, it is more likely that it will be approved in other ones, but this will depend on API-specific aspects, doses of PE, and acute or chronic administration needs [460]. There is however, value in understanding how abrasive substances interact with the intestinal epithelium to predict how PEs might behave in man.

At one extreme, mild mucosal damage occurs in 40–50% of patients taking low dose aspirin, and these patients are at an increased risk of GI bleeding [460], although the risk of bleeding is multi-factorial [461]. There is no evidence to suggest that PEs inhibit cyclooxygenase, but perturbation has the potential to impact cell viability. The response of the intestinal epithelium to transcellular PEs that cause mild mucosal perturbation has been studied in pre-clinical delivery models, which has given insight into the rate and extent of cell and tissue recovery prior to repeat dosing. The most relevant example to date was observed following histological assessment of rectal mucosae of patients administered Doktacillin™ suppositories containing C_{10} [83]. Mild and reversible histological damage to the mucosae was seen in patients, and there was association between enhancement and mucosal damage in the rectum, which the authors ascribed to a combination of C_{10} , the suppository base, and hyperosmolar conditions. Pre-clinical data has highlighted how exposure time and concentration impact epithelial integrity and viability, and also the extent to which the barrier can repair following removal of the PE. Studies reveal such information for bile salts [71, 462–464], ethoxylates [465] medium chain fatty acids [149], monoglycerides [71] and SDS [71].

Treatment of Caco-2 monolayers with C_{10} (8.5 mM) for 15, 30 or 60 min led to a reduction in TEER to 15%, 5% and 0% of initial respectively, but upon removal of the PE, TEER recovered to 100% after 2 h, 4 h and 7 h, respectively [149]. A number of toxicity metrics were evaluated during the recovery from the 60 min treatment with C_{10} . In high content image analysis, there was no change in cell number between treatment and 24 h recovery, and there was a progressive recovery in a range of cytotoxicity metrics to control levels including intracellular Ca^{2+} and

plasma membrane permeability. Electron microscopic analysis showed that damaged epithelium recovered after 4 and 24 h, and there was a time-dependent modulation in expression of inflammatory markers. The most noteworthy change was observed for interleukin-8 (IL-8) where expression changed by +11 (1 h), +26 (4 h), +3 (8 h) and –3 (24 h) recovery. Increased IL-8 expression was observed in inflamed mucosa of patients with IBD [466], and incubation of colonic epithelial cells with IL-8 was associated with recruitment of neutrophils and an increase in resealing of the mucosal barrier [467–469].

Compared to monolayers, recovery of intestinal epithelial integrity is better established in isolated intestinal tissue mucosae and in animal models. Short term exposure of isolated guinea pig mucosae to Triton® X-100 (0.06% w/v) led to denudation at the tips of 86% of ileal villi, which reversed after 60 min [465]. The perfusion of rat intestine for 15 min caused denuding of both enterocytes and goblet cells in the order of sodium deoxycholate (5 mM) > SLS (5 mM) > EDTA (25mM) > PEG 400 (50%) [470]. While extensive loss of epithelial cells (80%) recovered to only 5% after 2 h, complete recovery was noted after 24 h. Damage to rat intestinal mucosa was also observed following instillation of C_{10} (100 mM) and oral delivery of SDS (1–2%), and repair was noted by light microscopy between 30 and 60 min for C_{10} [62] and 1 h for SDS [75]. Pre-treatment of rat colonic mucosae with misoprostol reduced C_{10} -induced mucosal damage through stimulation of mucous secretion [149]. Misoprostol also attenuated C_{10} -induced damage to Caco-2 monolayers in a mechanism involving Ca^{2+} homeostasis and production of phospholipids to reinforce the plasma membrane [149]. The capacity of the intestinal mucosa to repair following injury has been studied in response to endogenous detergents [471], dietary agents, and xenobiotics [472], where prostaglandins [473], nitric oxide [471] and growth factors [474], play a protective role (reviewed in [452]). Injured intestinal epithelial cells are detached from the basement membrane and from cells that shouldering the injury, which leads to sloughing. The response to protect exposure of the basement membrane to luminal constituents involves villus contraction, epithelial restitution and resealing of complexes at the lateral membrane.

5.2. The bystander absorption argument

A concern regarding the use of PEs in oral peptide formulations is the possible absorption of bystanders (such as toxins, bacteria, viruses and allergens) during the period of temporary epithelial permeability enhancement. There have been no adverse local or systemic immunological responses reported in the scientific literature or clinical studies with any of the lead candidate PE-containing delivery systems. A relatively small number of pre-clinical studies have evaluated the uptake of potentially harmful substances during an enhancement window. C_{10} (10–100 mM) did not permit translocation of *E. coli* across Caco-2 monolayers because the surfactant exhibits strong antimicrobial activity [89], whereas translocation of *E. coli* across monolayers was increased in the presence of Triton® X-100. In isolated ileal mucosae, C_{10} reduced adhesion of *S. typhimurium* [475].

Concern also relates to induced permeability to inflammatory molecules that may lead to intestinal inflammation, which in turn may propagate more serious permeability alterations associated with IBD. In principle, this is a valid concern as chemical inducers of murine colitis including dextran sodium sulphate (DSS) are believed to damage the colonic epithelium resulting in uptake of bacterial products to the underlying immune cell-rich sub-mucosa. However, PEs have not been shown to cause the extensive mucosal damage associated with ingestion of large quantities of DSS in rodents. On the contrary, even the PEs in clinical trials have only a modest effect on permeability of peptides so they are unlikely to increase permeability to large bacterial toxins or endotoxin. Chitosan nanoparticles opened TJs and increased intestinal permeability of insulin, but co-administration with LPS (5 mg/kg) to rats for 7 days did not lead to an increase in hepatic necrosis [476]. In a similar study, co-administration of penetratin (5mM) or

C₁₀ (154 mM) with LPS for 7 days to mice had no effect on biomarkers of hepatic necrosis, however taurodeoxycholate (96 mM) increased plasma levels of aspartate transaminase and alanine transaminase [477]. It is noteworthy that uptake of LPS into the hepatic portal vein is not pathological as low level uptake is associated with maintaining a normal responsive kupffer cell population, but excessive levels may lead to hepatic dysfunction [478].

The necessity of intimate contact of PE and payload at the mucosal surface to cause temporary and reversible membrane permeability provides an additional argument that bystander molecules present will not have their permeation assisted, since they are dilute and not in direct association with the formulation.

It has been proposed [479] that surfactants derived from lipid digestion (including monoglycerides and free fatty acids [463,480]) can impair barrier integrity following ingestion of a high fat meal. Clinical manifestations can be diarrhoea or constipation. In more extreme cases, both endogenous and exogenous substances that alter barrier integrity are considered environmental contributors to intestinal diseases that are characterised by an over active immune response. A strong link is established between abnormal intestinal permeability and mucosal inflammation in IBD [481]. While there is no evidence to suggest that PEs increase regional antigen permeation, they should not be used in patients with IBD. Most toxicological concerns relating to the use of

PEs relate to transcellular perturbation, but the effects of paracellular PEs are not free from concerns relating to barrier integrity. Altered expression of claudin is associated with ulcerative colitis (claudin-1,2,3,4,7), Crohn's disease (claudin-2,3,5,8), celiac disease (claudin-2,3,5,7,15), irritable bowel disease (claudin-1,2,4) and various infectious diseases (reviewed in [482]).

5.3. Are paracellular PEs safer than transcellular PEs?

One of the driving forces for development of paracellular PEs is their apparent safety advantage relative to those that act transcellularly, but these agents are new chemical entities whose development comes with far more risk than PEs with established safety in man. The most advanced paracellular candidate is EDTA, but there are significant restrictions on this chelator that could impact the dose required to facilitate oral peptide delivery [483]. The maximum amount of EDTA in the FDA Inactive Ingredients List is 4 mg, significantly lower than the quantities used in pre-clinical testing and in oral peptide clinical trials. The discovery of ZoT for example was associated with elucidation of the toxicological actions of virulent strains of *V. cholera*. The native forms of ZoT, *C. perfringens* enterotoxin, and melittin are mainly tools to elucidate the structure and function of the TJ, but their analogues might be more suitable as PE candidates in oral peptide formulations. The FDA

Table 3

Classification for intestinal PEs based on criteria in preclinical models. White boxes: pass; grey boxes: fail.

Classification	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6
	Fast enhancement Strong enhancement Good recovery	Fast enhancement Strong enhancement Poor recovery	Fast enhancement Poor enhancement Good recovery	Slow enhancement Strong enhancement Good recovery	Slow enhancement Strong enhancement Poor recovery	Slow enhancement Poor enhancement Good recovery
Rate	Rate of enhancement I (in vitro) Time required for 80% drop in teer in caco-2	<15 min	<15 min	<15 min	>15 min	>15 min
	Rate of enhancement II (ex vivo) Time required for an 80% drop in teer in isolated intestinal mucosae in ussing chambers	<30 min	<30 min	<30 min	>30 min	>30 min
	Rate of enhancement III (in situ) TMAX fd4 in intestinal instillation	<30 min	<30 min	<30 min	>30 min	>30 min
Extent	Extent of enhancement I (in vitro) PAPP of mannitol in caco-2 monolayers isolated intestinal mucosae in ussing chambers over 2 h	>2 x 10 ⁻⁵ cm/s	>2 x 10 ⁻⁵ cm/s	<2 x 10 ⁻⁵ cm/s	>2 x 10 ⁻⁵ cm/s	<2 x 10 ⁻⁵ cm/s
	Extent of enhancement II (ex vivo) PAPP of marker in isolated intestinal mucosae in ussing chambers after two h	>1 x 10 ⁻⁵ cm/s	>1 x 10 ⁻⁵ cm/s	<1 x 10 ⁻⁵ cm/s	>1 x 10 ⁻⁵ cm/s	<1 x 10 ⁻⁵ cm/s
	Extent of enhancement III (in situ) FABS of fd4 in intestinal instillation or via oral delivery	>20% F	>20% F	<20% F	>20% F	<20% F
Recovery	Recovery I (in vitro) Histology score in isolated intestinal mucosae in ussing chambers in two h	1–to–3	4	1–to–3	1–to–3	4
	Recovery II (in situ) Histology score after 2 h in intestinal instillation	1–to–2	3–4	1–to–3	1–to–3	4
	Recovery III (in situ) Fabs of fd4 between 2–4 h post addition of enhancer is <25% of coadministration (1–2 h)	<25% of F ₀₋₂	>25% of F ₀₋₂	<25% of F ₀₋₂	<25% of F ₀₋₂	>25% of F ₀₋₂
	Safety (global rating of five) Allowed excipient, food additive, gras, no systemic toxicity, no activation of cellular signalling	+++++	-----	+++++	+++++	-----
	Comment	Ideal enhancer with strong development potential in oral peptide delivery	Unacceptable safety concerns, but other routes might be possible (e.g. topical)	Modest enhancer with good safety, possible delivery of potent low MW solutes	Strong enhancer that is more ideally suited to topical, rectal due to slow onset	Unacceptable safety concerns, but other routes might be possible (e.g. topical)

and International Council for Harmonisation (ICH) provides guidance documents for short, medium and long term pre-clinical safety testing of candidate pharmaceutical excipients. In the event that a candidate excipient is found to be pharmacologically active or where previous studies have reported toxicity, this can influence further development.

The difference in mode of action for 1st and 2nd generation TJ modulators tends to favour development of 2nd generation from a safety perspective. There is less likelihood that a 2nd generation short peptide sequence will be absorbed at sufficiently high concentration to reach a threshold concentration to alter TJs at other epithelia. This contrasts with first generation TJ modulators that target ubiquitous cell processes in opening of TJs (e.g. PKC [484]). The consequence of absorption of first generation molecules is far more difficult to predict in different tissues, organs and systems, and they therefore represent a significant development risk due to potential systemic toxicity. Nevertheless, regardless of the general consensus that 2nd generation PEs are less likely to have off-target toxicity, there have been cases where peptides that alter CAR sequences can increase permeability at other epithelia (e.g. kidney epithelia) and endothelia (e.g. blood brain barrier [485]) and, if the peptides are not metabolised prior to excretion, there is the possibility that they could be concentrated within the bladder to act on TJs. The Claudin binder, C-CPE, distributed to liver (24%) and kidney (9.5%) 10 min post i.v. injection in mice [486]. Hepatic levels returned to 3.2% after 3 h, but renal levels increased to 47% after 6 h. A C-CPE mutant that lacked the ability to bind claudin had lower distribution in the liver but similar distribution in kidney. Likewise, C-CPE increased biomarkers of hepatic injury. This PE class therefore still poses a development risk that justifies more detailed safety assessment.

To date there have been no disclosed clinical assessments involving specific paracellular PEs, which indicates that TJ modulators are not yet viewed as lead candidates. There is the argument that there is little incentive to develop innovative excipients and delivery systems due to the risk involved. In order to avoid the requirement of supplementary safety testing, delivery companies are prioritising development of PEs that have established safety record in man. Others have invested in pre-clinical safety testing that enables them to formally request GRAS status and/or provide regulators with safety information to mitigate risk [368].

6. PE developability classification system

The effectiveness of a PE for oral peptide delivery can be simplified to three key criteria (i) onset time, (ii) extent of enhancement and (iii) recoverability/safety. These criteria were first used to categorise PEs as Class I (strong and fast reactivity with fast recovery, e.g. C₁₀, acylcarnitines), Class II (moderate and fast reactivity with fast recovery e.g. salicylate) and Class III (strong reactivity with slow recovery e.g. EDTA and CA) [206]. Here, we offer a PE Developability Classification System (PEDCS) to assist identification of PEs that demonstrate adequate enhancement for further assessment in oral formulation (Table 3). The metrics were selected based on performance of C₁₀ in (i) Caco-2 monolayers (TEER, P_{app} of [¹⁴C]-mannitol), (ii) isolated rat colonic mucosae (TEER, P_{app} of [¹⁴C]-mannitol, histology score [487], and (iii) colonic instillation (F_{ABS} of FD4, enhancement kinetics, histology score). We also reviewed Compendium information and literature. A PE is required to obtain a score of >2 of 3 in order to be assigned “Fast Enhancement” status; >2 of 3 to gain “strong enhancement” and >3 of 4 to gain “recovery/safety” criteria. A fast onset of enhancement action is required to ensure permeation of peptides under normal GI transit. It is therefore optimal that a rapid decrease in TEER be recorded in cell monolayer and tissue mucosae models (Table 3). The most effective rate metric is a short T_{MAX} of FD4 in rat intestinal instillations. An optimal extent is measured by an increase in P_{app} of FD4 to >1 × 10^{−5} cm/s; a level predictive of high permeability within the BCS. In instillation, F_{ABS} of FD4 should be >20% in rats.

Safety is measured at effective concentrations by (i) tissue damage observed in 2 h incubation in isolated intestinal tissue mounted in Ussing chambers, (ii) tissue damage and histological recovery after 2 h in rat intestinal instillations, (iii) the capacity of the GI to recover from a transport induced state as determined by F_{ABS} of FD4 measured 2–4 h post administration of PE, and (iv) overall safety assessment is a global rating that must be substantiated with assessment of five metrics (excipient status, additive status, GRAS status, mode of enhancement action, and published systemic safety data). The ideal PE is designated Class 1 due to (i) fast enhancement to facilitate rapid uptake of peptide during GI transit, (ii) exhibition of potent and efficacious enhancement and (iii) demonstration of good safety/recovery. Class 2 PEs are likely to have strong enhancement action, but overall regional safety via oral delivery is a key consideration; although other routes may be possible. Class 3 PEs have modest enhancement and could be more effective in assisting permeation of selected small molecule drugs. Class 4 PEs have strong enhancement action but their slow rate of onset limits their application to alternative routes of administration, such as topical buccal or rectal. Class 5 PEs are slow to act, but despite strong enhancement action, their use is limited by safety concerns. Class 6 PEs have good safety, but this is accompanied by slow onset and modest to low enhancement action, which renders them unsuitable for oral peptide delivery.

7. Conclusions

Over 250 PEs have improved intestinal permeability of poorly absorbed drugs including peptides in every conceivable pre-clinical drug delivery model. Yet there is a relatively poor translation of PE-based delivery systems for oral peptides. The majority of pre-clinical data was achieved in epithelial monolayer cultures, isolated tissue and intestinal instillations, and, while these delivery models identify PEs that alter barrier integrity and improve flux/bioavailability of peptides, such systems do not address formulation considerations. Clinical evaluation has largely been limited to a small group of PEs that can be formulated into solid-dose formulations and which have established safety in man. Even then, bioavailability typically remains low and variable, leading to new strategies to formulate peptides, such as PEs in lipid-based systems and the application of nano-encapsulation strategies with PEs. Novel TJ modulators are a promising group of candidate PEs, but none have yet progressed to clinical testing. Convergence between delivery and formulation sciences will facilitate better understanding of the hurdles to translation between oral peptide delivery systems and optimal dosage forms for man.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.addr.2016.06.005>.

Acknowledgements

This research was funded in part by Science Foundation Ireland Centre for Medical Devices (CURAM), 13/RC/2073, by the European Union Seventh Framework Programme (FP7/2007–2013) under grant agreement no. 281035 (TRANS-INT), and by the RCSI School of Pharmacy. The authors acknowledge the assistance of Ms Emma Duke and Mr Barinédum Yorkuri (both RCSI Pharmacy) for assistance in populating Table 1 and Dr Ronan Kelly (RCSI Library) for assisting in the retrieval of archived research publications.

References

- [1] E.S. Swenson, W.J. Curatolo, (C) Means to enhance penetration: (2) intestinal permeability enhancement for proteins, peptides and other polar drugs: mechanisms and potential toxicity, *Adv. Drug Deliv. Rev.* 8 (1992) 39–92.
- [2] S. Maher, B. Ryan, A. Duffy, D.J. Brayden, Formulation strategies to improve oral peptide delivery, *Pharm. Pat. Anal.* 3 (2014) 313–336.
- [3] S. Tuvia, J. Atsmon, S.L. Teichman, S. Katz, P. Salama, D. Pelled, I. Landau, I. Karmeli, M. Bidlingmaier, C.J. Strasburger, D.L. Kleinberg, S. Melmed, R. Mamluk, Oral octreotide absorption in human subjects: comparable pharmacokinetics to

- parenteral octreotide and effective growth hormone suppression, *J. Clin. Endocrinol. Metab.* 97 (2012) 2362–2369.
- [4] T.A. Aguirre, D. Teijeiro-Osorio, M. Rosa, I.S. Coulter, M.J. Alonso, D.J. Brayden, Current status of selected oral peptide technologies in advanced preclinical development and in clinical trials, *Adv. Drug Deliv. Rev.* 106 (2016) 29–47.
 - [5] K. Fosgerau, T. Hoffmann, Peptide therapeutics: current status and future directions, *Drug Discov. Today* 20 (2015) 122–128.
 - [6] A.L. Lewis, J. Richard, Challenges in the delivery of peptide drugs: an industry perspective, *Ther. Deliv.* 6 (2015) 149–163.
 - [7] A. Patel, K. Cholkar, A.K. Mitra, Recent developments in protein and peptide parenteral delivery approaches, *Ther. Deliv.* 5 (2014) 337–365.
 - [8] D.R. Owens, Clinical evidence for the earlier initiation of insulin therapy in type 2 diabetes, *Diabetes Technol. Ther.* 15 (2013) 776–785.
 - [9] M.B. Davidson, Pro's and con's of the early use of insulin in the management of type 2 diabetes: a clinical evaluation, *Curr. Opin. Endocrinol. Diabetes Obes.* 16 (2009) 107–112.
 - [10] E. Arbit, M. Kidron, Oral insulin: the rationale for this approach and current developments, *J. Diabetes Sci. Technol.* 3 (2009) 562–567.
 - [11] D.R. Owens, New horizons—alternative routes for insulin therapy, *Nat. Rev. Drug Discov.* 1 (2002) 529–540.
 - [12] M. Lorenz, A. Evers, M. Wagner, Recent progress and future options in the development of GLP-1 receptor agonists for the treatment of diabetes, *Bioorg. Med. Chem. Lett.* 23 (2013) 4011–4018.
 - [13] J. Wang, V. Yadav, A.L. Smart, S. Tajiri, A.W. Basit, Toward oral delivery of biopharmaceuticals: an assessment of the gastrointestinal stability of 17 peptide drugs, *Mol. Pharm.* 12 (2015) 966–973.
 - [14] L.R. Brown, Commercial challenges of protein drug delivery, *Expert Opin. Drug Deliv.* 2 (2005) 29–42.
 - [15] R. Eldor, E. Arbit, A. Corcos, M. Kidron, Glucose-reducing effect of the ORMD-0801 oral insulin preparation in patients with uncontrolled type 1 diabetes: a pilot study, *PLoS One* 8 (2013) e59524.
 - [16] R. Eldor, R. Kidron, E. Arbit, A Single-Blind, Two-Period Study to Assess the Safety and Pharmacodynamics of an Orally Delivered GLP-1 Analog (Exenatide) in Healthy Subjects, 70th Meeting of the American Diabetes Association (Florida, USA) (2010).
 - [17] S. Thakral, N.K. Thakral, D.K. Majumdar, Eudragit: a technology evaluation, *Expert Opin. Drug Deliv.* 10 (2013) 131–149.
 - [18] C. Schiller, C.P. Frohlich, T. Giessmann, W. Siegmund, H. Monnikes, N. Hosten, W. Weitschies, Intestinal fluid volumes and transit of dosage forms as assessed by magnetic resonance imaging, *Aliment. Pharmacol. Ther.* 22 (2005) 971–979.
 - [19] E. Toutou, A. Rubinstein, Targeted enteral delivery of insulin to rats, *Int. J. Pharm.* 30 (1986) 95–99.
 - [20] M. Morishita, I. Morishita, K. Takayama, Y. Machida, T. Nagai, Site-dependent effect of aprotinin, sodium caprate, Na₂EDTA and sodium glycocholate on intestinal absorption of insulin, *Biol. Pharm. Bull.* 16 (1993) 68–72.
 - [21] J.F. Woodley, Enzymatic barriers for GLP peptide and protein delivery, *Crit. Rev. Ther. Drug Carrier Syst.* 11 (1994) 61–95.
 - [22] F. Gabor, C. Fillafer, L. Neutsc, G. Ratzinger, M. Wirth, Improving oral delivery, *Handb. Exp. Pharmacol.* (2010) 345–398.
 - [23] A. Yamamoto, T. Taniguchi, K. Rikyuu, T. Tsuji, T. Fujita, M. Murakami, S. Muranishi, Effects of various protease inhibitors on the intestinal absorption and degradation of insulin in rats, *Pharm. Res.* 11 (1994) 1496–1500.
 - [24] M. Werle, H. Takeuchi, Strategies to Overcome the Enzymatic Barrier, in: A. Bernkop-Schnurch (Ed.), *Oral Delivery of Macromolecular Drugs*, Springer, US 2009, pp. 65–83.
 - [25] S.H. Welling, F. Hubalek, J. Jacobsen, D.J. Brayden, U.L. Rahbek, S.T. Buckley, The role of citric acid in oral peptide and protein formulations: relationship between calcium chelation and proteolysis inhibition, *Eur. J. Pharm. Biopharm.* 86 (2014) 544–551.
 - [26] Oramed Ltd US20110046053 (2011).
 - [27] V. Agarwal, I.K. Reddy, M.A. Khan, Oral delivery of proteins: effect of chicken and duck ovomucoid on the stability of insulin in the presence of α -chymotrypsin and trypsin, *Pharm. Pharmacol. Commun.* 6 (2000) 223–227.
 - [28] H. Liu, R. Tang, W.S. Pan, Y. Zhang, H. Liu, Potential utility of various protease inhibitors for improving the intestinal absorption of insulin in rats, *J. Pharm. Pharmacol.* 55 (2003) 1523–1529.
 - [29] Select committee on GRAS substances opinion: soy protein isolated (1979, report number 101, ID Code 97076-84-8), <http://www.fda.gov/Food/IngredientsPackagingLabeling/GRAS/SCOGS/ucm261441.htm>.
 - [30] J.C. Smith, F.D. Wilson, P.V. Allen, D.L. Berry, Hypertrophy and hyperplasia of the rat pancreas produced by short-term dietary administration of soya-derived protein and soybean trypsin inhibitor, *J. Appl. Toxicol.* 9 (1989) 175–179.
 - [31] R.N. Melmed, A.A. El-Aaser, S.J. Holt, Hypertrophy and hyperplasia of the neonatal rat exocrine pancreas induced by orally administered soybean trypsin inhibitor, *Biochim. Biophys. Acta* 421 (1976) 280–288.
 - [32] M. Werle, B. Loretz, D. Entstrasser, F. Föger, Design and evaluation of a chitosan-aptinin conjugate for the peroral delivery of therapeutic peptides and proteins susceptible to enzymatic degradation, *J. Drug Target.* 15 (2007) 327–333.
 - [33] A. Bernkop-Schnurch, A. Krauland, C. Valenta, Development and in vitro evaluation of a drug delivery system based on chitosan-EDTA BBI conjugate, *J. Drug Target.* 6 (1998) 207–214.
 - [34] G.M. Pauletti, S. Gangwar, G.T. Knipp, M.M. Nerurkar, F.W. Okumu, K. Tamura, T.J. Siahaan, R.T. Borchardt, Structural requirements for intestinal absorption of peptide drugs, *J. Control. Release* 41 (1996) 3–17.
 - [35] P. Langguth, V. Bohnert, J. Heizmann, H.P. Merkle, S. Wolfram, G.L. Amidon, S. Yamashita, The challenge of proteolytic enzymes in intestinal peptide delivery, *J. Control. Release* 46 (1997) 39–57.
 - [36] V. Agarwal, S. Nazzari, I.K. Reddy, M.A. Khan, Transport studies of insulin across rat jejunum in the presence of chicken and duck ovomucoids, *J. Pharm. Pharmacol.* 53 (2001) 1131–1138.
 - [37] E. Ziv, O. Lior, M. Kidron, Absorption of protein via the intestinal wall. A quantitative model, *Biochem. Pharmacol.* 36 (1987) 1035–1039.
 - [38] A.N. Round, N.M. Rigby, A. Garcia de la Torre, A. Macierzanka, E.N. Mills, A.R. Mackie, Lamellar structures of MUC2-rich mucin: a potential role in governing the barrier and lubricating functions of intestinal mucus, *Biomacromolecules* 13 (2012) 3253–3261.
 - [39] S.K. Lai, Y.Y. Wang, J. Hanes, Mucus-penetrating nanoparticles for drug and gene delivery to mucosal tissues, *Adv. Drug Deliv. Rev.* 61 (2009) 158–171.
 - [40] M. Boegh, M. Garcia-Diaz, A. Mullertz, H.M. Nielsen, Steric and interactive barrier properties of intestinal mucus elucidated by particle diffusion and peptide permeation, *Eur. J. Pharm. Biopharm.* 95 (2015) 136–143.
 - [41] Y. Aoki, M. Morishita, K. Takayama, Role of the mucous/glycocalyx layers in insulin permeation across the rat ileal membrane, *Int. J. Pharm.* 297 (2005) 98–109.
 - [42] A.W. Larhed, P. Artursson, J. Grasjo, E. Bjork, Diffusion of drugs in native and purified gastrointestinal mucus, *J. Pharm. Sci.* 86 (1997) 660–665.
 - [43] S. Takatsuka, T. Kitazawa, T. Morita, Y. Horikiri, H. Yoshino, Enhancement of intestinal absorption of poorly absorbed hydrophilic compounds by simultaneous use of mucolytic agent and non-ionic surfactant, *Eur. J. Pharm. Biopharm.* 62 (2006) 52–58.
 - [44] S. Takatsuka, T. Morita, A. Koguchi, Y. Horikiri, H. Yamahara, H. Yoshino, Synergistic absorption enhancement of salmon calcitonin and reversible mucosal injury by applying a mucolytic agent and a non-ionic surfactant, *Int. J. Pharm.* 316 (2006) 124–130.
 - [45] A. Bernkop-Schnurch, C. Valenta, S.M. Dae, Peroral polypeptide delivery. A comparative in vitro study of mucolytic agents, *Arzneimittelforschung* 49 (1999) 799–803.
 - [46] H. Malhaire, J.C. Gimel, E. Roger, J.P. Benoit, F. Lagarce, How to design the surface of peptide-loaded nanoparticles for efficient oral bioavailability? *Adv. Drug Deliv. Rev.* 106 (2016) 126–142.
 - [47] C.A. Lipinski, F. Lombardo, B.W. Dominy, P.J. Feeney, Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings, *Adv. Drug Deliv. Rev.* 46 (2001) 3–26.
 - [48] G.L. Amidon, H. Lennernas, V.P. Shah, J.P. Crison, A theoretical basis for a biopharmaceutical drug classification: the correlation of in vitro drug product dissolution and in vivo bioavailability, *Pharm. Res.* 12 (1995) 413–420.
 - [49] C.Y. Wu, L.Z. Benet, Predicting drug disposition via application of BCS: transport/absorption/elimination interplay and development of a biopharmaceutics drug disposition classification system, *Pharm. Res.* 22 (2005) 11–23.
 - [50] J.M. Butler, J.B. Dressman, The developability classification system: application of biopharmaceutics concepts to formulation development, *J. Pharm. Sci.* 99 (2010) 4940–4954.
 - [51] D.J. McClements, F. Li, H. Xiao, The nutraceutical bioavailability classification scheme: classifying nutraceuticals according to factors limiting their oral bioavailability, *Annu. Rev. Food Sci. Technol.* 6 (2015) 299–327.
 - [52] C.A. Lipinski, Lead- and drug-like compounds: the rule-of-five revolution, *Drug Discov. Today Technol.* 1 (2004) 337–341.
 - [53] Y.H. Lee, B.A. Perry, J.P. Sutyak, W. Stern, P.J. Sinko, Regional differences in intestinal spreading and pH recovery and the impact on salmon calcitonin absorption in dogs, *Pharm. Res.* 17 (2000) 284–290.
 - [54] L.Z. Benet, G.L. Amidon, D.M. Barends, H. Lennernas, J.E. Polli, V.P. Shah, S.A. Stavchansky, L.X. Yu, The use of BDDCS in classifying the permeability of marketed drugs, *Pharm. Res.* 25 (2008) 483–488.
 - [55] L.Z. Benet, C.L. Cummins, The drug efflux-metabolism alliance: biochemical aspects, *Adv. Drug Deliv. Rev.* 50 (Suppl. 1) (2001) S3–11.
 - [56] A. Foradori, S. Mezzano, C. Videla, J. Pefaur, A. Elberg, Modification of the pharmacokinetics of cyclosporine A and metabolites by the concomitant use of Neoral and diltiazem or ketoconazole in stable adult kidney transplants, *Transplant. Proc.* 30 (1998) 1685–1687.
 - [57] H.J. Lee, G.L. Amidon, The effect of enzyme inhibitor and absorption site following [D-al₂, D-leu₅]enkephalin oral administration in rats, *Biopharm. Drug Dispos.* 23 (2002) 131–141.
 - [58] S.C. Sutton, E.L. McCluskey, K. Engle, J.D. Pipkin, J.A. Fix, Enhanced bioavailability of cefoxitin using palmitoylcarnitine. II. Use of directly compressed tablet formulations in the rat and dog, *Pharm. Res.* 10 (1993) 1516–1520.
 - [59] H. Junginger, Excipients as Absorption Enhancers, in: R. Krishna, L. Yu (Eds.), *Biopharmaceutics Applications in Drug Development*, Springer, US 2008, pp. 139–174.
 - [60] R.I. Mahato, A.S. Narang, L. Thoma, D.D. Miller, Emerging trends in oral delivery of peptide and protein drugs, *Crit. Rev. Ther. Drug Carrier Syst.* 20 (2003) 153–214.
 - [61] S.S. Davis, J.G. Hardy, J.W. Fara, Transit of pharmaceutical dosage forms through the small intestine, *Gut* 27 (1986) 886–892.
 - [62] X. Wang, S. Maher, D.J. Brayden, Restoration of rat colonic epithelium after in situ intestinal instillation of the absorption promoter, sodium caprate, *Ther. Deliv.* 1 (2010) 75–82.
 - [63] C.-M. Lehr, F.G.J. Poelma, H.E. Junginger, J.J. Tukker, An estimate of turnover time of intestinal mucus gel layer in the rat in situ loop, *Int. J. Pharm.* 70 (1991) 235–240.
 - [64] J.D. Smart, The basics and underlying mechanisms of mucoadhesion, *Adv. Drug Deliv. Rev.* 57 (2005) 1556–1568.
 - [65] R. Kumar, V.R. Sinha, Thiomers: A potential carrier for therapeutic delivery, *React. Funct. Polym.* 73 (2013) 1156–1166.
 - [66] A. Bernkop-Schnurch, Mucoadhesive systems in oral drug delivery, *Drug Discov. Today Technol.* 2 (2005) 83–87.
 - [67] G. Millotti, F. Laffeur, G. Perera, C. Vigil, K. Pickl, F. Sinner, A. Bernkop-Schnurch, In vivo evaluation of thiolated chitosan tablets for oral insulin delivery, *J. Pharm. Sci.* 103 (2014) 3165–3170.

- [68] F.A. Dorkoosh, J.C. Verhoef, J.H. Verheijden, M. Rafiee-Tehrani, G. Borchard, H.E. Junginger, Peroral absorption of octreotide in pigs formulated in delivery systems on the basis of superporous hydrogel polymers, *Pharm. Res.* 19 (2002) 1532–1536.
- [69] F.A. Dorkoosh, M.P. Stokkel, D. Blok, G. Borchard, M. Rafiee-Tehrani, J.C. Verhoef, H.E. Junginger, Feasibility study on the retention of superporous hydrogel composite polymer in the intestinal tract of man using scintigraphy, *J. Control. Release* 99 (2004) 199–206.
- [70] D.J. Brayden, E. Walsh, Efficacious intestinal permeation enhancement induced by the sodium salt of 10-undecylenic acid, a medium chain fatty acid derivative, *AAPS J.* 16 (2014) 1064–1076.
- [71] M. Shima, Y. Kimura, S. Adachi, R. Matsuno, Recovery of Caco-2 cell monolayers to normal from the transport-enhanced state induced by capric acid sodium salt and its monoacylglycerol, *Biosci. Biotechnol. Biochem.* 63 (1999) 680–687.
- [72] T.W. Leonard, J. Lynch, M.J. McKenna, D.J. Brayden, Promoting absorption of drugs in humans using medium-chain fatty acid-based solid dosage forms: GIPET, *Expert Opin. Drug Deliv.* 3 (2006) 685–692.
- [73] E.L. Lecluyse, S.C. Sutton, J.A. Fix, In vitro effects of long-chain acylcarnitines on the permeability, transepithelial electrical resistance and morphology of rat colonic mucosa, *J. Pharmacol. Exp. Ther.* 265 (1993) 955–962.
- [74] M. Baluom, M. Friedman, A. Rubinstein, The importance of intestinal residence time of absorption enhancer on drug absorption and implication on formulative considerations, *Int. J. Pharm.* 176 (1998) 21–30.
- [75] Y. Narkar, R. Burnette, R. Bleher, R. Albrecht, A. Kandela, J.R. Robinson, Evaluation of mucosal damage and recovery in the gastrointestinal tract of rats by a penetration enhancer, *Pharm. Res.* 25 (2008) 25–38.
- [76] M. Baluom, M. Friedman, P. Assaf, A.I. Haj-Yehia, A. Rubinstein, Synchronized release of sulpiride and sodium decanoate from HPMC matrices: a rational approach to enhance sulpiride absorption in the rat intestine, *Pharm. Res.* 17 (2000) 1071–1076.
- [77] M. Thanou, J.C. Verhoef, J.H. Verheijden, H.E. Junginger, Intestinal absorption of octreotide using trimethyl chitosan chloride: studies in pigs, *Pharm. Res.* 18 (2001) 823–828.
- [78] R.R.C. New, C.J. Kirby, Solubilisation of hydrophilic drugs in oily formulations, *Adv. Drug Deliv. Rev.* 25 (1997) 59–69.
- [79] R.G. Strickley, B.D. Anderson, Solid-state stability of human insulin. I. Mechanism and the effect of water on the kinetics of degradation in lyophiles from pH 2–5 solutions, *Pharm. Res.* 13 (1996) 1142–1153.
- [80] X. Cui, D. Cao, C. Qu, X. Zhang, A. Zheng, A study of the chemical and biological stability of vasoactive intestinal peptide, *Drug Dev. Ind. Pharm.* 39 (2013) 1907–1910.
- [81] A. Bak, D. Leung, S.E. Barrett, S. Forster, E.C. Minnihan, A.W. Leithead, J. Cunningham, N. Toussaint, L.S. Crocker, Physicochemical and formulation developability assessment for therapeutic peptide delivery—a primer, *AAPS J.* 17 (2015) 144–155.
- [82] P.J. Sinko, Y.H. Lee, V. Makhey, G.D. Leesman, J.P. Sutyak, H. Yu, B. Perry, C.L. Smith, P. Hu, E.J. Wagner, L.M. Falzone, L.T. McWhorter, J.P. Gilligan, W. Stern, Biopharmaceutical approaches for developing and assessing oral peptide delivery strategies and systems: in vitro permeability and in vivo oral absorption of salmon calcitonin (sCT), *Pharm. Res.* 16 (1999) 527–533.
- [83] T. Lindmark, J.D. Soderholm, G. Olaison, G. Alvan, G. Ocklind, P. Artursson, Mechanism of absorption enhancement in humans after rectal administration of ampicillin in suppositories containing sodium caprate, *Pharm. Res.* 14 (1997) 930–935.
- [84] A. Maroni, M.D. Del Curto, L. Zema, A. Foppoli, A. Gazzaniga, Film coatings for oral colon delivery, *Int. J. Pharm.* 457 (2013) 372–394.
- [85] A. Rubinstein, B. Tirosh, M. Baluom, T. Nassar, A. David, R. Radaï, I. Gliko-Kabir, M. Friedman, The rationale for peptide drug delivery to the colon and the potential of polymeric carriers as effective tools, *J. Control. Release* 46 (1997) 59–73.
- [86] G. Fethi, S. Lindberg, K. Itoh, N. Okada, T. Fujita, F. Habib, P. Artersson, M. Attia, A. Yamamoto, Improvement of absorption enhancing effects of n-dodecyl-beta-D-maltopyranoside by its colon-specific delivery using chitosan capsules, *Int. J. Pharm.* 293 (2005) 127–135.
- [87] T.Y. Ma, D. Nguyen, V. Bui, H. Nguyen, N. Hoa, Ethanol modulation of intestinal epithelial tight junction barrier, *Am. J. Phys.* 276 (1999) G965–G974.
- [88] M. Tomita, M. Hayashi, S. Awazu, Comparison of absorption-enhancing effect between sodium caprate and disodium ethylenediaminetetraacetate in Caco-2 cells, *Biol. Pharm. Bull.* 17 (1994) 753–755.
- [89] S. Maher, T.W. Leonard, J. Jacobsen, D.J. Brayden, Safety and efficacy of sodium caprate in promoting oral drug absorption: from in vitro to the clinic, *Adv. Drug Deliv. Rev.* 61 (2009) 1427–1449.
- [90] G. Fricker, A. Fahr, C. Beglinger, T. Kissel, G. Reiter, J. Drewe, Permeation enhancement of octreotide by specific bile salts in rats and human subjects: in vitro, in vivo correlations, *Br. J. Pharmacol.* 117 (1996) 217–223.
- [91] A. Fukui, Y. Naito, O. Handa, M. Kugai, T. Tsuji, H. Yoriki, Y. Qin, S. Adachi, Y. Higashimura, K. Mizushima, K. Kamada, K. Katada, K. Uchiyama, T. Ishikawa, T. Takagi, N. Yagi, S. Kokura, T. Yoshikawa, Acetyl salicylic acid induces damage to intestinal epithelial cells by oxidation-related modifications of ZO-1, *Am. J. Physiol. Gastrointest. Liver Physiol.* 303 (2012) G927–G936.
- [92] M. Thanou, J.C. Verhoef, H.E. Junginger, Oral drug absorption enhancement by chitosan and its derivatives, *Adv. Drug Deliv. Rev.* 52 (2001) 117–126.
- [93] M. Kondoh, A. Masuyama, A. Takahashi, N. Asano, H. Mizuguchi, N. Koizumi, M. Fujii, T. Hayakawa, Y. Horiguchi, Y. Watanabe, A novel strategy for the enhancement of drug absorption using a claudin modulator, *Mol. Pharmacol.* 67 (2005) 749–756.
- [94] E. Windsor, G.E. Cronheim, Gastro-intestinal absorption of heparin and synthetic heparinoids, *Nature* 190 (1961) 263–264.
- [95] E. Windsor, L. Freeman, An investigation of routes of administration of heparin other than injection, *Am. J. Med.* 37 (1964) 408–416.
- [96] J.L. Madara, R. Moore, S. Carlson, Alteration of intestinal tight junction structure and permeability by cytoskeletal contraction, *Am. J. Phys.* 253 (1987) C854–C861.
- [97] J.L. Madara, J. Stafford, D. Barenberg, S. Carlson, Functional coupling of tight junctions and microfilaments in T84 monolayers, *Am. J. Phys.* 254 (1988) G416–G423.
- [98] A. Fasano, S. Uzzau, Modulation of intestinal tight junctions by Zonula occludens toxin permits enteric administration of insulin and other macromolecules in an animal model, *J. Clin. Invest.* 99 (1997) 1158–1164.
- [99] A. Yamamoto, H. Tatsumi, M. Maruyama, T. Uchiyama, N. Okada, T. Fujita, Modulation of intestinal permeability by nitric oxide donors: implications in intestinal delivery of poorly absorbable drugs, *J. Pharmacol. Exp. Ther.* 296 (2001) 84–90.
- [100] I.T. Makgiansar, M. Avery, Y. Hu, K.L. Audus, T.J. Siahaan, Improving the selectivity of HAV-peptides in modulating E-cadherin-E-cadherin interactions in the intercellular junction of MDCK cell monolayers, *Pharm. Res.* 18 (2001) 446–453.
- [101] S. Tavelin, K. Hashimoto, J. Malkinson, L. Lazorova, I. Toth, P. Artursson, A new principle for tight junction modulation based on occludin peptides, *Mol. Pharmacol.* 64 (2003) 1530–1540.
- [102] N. Sonoda, M. Furuse, H. Sasaki, S. Yonemura, J. Katahira, Y. Horiguchi, S. Tsukita, Clostridium perfringens enterotoxin fragment removes specific claudins from tight junction strands: evidence for direct involvement of claudins in tight junction barrier, *J. Cell Biol.* 147 (1999) 195–204.
- [103] L. Gonzalez-Mariscal, S. Hernandez, J. Vega, Interventions designed to enhance drug delivery across epithelial and endothelial cells through the paracellular pathway, *Recent Pat. Drug Deliv. Formul.* 2 (2008) 145–176.
- [104] A. Fasano, J.P. Nataro, Intestinal epithelial tight junctions as targets for enteric bacteria-derived toxins, *Adv. Drug Deliv. Rev.* 56 (2004) 795–807.
- [105] T.Y. Ma, J.M. Anderson, CHAPTER 61 — Tight Junctions and the Intestinal Barrier, in: L.R. Johnson (Ed.), *Physiology of the Gastrointestinal Tract* (Fourth Edition), Academic Press, Burlington 2006, pp. 1559–1594.
- [106] University of Maryland at Baltimore US5665389 (1997).
- [107] A. Fasano, Zonulin and its regulation of intestinal barrier function: the biological door to inflammation, autoimmunity, and cancer, *Physiol. Rev.* 91 (2011) 151–175.
- [108] S. Gopalakrishnan, A. Tripathi, A.P. Tamiz, S.S. Alkan, N.B. Pandey, Larazotide acetate promotes tight junction assembly in epithelial cells, *Peptides* 35 (2012) 95–101.
- [109] C.P. Kelly, P.H. Green, J.A. Murray, A. Dimarino, A. Colatella, D.A. Leffler, T. Alexander, R. Arsenescu, F. Leon, J.G. Jiang, L.A. Arterburn, B.M. Paterson, R.N. Fedorak, G. Larazotide Acetate Celiac Disease Study, Larazotide acetate in patients with celiac disease undergoing a gluten challenge: a randomised placebo-controlled study, *Aliment. Pharmacol. Ther.* 37 (2013) 252–262.
- [110] D.A. Leffler, C.P. Kelly, H.Z. Abdallah, A.M. Colatella, L.A. Harris, F. Leon, L.A. Arterburn, B.M. Paterson, Z.H. Lan, J.A. Murray, A randomized, double-blind study of larazotide acetate to prevent the activation of celiac disease during gluten challenge, *Am. J. Gastroenterol.* 107 (2012) 1554–1562.
- [111] S. Gopalakrishnan, N. Pandey, A.P. Tamiz, J. Vere, R. Carrasco, R. Somerville, A. Tripathi, M. Ginski, B.M. Paterson, S.S. Alkan, Mechanism of action of ZOT-derived peptide AT-1002, a tight junction regulator and absorption enhancer, *Int. J. Pharm.* 365 (2009) 121–130.
- [112] S. Maher, M. Devocelle, S. Ryan, S. McClean, D.J. Brayden, Impact of amino acid replacements on in vitro permeation enhancement and cytotoxicity of the intestinal absorption promoter, melittin, *Int. J. Pharm.* 387 (2010) 154–160.
- [113] A. Fasano, G. Fiorentini, G. Donelli, S. Uzzau, J.B. Kaper, K. Margareten, X. Ding, S. Guandalini, L. Comstock, S.E. Goldblum, Zonula occludens toxin modulates tight junctions through protein kinase C-dependent actin reorganization, in vitro, *J. Clin. Invest.* 96 (1995) 710–720.
- [114] M. Di Pietro, R. Lu, S. Uzzau, W. Wang, K. Margareten, C. Pazzani, F. Maimone, A. Fasano, Zonula occludens toxin structure-function analysis. Identification of the fragment biologically active on tight junctions and of the zonulin receptor binding domain, *J. Biol. Chem.* 276 (2001) 19160–19165.
- [115] K.H. Song, A. Fasano, N.D. Eddington, Effect of the six-mer synthetic peptide (AT1002) fragment of zonula occludens toxin on the intestinal absorption of cyclosporin A, *Int. J. Pharm.* 351 (2008) 8–14.
- [116] M. Kataoka, J.F. Head, B.A. Seaton, D.M. Engelman, Melittin binding causes a large calcium-dependent conformational change in calmodulin, *Proc. Natl. Acad. Sci. U. S. A.* 86 (1989) 6944–6948.
- [117] S. Maher, S. McClean, Investigation of the cytotoxicity of eukaryotic and prokaryotic antimicrobial peptides in intestinal epithelial cells in vitro, *Biochem. Pharmacol.* 71 (2006) 1289–1298.
- [118] S. Maher, X. Wang, V. Bzik, S. McClean, D.J. Brayden, Evaluation of intestinal absorption and mucosal toxicity using two promoters. II. Rat instillation and perfusion studies, *Eur. J. Pharm. Sci.* 38 (2009) 301–311.
- [119] S. Maher, R. Kennelly, V.A. Bzik, A.W. Baird, X. Wang, D. Winter, D.J. Brayden, Evaluation of intestinal absorption enhancement and local mucosal toxicity of two promoters. I. Studies in isolated rat and human colonic mucosae, *Eur. J. Pharm. Sci.* 38 (2009) 291–300.
- [120] P. Nava, S. Lopez, C.F. Arias, S. Islas, L. Gonzalez-Mariscal, The rotavirus surface protein VP8 modulates the gate and fence function of tight junctions in epithelial cells, *J. Cell Sci.* 117 (2004) 5509–5519.
- [121] Mariscal-Gonzalez L, Nava-Dominguez, P. US2006/0251663 (2006).
- [122] A. Veshnyakova, J. Protze, J. Rossa, I.E. Blasig, G. Krause, J. Piontek, On the interaction of *Clostridium perfringens* enterotoxin with claudins, *Toxins* (Basel) 2 (2010) 1336–1356.
- [123] R.J. Msrny, G.T. Brown, K. Gerner-Smidt, A.G. Buret, J.B. Meddings, C. Quan, M. Koval, A. Nusrat, A key claudin extracellular loop domain is critical for epithelial barrier integrity, *Am. J. Pathol.* 172 (2008) 905–915.
- [124] C. Staat, C. Coisne, S. Dabrowski, S.M. Stamatovic, A.V. Andjelkovic, H. Wolburg, B. Engelhardt, I.E. Blasig, Mode of action of claudin peptidomimetics in the transient opening of cellular tight junction barriers, *Biomaterials* 54 (2015) 9–20.

- [125] Y. Saitoh, H. Suzuki, K. Tani, K. Nishikawa, K. Irie, Y. Ogura, A. Tamura, S. Tsukita, Y. Fujiyoshi, Tight junctions. Structural insight into tight junction disassembly by Clostridium perfringens enterotoxin, *Science* 347 (2015) 775–778.
- [126] H. Uchida, M. Kondoh, T. Hanada, A. Takahashi, T. Hamakubo, K. Yagi, A claudin-4 modulator enhances the mucosal absorption of a biologically active peptide, *Biochem. Pharmacol.* 79 (2010) 1437–1444.
- [127] A. Takahashi, Y. Saito, M. Kondoh, K. Matsushita, S.M. Krug, H. Suzuki, H. Tsujino, X. Li, H. Aoyama, K. Matsuhisa, T. Uno, M. Fromm, T. Hamakubo, K. Yagi, Creation and biochemical analysis of a broad-specific claudin binder, *Biomaterials* 33 (2012) 3464–3474.
- [128] J.M. Anderson, C.M. Van Itallie, Tight junctions and the molecular basis for regulation of paracellular permeability, *Am. J. Phys.* 269 (1995) G467–G475.
- [129] T. Lindmark, Y. Kimura, P. Artursson, Absorption enhancement through intracellular regulation of tight junction permeability by medium chain fatty acids in Caco-2 cells, *J. Pharmacol. Exp. Ther.* 284 (1998) 362–369.
- [130] M. Hayashi, M. Tomita, Mechanistic analysis for drug permeation through intestinal membrane, *Drug Metab. Pharmacokinet.* 22 (2007) 67–77.
- [131] C.B. Collares-Buzato, G.T. McEwan, M.A. Jepson, N.L. Simmons, B.H. Hirst, Paracellular barrier and junctional protein distribution depend on basolateral extracellular Ca^{2+} in cultured epithelia, *Biochim. Biophys. Acta* 1222 (1994) 147–158.
- [132] A.B.J. Noach, Y. Kurosaki, M.C.M. Blom-Rosemalen, A.G. de Boer, D.D. Breimer, Cell-polarity dependent effect of chelation on the paracellular permeability of confluent caco-2 cell monolayers, *Int. J. Pharm.* 90 (1993) 229–237.
- [133] P. Artursson, C. Magnusson, Epithelial transport of drugs in cell culture. II: Effect of extracellular calcium concentration on the paracellular transport of drugs of different lipophilicities across monolayers of intestinal epithelial (Caco-2) cells, *J. Pharm. Sci.* 79 (1990) 595–600.
- [134] A.S. Yap, W.M. Briehner, B.M. Gumbiner, Molecular and functional analysis of cadherin-based adherens junctions, *Annu. Rev. Cell Dev. Biol.* 13 (1997) 119–146.
- [135] T.Y. Ma, D. Tran, N. Hoa, D. Nguyen, M. Merryfield, A. Tarnawski, Mechanism of extracellular calcium regulation of intestinal epithelial tight junction permeability: role of cytoskeletal involvement, *Microsc. Res. Tech.* 51 (2000) 156–168.
- [136] M. Tomita, M. Hayashi, S. Awazu, Absorption-enhancing mechanism of EDTA, caprate, and decanoylcarnitine in Caco-2 cells, *J. Pharm. Sci.* 85 (1996) 608–611.
- [137] L. Gonzalez-Mariscal, R. Tapia, D. Chamorro, Crosstalk of tight junction components with signaling pathways, *Biochim. Biophys. Acta* 1778 (2008) 729–756.
- [138] J.R. Turner, J.M. Angle, E.D. Black, J.L. Joyal, D.B. Sacks, J.L. Madara, PKC-dependent regulation of transepithelial resistance: roles of MLC and MLC kinase, *Am. J. Phys.* 277 (1999) C554–C562.
- [139] Oramed Ltd. US20110014247 (2011).
- [140] R. Eldor, M. Kidron, E. Arbit, Open-label study to assess the safety and pharmacodynamics of five oral insulin formulations in healthy subjects, *Diabetes Obes. Metab.* 12 (2010) 219–223.
- [141] University of Kansas, WO 2002034880, 2002.
- [142] V. Wong, B.M. Gumbiner, A synthetic peptide corresponding to the extracellular domain of occludin perturbs the tight junction permeability barrier, *J. Cell Biol.* 136 (1997) 399–409.
- [143] J.R. Turner, 'Putting the squeeze' on the tight junction: understanding cytoskeletal regulation, *Semin. Cell Dev. Biol.* 11 (2000) 301–308.
- [144] S.A. Blair, S.V. Kane, D.R. Clayburgh, J.R. Turner, Epithelial myosin light chain kinase expression and activity are upregulated in inflammatory bowel disease, *Lab. Invest.* 86 (2006) 191–201.
- [145] T.Y. Ma, N.T. Hoa, D.D. Tran, V. Bui, A. Pedram, S. Mills, M. Merryfield, Cytochalasin B modulation of Caco-2 tight junction barrier: role of myosin light chain kinase, *Am. J. Physiol. Gastrointest. Liver Physiol.* 279 (2000) G875–G885.
- [146] L.M. Feighery, S.W. Cochrane, R. Quinn, A.W. Baird, D. O'Toole, S.E. Owens, D. O'Donoghue, R.J. Msrny, D.J. Brayden, Myosin light chain kinase inhibition: correction of increased intestinal epithelial permeability in vitro, *Pharm. Res.* 25 (2008) 1377–1386.
- [147] A. Tavernier, R. Dondi, K. Almansour, F. Laurent, S.E. Owens, I.M. Eggleston, N. Fotaki, R.J. Msrny, Enhanced paracellular transport of insulin can be achieved via transient induction of myosin light chain phosphorylation, *J. Control. Release* 210 (2015) 189–197.
- [148] D.M. Small, A classification of biologic lipids based upon their interaction in aqueous systems, *J. Am. Oil Chem. Soc.* 45 (1968) 108–119.
- [149] D.J. Brayden, S. Maher, B. Bahar, E. Walsh, Sodium caprate-induced increases in intestinal permeability and epithelial damage are prevented by misoprostol, *Eur. J. Pharm. Biopharm.* 94 (2015) 194–206.
- [150] D.J. Brayden, J. Gleeson, E.G. Walsh, A head-to-head multi-parametric high content analysis of a series of medium chain fatty acid intestinal permeation enhancers in Caco-2 cells, *Eur. J. Pharm. Biopharm.* 88 (2014) 830–839.
- [151] A. Helenius, K. Simons, Solubilization of membranes by detergents, *Biochim. Biophys. Acta* 415 (1975) 29–79.
- [152] D. Lichtenberg, R.J. Robson, E.A. Dennis, Solubilization of phospholipids by detergents. Structural and kinetic aspects, *Biochim. Biophys. Acta* 737 (1983) 285–304.
- [153] M.N. Jones, Surfactants in membrane solubilisation, *Int. J. Pharm.* 177 (1999) 137–159.
- [154] S.Y. Lee, R. Welbourn, S.M. Clarke, M.W. Skoda, L. Clifton, A. Zarbakhsh, Adsorption of sodium hexanoate on alpha-alumina, *J. Colloid Interface Sci.* 407 (2013) 348–353.
- [155] N.L. Prisle, T. Raatikainen, R. Sorjamaa, B. Svenningsson, A.R.I. Laaksonen, M. Bilde, Surfactant partitioning in cloud droplet activation: a study of C8, C10, C12 and C14 normal fatty acid sodium salts, *Tellus B* 60 (2008) 416–431.
- [156] Human health risk assessment, Environmental Risk Assessment on ingredients of European household cleaning products, *Fat. Acid Salts* (2012) 1–61.
- [157] U. Subudhi, A.K. Mishra, Micellization of bile salts in aqueous medium: a fluorescence study, *Colloids Surf. B Biointerfaces* 57 (2007) 102–107.
- [158] P. Sharma, M.V. Varma, H.P. Chawla, R. Panchagnula, Absorption enhancement, mechanistic and toxicity studies of medium chain fatty acids, cyclodextrins and bile salts as peroral absorption enhancers, *Farmaco* 60 (2005) 884–893.
- [159] K. Gradauer, A. Nishiumi, K. Unrinin, H. Higashino, M. Kataoka, B.L. Pedersen, S.T. Buckley, S. Yamashita, Interaction with mixed micelles in the intestine attenuates the permeation enhancing potential of alkyl-maltosides, *Mol. Pharm.* 12 (2015) 2245–2253.
- [160] C.J. Porter, N.L. Trevaskis, W.N. Charman, Lipids and lipid-based formulations: optimizing the oral delivery of lipophilic drugs, *Nat. Rev. Drug Discov.* 6 (2007) 231–248.
- [161] S. Mori, A. Matsuura, Y.V. Rama Prasad, K. Takada, Studies on the intestinal absorption of low molecular weight heparin using saturated fatty acids and their derivatives as an absorption enhancer in rats, *Biol. Pharm. Bull.* 27 (2004) 418–421.
- [162] E.K. Anderberg, T. Lindmark, P. Artursson, Sodium caprate elicits dilatations in human intestinal tight junctions and enhances drug absorption by the paracellular route, *Pharm. Res.* 10 (1993) 857–864.
- [163] M.J. Govers, D.S. Termont, J.A. Lapre, J.H. Kleibeuker, R.J. Vonk, R. Van der Meer, Calcium in milk products precipitates intestinal fatty acids and secondary bile acids and thus inhibits colonic cytotoxicity in humans, *Cancer Res.* 56 (1996) 3270–3275.
- [164] V. Gupta, B.H. Hwang, N. Doshi, S. Mitragotri, A permeation enhancer for increasing transport of therapeutic macromolecules across the intestine, *J. Control. Release* 172 (2013) 541–549.
- [165] D. Attwood, A.T. Florence, Biological Implications Of Surfactant Presence In Formulations, Surfactant Systems, Springer, Netherlands, 1983 388–468.
- [166] K. Nishimura, Y. Nozaki, A. Yoshimi, S. Nakamura, N. Kitagawa, N. Kakeya, K. Kitao, Studies on the promoting effects of carboxylic acid derivatives on the rectal absorption of beta-lactam antibiotics in rats, *Chem. Pharm. Bull.* 33 (1985) 282–291.
- [167] D.A. Brown, Lipid rafts, detergent-resistant membranes, and raft targeting signals, *Physiology (Bethesda)* 21 (2006) 430–439.
- [168] F.A. Kummerow, Modification of cell membrane composition by dietary lipids and its implications for atherosclerosis, *Ann. N. Y. Acad. Sci.* 414 (1983) 29–43.
- [169] D. Lichtenberg, S. Yedgar, G. Cooper, S. Gatt, Studies on the molecular packing of mixed dispersions of Triton X-100 and sphingomyelin and its dependence on temperature and cloud point, *Biochemistry* 18 (1979) 2574–2582.
- [170] Y. Barenholz, T.E. Thompson, Sphingomyelins in bilayers and biological membranes, *Biochim. Biophys. Acta* 604 (1980) 129–158.
- [171] T.J. McIntosh, S.A. Simon, D. Needham, C.H. Huang, Structure and cohesive properties of sphingomyelin/cholesterol bilayers, *Biochemistry* 31 (1992) 2012–2020.
- [172] Y. Barenholz, S. Gatt, Sphingomyelin: metabolism, chemical synthesis, chemical and physical properties, *Phospholipids*, Elsevier 1982, pp. 129–178.
- [173] M.L. Rogerson, B.H. Robinson, S. Bucak, P. Walde, Kinetic studies of the interaction of fatty acids with phosphatidylcholine vesicles (liposomes), *Colloids Surf. B Biointerfaces* 48 (2006) 24–34.
- [174] M.L. Jackson, C.F. Schmidt, D. Lichtenberg, B.J. Litman, A.D. Albert, Solubilization of phosphatidylcholine bilayers by octyl glucoside, *Biochemistry* 21 (1982) 4576–4582.
- [175] D. Lichtenberg, Y. Zilberman, P. Greenzaid, S. Zamir, Structural and kinetic studies on the solubilization of lecithin by sodium deoxycholate, *Biochemistry* 18 (1979) 3517–3525.
- [176] M. Tomita, T. Sawada, T. Ogawa, H. Ouchi, M. Hayashi, S. Awazu, Differences in the enhancing effects of sodium caprate on colonic and jejunal drug absorption, *Pharm. Res.* 9 (1992) 648–653.
- [177] M. Tomita, M. Hayashi, T. Horie, T. Ishizawa, S. Awazu, Enhancement of colonic drug absorption by the transcellular permeation route, *Pharm. Res.* 5 (1988) 786–789.
- [178] E.A. Dennis, Formation and characterization of mixed micelles of the nonionic surfactant Triton X-100 with egg, dipalmitoyl, and dimyristoyl phosphatidylcholines, *Arch. Biochem. Biophys.* 165 (1974) 764–773.
- [179] T. Sawada, T. Ogawa, M. Tomita, M. Hayashi, S. Awazu, Role of paracellular pathway in nonelectrolyte permeation across rat colon epithelium enhanced by sodium caprate and sodium caprylate, *Pharm. Res.* 8 (1991) 1365–1371.
- [180] A.A. Raoof, P. Chiu, Z. Ramtoola, I.K. Cumming, C. Teng, S.P. Weinbach, G.E. Hardee, A.L. Levin, R.S. Geary, Oral bioavailability and multiple dose tolerability of an antisense oligonucleotide tablet formulated with sodium caprate, *J. Pharm. Sci.* 93 (2004) 1431–1439.
- [181] E.G. Walsh, B.E. Adamczyk, K.B. Chalasani, S. Maher, E.B. O'Toole, J.S. Fox, T.W. Leonard, D.J. Brayden, Oral delivery of macromolecules: rationale underpinning Gastrointestinal Permeation Enhancement Technology (GIPET), *Ther. Deliv.* 2 (2011) 1595–1610.
- [182] J.K. Amory, T.W. Leonard, S.T. Page, E. O'Toole, M.J. McKenna, W.J. Bremner, Oral administration of the GnRH antagonist acylate, in a GIPET-enhanced tablet form, acutely suppresses serum testosterone in normal men: single-dose pharmacokinetics and pharmacodynamics, *Cancer Chemother. Pharmacol.* 64 (2009) 641–645.
- [183] K.I. Cumming, Elan Corp Plc, Z. Ramtoola, WO2000050012, 2000.
- [184] T. Uchiyama, T. Sugiyama, Y.S. Quan, A. Kotani, N. Okada, T. Fujita, S. Muranishi, A. Yamamoto, Enhanced permeability of insulin across the rat intestinal membrane by various absorption enhancers: their intestinal mucosal toxicity and absorption-enhancing mechanism of n-lauryl-beta-D-maltopyranoside, *J. Pharm. Pharmacol.* 51 (1999) 1241–1250.
- [185] N. Muranushi, E. Mack, S.W. Kim, The effects of fatty acids and their derivatives on the intestinal absorption of insulin in rat, *Drug Dev. Ind. Pharm.* 19 (1993) 929–941.
- [186] A.A. Raoof, Z. Ramtoola, B. McKenna, R.Z. Yu, G. Hardee, R.S. Geary, Effect of sodium caprate on the intestinal absorption of two modified antisense oligonucleotides in pigs, *Eur. J. Pharm. Sci.* 17 (2002) 131–138.

- [187] L.G. Tillman, R.S. Geary, G.E. Hardee, Oral delivery of antisense oligonucleotides in man, *J. Pharm. Sci.* 97 (2008) 225–236.
- [188] D.L. Burcham, B.A. Aungst, M. Hussain, M.A. Gorko, C.Y. Quon, S.M. Huang, The effect of absorption enhancers on the oral absorption of the GP IIb/IIIa receptor antagonist, DMP 728, in rats and dogs, *Pharm. Res.* 12 (1995) 2065–2070.
- [189] S.H. Yalkowsky, G. Zografi, Some micellar properties of long-chain acylcarnitines, *J. Colloid Interface Sci.* 34 (1970) 525–533.
- [190] J.A. Fix, K. Engle, P.A. Porter, P.S. Leppert, S.J. Selk, C.R. Gardner, J. Alexander, Acylcarnitines: drug absorption-enhancing agents in the gastrointestinal tract, *Am. J. Phys.* 251 (1986) G332–G340.
- [191] Enteris BioPharma, Introduction to Enteris BioPharm, <http://enterisbiopharma.com/beta/wp-content/uploads/2013/09/Introduction-to-Enteris.pdf>2013.
- [192] N. Binkley, M. Bolognese, A. Sidorowicz-Bialynicka, T. Vally, R. Trout, C. Miller, C.E. Buben, J.P. Gilligan, D.S. Krause, A phase 3 trial of the efficacy and safety of oral recombinant calcitonin: the Oral Calcitonin in Postmenopausal Osteoporosis (ORACAL) trial, *J. Bone Miner. Res. Off. J. Am. Soc. Bone Miner. Res.* 27 (2012) 1821–1829.
- [193] A. Sturmer, N. Mehta, J. Giacchi, T. Catagay, R. Tavakkol, S. Mitta, L. Fitzpatrick, J. Wald, J. Trang, W. Stern, Pharmacokinetics of oral recombinant human parathyroid hormone [rhPTH(1–31)NH₂)] in postmenopausal women with osteoporosis, *Clin. Pharmacokinet.* 52 (2013) 995–1004.
- [194] Unigene Laboratories Ltd. US6086918 (2000).
- [195] F.J. Varum, G.B. Hutton, A.W. Basit, Food, physiology and drug delivery, *Int. J. Pharm.* 457 (2013) 446–460.
- [196] Unigene Laboratories Ltd. US5912014 (1999).
- [197] Y.H. Lee, P.J. Sinko, Oral delivery of salmon calcitonin, *Adv. Drug Deliv. Rev.* 42 (2000) 225–238.
- [198] Y.H. Lee, B.A. Perry, S. Labruno, H.S. Lee, W. Stern, L.M. Falzone, P.J. Sinko, Impact of regional intestinal pH modulation on absorption of peptide drugs: oral absorption studies of salmon calcitonin in beagle dogs, *Pharm. Res.* 16 (1999) 1233–1239.
- [199] Y. Hee Lee, G.D. Leesman, V. Makhey, H. Yu, P. Hu, B. Perry, J.P. Sutyak, E.J. Wagner, L.M. Falzone, W. Stern, P.J. Sinko, Regional oral absorption, hepatic first-pass effect, and non-linear disposition of salmon calcitonin in beagle dogs, *Eur. J. Pharm. Biopharm.* 50 (2000) 205–211.
- [200] N. Binkley, H. Bone, J.P. Gilligan, D.S. Krause, Efficacy and safety of oral recombinant calcitonin tablets in postmenopausal women with low bone mass and increased fracture risk: a randomized, placebo-controlled trial, *Osteoporos. Int.* 25 (2014) 2649–2656.
- [201] L. Wu, G. Zhang, Q. Lu, Q. Sun, M. Wang, N. Li, Z. Gao, Y. Sun, T. Li, D. Han, X. Yu, L. Wang, W. Sun, D. Zhao, Y. Wu, Y. Lu, X. Chen, Evaluation of salmon calcitonin (sCT) enteric-coated capsule for enhanced absorption and GI tolerability in rats, *Drug Dev. Ind. Pharm.* 36 (2010) 362–370.
- [202] M. Hayashi, T. Sakai, Y. Hasegawa, T. Nishikawahara, H. Tomioka, A. Iida, N. Shimizu, M. Tomita, S. Awazu, Physiological mechanism for enhancement of paracellular drug transport, *J. Control. Release* 62 (1999) 141–148.
- [203] S. Guarini, W. Ferrari, Structural restriction in bile acids and non-ionic detergents for promotion of heparin absorption from rat gastro-intestinal tract, *Arch. Int. Pharmacodyn. Ther.* 271 (1984) 4–10.
- [204] A.T. Florence, Surfactant interactions with biomembranes and drug absorption, *Pure Appl. Chem.* (1981) 2057.
- [205] K.A. Walters, P.H. Dugard, A.T. Florence, Non-ionic surfactants and gastric mucosal transport of paraquat, *J. Pharm. Pharmacol.* 33 (1981) 207–213.
- [206] S. Muranishi, Absorption enhancers, *Crit. Rev. Ther. Drug Carrier Syst.* 7 (1990) 1–33.
- [207] V. Jannin, J. Musakhanian, D. Marchaud, Approaches for the development of solid and semi-solid lipid-based formulations, *Adv. Drug Deliv. Rev.* 60 (2008) 734–746.
- [208] B.J. Aungst, N.J. Rogers, Site dependence of absorption-promoting actions of laurth-9, Na salicylate, Na₂EDTA, and aprotinin on rectal, nasal, and buccal insulin delivery, *Pharm. Res.* 5 (1988) 305–308.
- [209] T. Ishizawa, M. Hayashi, S. Awazu, Enhancement of jejunal and colonic absorption of fosfomycin by promoters in the rat, *J. Pharm. Pharmacol.* 39 (1987) 892–895.
- [210] M. Shichiri, Y. Yamasaki, R. Kawamori, M. Kikuchi, J. Abe, Increased intestinal absorption of insulin: an insulin suppository, *J. Pharm. Pharmacol.* 30 (1978) 806–808.
- [211] E.A. Hosny, H.I. Al-Shora, M.M. Elmazar, Relative hypoglycemic effect of insulin suppositories in diabetic beagle dogs: optimization of various concentrations of sodium salicylate and polyoxyethylene-9-lauryl ether, *Biol. Pharm. Bull.* 24 (2001) 1294–1297.
- [212] K. Morimoto, H. Akatsuchi, K. Morisaka, A. Kamada, Effect of non-ionic surfactants in a polyacrylic acid gel base on the rectal absorption of [Asu1,7]-eel calcitonin in rats, *J. Pharm. Pharmacol.* 37 (1985) 759–760.
- [213] B. Ersson, L. Rydén, J.-C. Janson, Introduction to Protein Purification, Protein Purification, John Wiley & Sons, Inc., 2011 1–22.
- [214] E.S. Swenson, W.B. Milisen, W. Curatolo, Intestinal permeability enhancement: structure-activity and structure-toxicity relationships for nonylphenoxypolyoxyethylene surfactant permeability enhancers, *Pharm. Res.* 11 (1994) 1501–1504.
- [215] R.G. Strickley, Solubilizing excipients in oral and injectable formulations, *Pharm. Res.* 21 (2004) 201–230.
- [216] E.K. Anderberg, C. Nystrom, P. Artursson, Epithelial transport of drugs in cell culture, VII: effects of pharmaceutical surfactant excipients and bile acids on transepithelial permeability in monolayers of human intestinal epithelial (Caco-2) cells, *J. Pharm. Sci.* 81 (1992) 879–887.
- [217] D. Dimitrijevic, A.J. Shaw, A.T. Florence, Effects of some non-ionic surfactants on transepithelial permeability in Caco-2 cells, *J. Pharm. Pharmacol.* 52 (2000) 157–162.
- [218] K. Nakanishi, M. Masada, T. Nadai, Effect of pharmaceutical adjuvants on the rectal permeability of drugs. II. Effect of Tween-type surfactants on the permeability of drugs in the rat rectum, *Chem. Pharm. Bull.* 31 (1983) 3255–3263.
- [219] A. Kosior, Investigation of physical and hypoglycaemic properties of rectal suppositories with chosen insulin, *Acta Pol. Pharm.* 59 (2002) 353–358.
- [220] A. Szuts, P. Szabo-Revesz, Sucrose esters as natural surfactants in drug delivery systems—a mini-review, *Int. J. Pharm.* 433 (2012) 1–9.
- [221] A. Yamamoto, H. Katsumi, K. Kusamori, T. Sakane, Improvement of intestinal absorption of poorly absorbable drugs by various sugar esters, *Yakugaku Zasshi* 134 (2014) 47–53.
- [222] H. Onishi, Y. Imura, M. Uchida, Y. Machida, Enhancement potential of sucrose laurate (L-1695) on intestinal absorption of water-soluble high molecular weight compounds, *Curr. Drug Deliv.* 9 (2012) 487–494.
- [223] K. Chansanroj, G. Betz, Sucrose esters with various hydrophilic-lipophilic properties: novel controlled release agents for oral drug delivery matrix tablets prepared by direct compaction, *Acta Biomater.* 6 (2010) 3101–3109.
- [224] M. Masahiro, K. Yoko, T. Kanji, M. Shozo, Assessment of enhancing ability of medium-chain alkyl saccharides as new absorption enhancers in rat rectum, *Int. J. Pharm.* 79 (1992) 159–169.
- [225] B.J. Aungst, Site-dependence and structure-effect relationships for alkylglycosides as transmucosal absorption promoters for insulin, *Int. J. Pharm.* 105 (1994) 219–225.
- [226] E.T. Maggio, Intravital: highly effective intranasal delivery of peptide and protein drugs, *Expert Opin. Drug Deliv.* 3 (2006) 529–539.
- [227] S.B. Petersen, G. Nolan, S. Maher, U.L. Rahbek, M. Guldbrandt, D.J. Brayden, Evaluation of alkylmaltoosides as intestinal permeation enhancers: comparison between rat intestinal mucosal sheets and Caco-2 monolayers, *Eur. J. Pharm. Sci.* 47 (2012) 701–712.
- [228] S.B. Petersen, L.G. Nielsen, U.L. Rahbek, M. Guldbrandt, D.J. Brayden, Colonic absorption of salmon calcitonin using tetradecyl maltoside (TDM) as a permeation enhancer, *Eur. J. Pharm. Sci.* 48 (2013) 726–734.
- [229] M. Katsuma, S. Watanabe, H. Kawai, S. Takemura, K. Sako, Effects of absorption promoters on insulin absorption through colon-targeted delivery, *Int. J. Pharm.* 307 (2006) 156–162.
- [230] L. Hovgaard, E.J. Mack, S.W. Kim, Insulin stabilization and GI absorption, *J. Control. Release* 19 (1992) 99–108.
- [231] I. Sasaki, H. Tozaki, K. Matsumoto, Y. Ito, T. Fujita, M. Murakami, S. Muranishi, A. Yamamoto, Development of an oral formulation of azetirelin, a new thyrotropin-releasing hormone (TRH) analogue, using n-lauryl-beta-D-maltopyranoside as an absorption enhancer, *Biol. Pharm. Bull.* 22 (1999) 611–615.
- [232] E.T. Maggio, P. Grasso, Oral delivery of octreotide acetate in Intravital(R) improves uptake, half-life, and bioavailability over subcutaneous administration in male Swiss webster mice, *Regul. Pept.* 167 (2011) 233–238.
- [233] Aegis Therapeutics Llc. US20130034597 (2013).
- [234] D.W. Lee, M.C. Leinung, P. Grasso, Oral delivery of mouse [D-Leu-4]-OB3, a synthetic peptide amide with leptin-like activity, in male Swiss Webster mice: a study comparing the pharmacokinetics of oral delivery to intraperitoneal, subcutaneous, intramuscular, and intranasal administration, *Regul. Pept.* 160 (2010) 129–132.
- [235] P. Pavlidis, N. Powell, R.P. Vincent, D. Ehrlich, I. Bjarnason, B. Hayee, Systematic review: bile acids and intestinal inflammation-luminal aggressors or regulators of mucosal defence? *Aliment. Pharmacol. Ther.* 42 (2015) 802–817.
- [236] M.J. Monte, J.J. Marin, A. Antelo, J. Vazquez-Tato, Bile acids: chemistry, physiology, and pathophysiology, *World J. Gastroenterol.* 15 (2009) 804–816.
- [237] J.P.F. Bai, Effects of bile salts on brush-border and cytosolic proteolytic activities of intestinal enterocytes, *Int. J. Pharm.* 111 (1994) 147–152.
- [238] E. Ziv, A. Eldor, Y. Kleinman, H. Bar-On, M. Kidron, Bile salts facilitate the absorption of heparin from the intestine, *Biochem. Pharmacol.* 32 (1983) 773–776.
- [239] J. Hastewella, S. Lynch, R. Fox, I. Williamson, P. Skelton-Stroud, M. Mackay, Enhancement of human calcitonin absorption across the rat colon in vivo, *Int. J. Pharm.* 101 (1994) 115–120.
- [240] E. Ziv, M. Kidron, E.M. Berry, H. Bar-On, Bile salts promote the absorption of insulin from the rat colon, *Life Sci.* 29 (1981) 803–809.
- [241] T. Fujita, T. Fujita, K. Morikawa, H. Tanaka, O. Iemura, A. Yamamoto, S. Muranishi, Improvement of intestinal absorption of human calcitonin by chemical modification with fatty acids: synergistic effects of acylation and absorption enhancers, *Int. J. Pharm.* 134 (1996) 47–57.
- [242] W. Kamm, A. Jonczyk, T. Jung, G. Luckenbach, P. Raddatz, T. Kissel, Evaluation of absorption enhancement for a potent cyclopeptide alpha(nu)beta(3)-antagonist in a human intestinal cell line (Caco-2), *Eur. J. Pharm. Sci.* 10 (2000) 205–214.
- [243] I.W. Kim, H.J. Yoo, I.S. Song, Y.B. Chung, D.C. Moon, S.J. Chung, C.K. Shim, Effect of excipients on the stability and transport of recombinant human epidermal growth factor (rhEGF) across Caco-2 cell monolayers, *Arch. Pharm. Res.* 26 (2003) 330–337.
- [244] V. Bocci, A. Naldini, F. Corradeschi, E. Lencioni, Colorectal administration of human interferon- α , *Int. J. Pharm.* 24 (1985) 109–114.
- [245] A. Yamamoto, T. Okagawa, A. Kotani, T. Uchiyama, T. Shimura, S. Tabata, S. Kondo, S. Muranishi, Effects of different absorption enhancers on the permeation of ebitrate, an ACTH analogue, across intestinal membranes, *J. Pharm. Pharmacol.* 49 (1997) 1057–1061.
- [246] A. Mizuno, M. Ueda, G. Kawanishi, Effects of salicylate and other enhancers on rectal absorption of erythropoietin in rats, *J. Pharm. Pharmacol.* 44 (1992) 570–573.
- [247] Genexer Pharmaceutical Inc. US7255102 (2007).
- [248] Biocon Ltd. US7196059 (2007).
- [249] Aegis Therapeutics Llc. US7998927B2 (2011).
- [250] William L. Lu F. US20120128767 (2012).
- [251] I. Raz, H. Bar-On, M. Kidron, E. Ziv, Rectal administration of insulin, *Isr. J. Med. Sci.* 20 (1984) 173–175.
- [252] A. Khedkar, H. Iyer, A. Anand, M. Verma, S. Krishnamurthy, S. Savale, A. Atignat, A dose range finding study of novel oral insulin (IN-105) under fed conditions in type 2 diabetes mellitus subjects, *Diabetes Obes. Metab.* 12 (2010) 659–664.

- [253] Biocon Ltd. US2010/0285143 (2010).
- [254] S. Feldman, M. Reinhard, Interaction of sodium alkyl sulfates with everted rat small intestinal membrane, *J. Pharm. Sci.* 65 (1976) 1460–1462.
- [255] E.K. Anderberg, P. Artursson, Epithelial transport of drugs in cell culture. VIII: Effects of sodium dodecyl sulfate on cell membrane and tight junction permeability in human intestinal epithelial (Caco-2) cells, *J. Pharm. Sci.* 82 (1993) 392–398.
- [256] T. Hargreaves, Surfactants in Action, in: T. Hargreaves (Ed.) In: Chemical Formulation: An Overview of Surfactant Based Chemical Preparations Used in Everyday Life, The Royal Society of Chemistry 2003, pp. P007–P012.
- [257] M. Shima, Y. Kimura, S. Adachi, R. Matsuno, The relationship between transport-enhancement effects and cell viability by capric acid sodium salt, monocaprin, and dicaprin, *Biosci. Biotechnol. Biochem.* 62 (1998) 83–86.
- [258] K. Higaki, I. Kishimoto, H. Komatsu, M. Hashida, H. Sezaki, Effect of medium-chain glycerides (MGK) on the intestinal absorption and the hepatobiliary transport of bromthymol blue, *J. Pharmacobiodyn.* 9 (1986) 532–539.
- [259] C.W. Pouton, Lipid formulations for oral administration of drugs: non-emulsifying, self-emulsifying and 'self-microemulsifying' drug delivery systems, *Eur. J. Pharm. Sci.* 11 (Suppl. 2) (2000) S93–S98.
- [260] D.B. Warren, D.K. Chalmers, C.W. Pouton, Structure and dynamics of glyceride lipid formulations, with propylene glycol and water, *Mol. Pharm.* 6 (2009) 604–614.
- [261] P.P. Constantinides, S.H. Yiv, Particle size determination of phase-inverted water-in-oil microemulsions under different dilution and storage conditions, *Int. J. Pharm.* 115 (1995) 225–234.
- [262] C.L. Li, Y.J. Deng, Oil-based formulations for oral delivery of insulin, *J. Pharm. Pharmacol.* 56 (2004) 1101–1107.
- [263] T. Wang, N. Wang, A. Hao, X. He, T. Li, Y. Deng, Lyophilization of water-in-oil emulsions to prepare phospholipid-based anhydrous reverse micelles for oral peptide delivery, *Eur. J. Pharm. Sci.* 39 (2010) 373–379.
- [264] Dor Biopharm Inc., EP1460992B1, 2009.
- [265] T. Nishihata, J.H. Rytting, Absorption-promoting adjuvants: enhancing action on rectal absorption, *Adv. Drug Deliv. Rev.* 28 (1997) 205–228.
- [266] M. Sekine, K. Sasahara, T. Kojima, K. Hasegawa, R. Okada, S. Awazu, Improvement of bioavailability of poorly absorbed drugs. I. Effect of medium chain glyceride base on the rectal absorption of cefmetazole sodium in rats, *J. Pharmacobiodyn.* 7 (1984) 856–863.
- [267] Y. Watanabe, E.J. van Hoogdalem, A.G. de Boer, D.D. Breimer, Absorption enhancement of rectally infused cefoxitin by medium chain monoglycerides in conscious rats, *J. Pharm. Sci.* 77 (1988) 847–849.
- [268] C.W. Pouton, Formulation of poorly water-soluble drugs for oral administration: physicochemical and physiological issues and the lipid formulation classification system, *Eur. J. Pharm. Sci.* 29 (2006) 278–287.
- [269] H.N. Prajapati, D.M. Dalrymple, A.T. Serajuddin, A comparative evaluation of mono-, di- and triglyceride of medium chain fatty acids by lipid/surfactant/water phase diagram, solubility determination and dispersion testing for application in pharmaceutical dosage form development, *Pharm. Res.* 29 (2012) 285–305.
- [270] M. Sekine, H. Terashima, K. Sasahara, K. Nishimura, R. Okada, S. Awazu, Improvement of bioavailability of poorly absorbed drugs. II. Effect of medium chain glyceride base on the intestinal absorption of cefmetazole sodium in rats and dogs, *J. Pharmacobiodyn.* 8 (1985) 286–295.
- [271] E.J. van Hoogdalem, C.D. Heijligers-Feijen, A.G. de Boer, J.C. Verhoef, D.D. Breimer, Rectal absorption enhancement of des-enkephalin-gamma-endorphin (DE gamma E) by medium-chain glycerides and EDTA in conscious rats, *Pharm. Res.* 6 (1989) 91–95.
- [272] M. Sekine, K. Sasahara, R. Okada, S. Awazu, Improvement of bioavailability of poorly absorbed drugs. IV. Mechanism of the promoting effect of medium chain glyceride on the rectal absorption of water soluble drugs, *J. Pharmacobiodyn.* 8 (1985) 645–652.
- [273] Y. Matsumoto, Y. Watanabe, N. Hori, M. Matsumoto, Duration of absorption-enhancing effect of sodium octanoate, sodium hexanoate or glyceryl-1-monooctanoate on rectal absorption of gentamicin in rabbits, *J. Pharmacobiodyn.* 13 (1990) 591–596.
- [274] H. Yoshitomi, T. Nishihata, G. Frederick, M. Dillsaver, T. Higuchi, Effect of triglyceride on small intestinal absorption of cefoxitin in rats, *J. Pharm. Pharmacol.* 39 (1987) 887–891.
- [275] S. RG, Currently Marketed Oral Lipid Based Dosage Forms: Drug Products And Excipients, Oral Lipid-Based Formulations: Enhancing The Bioavailability Of Poorly Water-Soluble Drugs, 2007.
- [276] S. Tenjarla, Microemulsions: an overview and pharmaceutical applications, *Crit. Rev. Ther. Drug Carrier Syst.* 16 (1999) 461–521.
- [277] G.S. Robert, Currently Marketed Oral Lipid-Based Dosage Forms, Oral Lipid-Based Formulations, CRC Press, 2007 1–31.
- [278] S.P. Moulik, B.K. Paul, Structure, dynamics and transport properties of microemulsions, *Adv. Colloid Interf. Sci.* 78 (1998) 99–195.
- [279] Y.V. Rama Prasad, T. Minamimoto, Y. Yoshikawa, N. Shibata, S. Mori, A. Matsura, K. Takada, In situ intestinal absorption studies on low molecular weight heparin in rats using labrasol as absorption enhancer, *Int. J. Pharm.* 271 (2004) 225–232.
- [280] B. Godin, E. Toutou, Enhancers For Enteral Delivery Of Macromolecules With Emphasis On Surfactants, in: E. Toutou, B.W. Barry (Eds.), Enhancement in Drug Delivery, Taylor and Francis 2011, pp. 37–56.
- [281] M.M. Nerurkar, P.S. Burton, R.T. Borchardt, The use of surfactants to enhance the permeability of peptides through Caco-2 cells by inhibition of an apically polarized efflux system, *Pharm. Res.* 13 (1996) 528–534.
- [282] B.J. Aungst, H. Saitoh, D.L. Burcham, S.-M. Huang, S.A. Mousa, M.A. Hussain, Enhancement of the intestinal absorption of peptides and nonpeptides, *J. Control. Release* 41 (1996) 19–31.
- [283] X. Sha, G. Yan, Y. Wu, J. Li, X. Fang, Effect of self-microemulsifying drug delivery systems containing Labrasol on tight junctions in Caco-2 cells, *Eur. J. Pharm. Sci.* 24 (2005) 477–486.
- [284] N. Venkatesan, K. Uchino, K. Amagase, Y. Ito, N. Shibata, K. Takada, Gastro-intestinal patch system for the delivery of erythropoietin, *J. Control. Release* 111 (2006) 19–26.
- [285] K.A. Hamid, H. Katsumi, T. Sakane, A. Yamamoto, The effects of common solubilizing agents on the intestinal membrane barrier functions and membrane toxicity in rats, *Int. J. Pharm.* 379 (2009) 100–108.
- [286] Novo Nordisk A/S US20120196800 (2012).
- [287] S. Eaimtrakarn, Y.V. Rama Prasad, T. Ohno, T. Konishi, Y. Yoshikawa, N. Shibata, K. Takada, Absorption enhancing effect of labrasol on the intestinal absorption of insulin in rats, *J. Drug Target.* 10 (2002) 255–260.
- [288] N. Venkatesan, J. Yoshimitsu, Y. Ito, N. Shibata, K. Takada, Liquid filled nanoparticles as a drug delivery tool for protein therapeutics, *Biomaterials* 26 (2005) 7154–7163.
- [289] Y. Ito, T. Kusawake, Y.V. Prasad, N. Sugioka, N. Shibata, K. Takada, Preparation and evaluation of oral solid heparin using emulsifier and adsorbent for in vitro and in vivo studies, *Int. J. Pharm.* 317 (2006) 114–119.
- [290] L. Guo, E. Ma, H. Zhao, Y. Long, C. Zheng, M. Duan, Preliminary evaluation of a novel oral delivery system for rhPTH1-34: in vitro and in vivo, *Int. J. Pharm.* 420 (2011) 172–179.
- [291] K. Koga, N. Takarada, K. Takada, Nano-sized water-in-oil-in-water emulsion enhances intestinal absorption of calcein, a high solubility and low permeability compound, *Eur. J. Pharm. Biopharm.* 74 (2010) 223–232.
- [292] Novo Nordisk A/S US2011/0144010 (2011).
- [293] Novo Nordisk A/S US2013/0058999 (2013).
- [294] Novo Nordisk A/S US20110293714 (2011).
- [295] T.A. Aguirre, M. Rosa, I.S. Coulter, D.J. Brayden, In vitro and in vivo preclinical evaluation of a minisphere emulsion-based formulation (SmPill(R)) of salmon calcitonin, *Eur. J. Pharm. Sci.* 79 (2015) 102–111.
- [296] P.P. Constantinides, J.-P. Scalart, Formulation and physical characterization of water-in-oil microemulsions containing long- versus medium-chain glycerides, *Int. J. Pharm.* 158 (1997) 57–68.
- [297] A. Cilek, N. Celebi, F. Tirnaksiz, Lecithin-based microemulsion of a peptide for oral administration: preparation, characterization, and physical stability of the formulation, *Drug Deliv.* 13 (2006) 19–24.
- [298] A. Cilek, N. Celebi, F. Tirnaksiz, A. Tay, A lecithin-based microemulsion of rh-insulin with aprotinin for oral administration: Investigation of hypoglycemic effects in non-diabetic and STZ-induced diabetic rats, *Int. J. Pharm.* 298 (2005) 176–185.
- [299] J. Luo, Macrolin Cortecs Ltd, IDrugs 1 (1998) 712–714.
- [300] M.E. Kraeling, W.A. Ritschel, Development of a colonic release capsule dosage form and the absorption of insulin, *Methods Find. Exp. Clin. Pharmacol.* 14 (1992) 199–209.
- [301] G. Sharma, K. Wilson, C.F. van der Walle, N. Sattar, J.R. Petrie, M.N. Ravi Kumar, Microemulsions for oral delivery of insulin: design, development and evaluation in streptozotocin induced diabetic rats, *Eur. J. Pharm. Biopharm.* 76 (2010) 159–169.
- [302] R.W. Greiner, D.F. Evans, Spontaneous formation of a water-continuous emulsion from a w/o microemulsion, *Langmuir* 6 (1990) 1793–1796.
- [303] E.L. McConnell, A.W. Basit, S. Murdan, Measurements of rat and mouse gastrointestinal pH, fluid and lymphoid tissue, and implications for in-vivo experiments, *J. Pharm. Pharmacol.* 60 (2008) 63–70.
- [304] D. Liu, T. Kobayashi, S. Russo, F. Li, S.E. Plevy, T.M. Gambling, J.L. Carson, R.J. Mumper, In vitro and in vivo evaluation of a water-in-oil microemulsion system for enhanced peptide intestinal delivery, *AAPS J.* 15 (2013) 288–298.
- [305] P.P. Constantinides, C.M. Lancaster, J. Marcello, D.C. Chiosso, D. Orner, I. Hidalgo, P.L. Smith, A.B. Sarkisian, S.H. Yiv, A.J. Owen, Enhanced intestinal absorption of an RGD peptide from water-in-oil microemulsions of different composition and particle size, *J. Control. Release* 34 (1995) 109–116.
- [306] P.P. Constantinides, G. Welzel, H. Ellens, P.L. Smith, S. Sturgis, S.H. Yiv, A.B. Owen, Water-in-oil microemulsions containing medium-chain fatty acids/salts: formulation and intestinal absorption enhancement evaluation, *Pharm. Res.* 13 (1996) 210–215.
- [307] M.L. Shively, Multiple emulsions for the delivery of proteins, *Pharm. Biotechnol.* 10 (1997) 199–211.
- [308] R.H. Engel, S.J. Riggi, M.J. Fahrenbach, Insulin: intestinal absorption as water-in-oil-in-water emulsions, *Nature* 219 (1968) 856–857.
- [309] M. Shichiri, Y. Shimizu, Y. Yoshida, R. Kawamori, M. Fukuchi, Y. Shigeta, H. Abe, Enteral absorption of water-in-oil-in-water insulin emulsions in rabbits, *Diabetologia* 10 (1974) 317–321.
- [310] S.T. Dogru, S. Calis, F. Oner, Oral multiple w/o/w emulsion formulation of a peptide salmon calcitonin: in vitro-in vivo evaluation, *J. Clin. Pharm. Ther.* 25 (2000) 435–443.
- [311] J.M. Barichello, M. Morishita, K. Takayama, T. Nagai, Absorption of insulin from pluronic F-127 gels following subcutaneous administration in rats, *Int. J. Pharm.* 184 (1999) 189–198.
- [312] M. Shichiri, R. Kawamori, Y. Goriya, M. Kikuchi, Y. Yamasaki, Y. Shigeta, H. Abe, Increased intestinal absorption of insulin in a micellar solution: water-in-oil-in-water insulin micelles, *Acta Diabetol. Lat.* 15 (1978) 175–183.
- [313] A. Silva-Cunha, J.L. Grossiord, F. Puisieux, M. Seiller, W/O/W multiple emulsions of insulin containing a protease inhibitor and an absorption enhancer: preparation, characterization and determination of stability towards proteases in vitro, *Int. J. Pharm.* 158 (1997) 79–89.
- [314] A. Suzuki, M. Morishita, M. Kajita, K. Takayama, K. Isowa, Y. Chiba, S. Tokiwa, T. Nagai, Enhanced colonic and rectal absorption of insulin using a multiple emulsion containing eicosapentaenoic acid and docosahexaenoic acid, *J. Pharm. Sci.* 87 (1998) 1196–1202.

- [315] N. Oba, H. Sugimura, Y. Umehara, M. Yoshida, T. Kimura, T. Yamaguchi, Evaluation of an oleic acid water-in-oil-in-water-type multiple emulsion as potential drug carrier via the enteral route, *Lipids* 27 (1992) 701–705.
- [316] M. Morishita, M. Kajita, A. Suzuki, K. Takayama, Y. Chiba, S. Tokiwa, T. Nagai, The dose-related hypoglycemic effects of insulin emulsions incorporating highly purified EPA and DHA, *Int. J. Pharm.* 201 (2000) 175–185.
- [317] M. Shima, M. Tanaka, Y. Kimura, S. Adachi, R. Matsuno, Hydrolysis of the oil phase of a W/O/W emulsion by pancreatic lipase, *J. Control. Release* 94 (2004) 53–61.
- [318] M. Shima, M. Tanaka, T. Fujii, K. Egawa, Y. Kimura, S. Adachi, R. Matsuno, Oral administration of insulin included in fine W/O/W emulsions to rats, *Food Hydrocoll.* 20 (2006) 523–531.
- [319] F. Courmarie, V. Rosilio, M. Chéron, C. Vauthier, B. Lacour, J.-L. Grossiord, M. Seiller, Improved formulation of W/O/W multiple emulsion for insulin encapsulation. Influence of the chemical structure of insulin, *Colloid Polym. Sci.* 282 (2004) 562–568.
- [320] A. Matsuzawa, M. Morishita, K. Takayama, T. Nagai, Absorption of insulin using water-in-oil-in-water emulsion from an enteral loop in rats, *Biol. Pharm. Bull.* 18 (1995) 1718–1723.
- [321] S. Marquette, C. Peerboom, A. Yates, L. Denis, J. Goole, K. Amighi, Encapsulation of immunoglobulin G by solid-in-oil-in-water: effect of process parameters on microsphere properties, *Eur. J. Pharm. Biopharm.* 86 (2014) 393–403.
- [322] E. Toorisaka, H. Ono, K. Arimori, N. Kamiya, M. Goto, Hypoglycemic effect of surfactant-coated insulin solubilized in a novel solid-in-oil-in-water (S/O/W) emulsion, *Int. J. Pharm.* 252 (2003) 271–274.
- [323] E. Toorisaka, M. Hashida, N. Kamiya, H. Ono, Y. Kokazu, M. Goto, An enteric-coated dry emulsion formulation for oral insulin delivery, *J. Control. Release* 107 (2005) 91–96.
- [324] A. Elsayed, M.A. Remawi, N. Qinna, A. Farouk, A. Badwan, Formulation and characterization of an oily-based system for oral delivery of insulin, *Eur. J. Pharm. Biopharm.* 73 (2009) 269–279.
- [325] S.A. Ben-Sasson, Chiasma Ltd., E. Cohen, WO2005094785, 2005.
- [326] Salama P, Mamluk R, Marom K, Weinstein I, Tzabari M. US20120009229 (2012).
- [327] Salama P, Karmeli I, Tuvia S, Marom K, Harush-Frenkel O, Gelbaum D, Mamluk R, Price FD, Peikin SR, US20110311621 (2011).
- [328] Chiasma Inc., WO2011116139, 2011.
- [329] Salama P, Mamluk R, Marom K, Weinstein I, Tzabari M, US20100105627 (2010).
- [330] Chiasma Inc., US8241670 (2012).
- [331] S. Tuvia, D. Pelled, K. Marom, P. Salama, M. Levin-Arama, I. Karmeli, G.H. Idelson, I. Landau, R. Mamluk, A novel suspension formulation enhances intestinal absorption of macromolecules via transient and reversible transport mechanisms, *Pharm. Res.* 31 (2014) 2010–2021.
- [332] S. Melmed, V. Popovic, M. Bidlingmaier, M. Mercado, A.J. van der Lely, N. Biermasz, M. Bolanowski, M. Coculescu, J. Schopohl, K. Racz, B. Glaser, M. Goth, Y. Greenman, P. Trainer, E. Mezosi, I. Shimon, A. Giustina, M. Korbonits, M.D. Bronstein, D. Kleinberg, S. Teichman, I. Gliko-Kabir, R. Mamluk, A. Haviv, C. Strasburger, Safety and efficacy of oral octreotide in acromegaly: results of a multicenter phase III trial, *J. Clin. Endocrinol. Metab.* 100 (2015) 1699–1708.
- [333] E.J. van Hoogdalem, M.A. Hardens, A.G. de Boer, D.D. Breimer, Absorption enhancement of rectally infused cefoxitin sodium by medium-chain fatty acids in conscious rats: concentration–effect relationship, *Pharm. Res.* 5 (1988) 453–456.
- [334] E.J. Van Hoogdalem, A.T. Wackwitz, A.G. De Boer, A.F. Cohen, D.D. Breimer, Rate-controlled rectal absorption enhancement of cefoxitin by co-administration of sodium salicylate or sodium octanoate in healthy volunteers, *Br. J. Clin. Pharmacol.* 27 (1989) 75–81.
- [335] J.W. Park, O.C. Jeon, S.K. Kim, T.A. Al-Hilal, H.T. Moon, C.Y. Kim, Y. Byun, Anticoagulant efficacy of solid oral formulations containing a new heparin derivative, *Mol. Pharm.* 7 (2010) 836–843.
- [336] J.H. Jonkman, C.A. Hunt, Ion pair absorption of ionized drugs—fact or fiction? *Pharm. Weekbl. Sci.* 5 (1983) 41–48.
- [337] S.K. Kim, S. Lee, S. Jin, H.T. Moon, O.C. Jeon, D.Y. Lee, Y. Byun, Diabetes correction in pancreatized canines by orally absorbable insulin-deoxycholate complex, *Mol. Pharm.* 7 (2010) 708–717.
- [338] L.E. Bromberg, A.M. Klibanov, Detergent-enabled transport of proteins and nucleic acids through hydrophobic solvents, *Proc. Natl. Acad. Sci. U. S. A.* 91 (1994) 143–147.
- [339] P.O. Hegg, Precipitation of egg white proteins below their isoelectric points by sodium dodecyl sulphate and temperature, *Biochim. Biophys. Acta* 579 (1979) 73–87.
- [340] J.D. Meyer, M.C. Manning, Hydrophobic ion pairing: altering the solubility properties of biomolecules, *Pharm. Res.* 15 (1998) 188–193.
- [341] C.A. Nelson, The binding of detergents to proteins. I. The maximum amount of dodecyl sulfate bound to proteins and the resistance to binding of several proteins, *J. Biol. Chem.* 246 (1971) 3895–3901.
- [342] P. Mukerjee, K.J. Mysels, Critical Micelle Concentrations of Aqueous Surfactant Systems (National Bureau of Standards 36, Washington, D.C.), 1971.
- [343] J. Matsuura, M.E. Powers, M.C. Manning, E. Shefter, Structure and stability of insulin dissolved in 1-octanol, *J. Am. Chem. Soc.* 115 (1993) 1261–1264.
- [344] P. Li, H.M. Nielsen, M. Fano, A. Mullertz, Preparation and characterization of insulin-surfactant complexes for loading into lipid-based drug delivery systems, *J. Pharm. Sci.* 102 (2013) 2689–2698.
- [345] Mediplex Corp., US7906137 (2011).
- [346] S. Lee, J. Lee, D.Y. Lee, S.K. Kim, Y. Lee, Y. Byun, A new drug carrier, Nalphe-deoxycholy-l-lysyl-methylester, for enhancing insulin absorption in the intestine, *Diabetologia* 48 (2005) 405–411.
- [347] S. Lee, S.K. Kim, D.Y. Lee, S.Y. Chae, Y. Byun, Pharmacokinetics of a new, orally available ceftriaxone formulation in physical complexation with a cationic analogue of bile acid in rats, *Antimicrob. Agents Chemother.* 50 (2006) 1869–1871.
- [348] D.Y. Lee, J. Lee, S. Lee, S.K. Kim, Y. Byun, Liphophilic complexation of heparin based on bile acid for oral delivery, *J. Control. Release* 123 (2007) 39–45.
- [349] J.W. Park, Y. Byun, Ionic complex of risenedronate with positively charged deoxycholic acid derivative: evaluation of physicochemical properties and enhancement of intestinal absorption in rats, *Arch. Pharm. Res.* 37 (2014) 1560–1569.
- [350] S. Sun, N. Liang, Y. Kawashima, D. Xia, F. Cui, Hydrophobic ion pairing of an insulin-sodium deoxycholate complex for oral delivery of insulin, *Int. J. Nanomedicine* 6 (2011) 3049–3056.
- [351] F. Cui, K. Shi, L. Zhang, A. Tao, Y. Kawashima, Biodegradable nanoparticles loaded with insulin-phospholipid complex for oral delivery: preparation, in vitro characterization and in vivo evaluation, *J. Control. Release* 114 (2006) 242–250.
- [352] R.D. Vaishya, A. Mandal, M. Gokulgandhi, S. Patel, A.K. Mitra, Reversible hydrophobic ion-pairing complex strategy to minimize acylation of octreotide during long-term delivery from PLGA microparticles, *Int. J. Pharm.* 489 (2015) 237–245.
- [353] Emisphere Technologies Inc., US5401516 (1995).
- [354] A. Leone-Bay, N. Santiago, D. Achan, K. Chaudhary, F. DeMorin, L. Falzarano, S. Haas, S. Kalbag, D. Kaplan, H. Leipold, et al., N-acylated alpha-amino acids as novel oral delivery agents for proteins, *J. Med. Chem.* 38 (1995) 4263–4269.
- [355] A. Leone-Bay, C. McInnes, N. Wang, F. DeMorin, D. Achan, C. Lercara, D. Sarubbi, S. Haas, J. Press, E. Barantsevich, et al., Microsphere formation in a series of derivatized alpha-amino acids: properties, molecular modeling, and oral delivery of salmon calcitonin, *J. Med. Chem.* 38 (1995) 4257–4262.
- [356] A. Leone-Bay, D.R. Paton, J. Freeman, C. Lercara, D. O'Toole, D. Gschneidner, E. Wang, E. Harris, C. Rosado, T. Rivera, A. DeVincent, M. Tai, F. Mercogliano, R. Agarwal, H. Leipold, R.A. Baughman, Synthesis and evaluation of compounds that facilitate the gastrointestinal absorption of heparin, *J. Med. Chem.* 41 (1998) 1163–1171.
- [357] A. Leone-Bay, H. Leipold, D.R. Paton, S.J. Milstein, R.A. Baughman, Oral delivery of rhGH: preliminary mechanistic considerations, *Drug News Perspect.* 9 (10) (1996) 586–591.
- [358] A. Leone-Bay, K.K. Ho, R. Agarwal, R.A. Baughman, K. Chaudhary, F. DeMorin, L. Genoble, C. McInnes, C. Lercara, S. Milstein, D. O'Toole, D. Sarubbi, B. Variano, D.R. Paton, 4-[4-[(2-Hydroxybenzoyl)amino]phenyl]butyric acid as a novel oral delivery agent for recombinant human growth hormone, *J. Med. Chem.* 39 (1996) 2571–2578.
- [359] A. Leone-Bay, H. Leipold, D. Sarubbi, B. Variano, T. Rivera, R.A. Baughman, Oral delivery of sodium cromolyn: preliminary studies in vivo and in vitro, *Pharm. Res.* 13 (1996) 222–226.
- [360] D. Brayden, E. Creed, A. O'Connell, H. Leipold, R. Agarwal, A. Leone-Bay, Heparin absorption across the intestine: effects of sodium N-[8-(2-hydroxybenzoyl)amino]caprylate in rat in situ intestinal instillations and in Caco-2 monolayers, *Pharm. Res.* 14 (1997) 1772–1779.
- [361] T.M. Rivera, A. Leone-Bay, D.R. Paton, H.R. Leipold, R.A. Baughman, Oral delivery of heparin in combination with sodium N-[8-(2-hydroxybenzoyl)amino]caprylate: pharmacological considerations, *Pharm. Res.* 14 (1997) 1830–1834.
- [362] A. Leone-Bay, D.R. Paton, B. Variano, H. Leipold, T. Rivera, J. Miura-Fraboni, R.A. Baughman, N. Santiago, Acylated non-alpha-amino acids as novel agents for the oral delivery of heparin sodium, *USP, J. Control. Release* 50 (1998) 41–49.
- [363] D. Malkov, R. Angelo, H.Z. Wang, E. Flanders, H. Tang, I. Gomez-Orellana, Oral delivery of insulin with the eligen technology: mechanistic studies, *Curr. Drug Deliv.* 2 (2005) 191–197.
- [364] S. Hess, V. Rothschild, A. Hoffman, Investigation of the enhancing mechanism of sodium N-[8-(2-hydroxybenzoyl)amino]caprylate effect on the intestinal permeability of polar molecules utilizing a voltage clamp method, *Eur. J. Pharm. Sci.* 25 (2005) 307–312.
- [365] A.W. Alani, J.R. Robinson, Mechanistic understanding of oral drug absorption enhancement of cromolyn sodium by an amino acid derivative, *Pharm. Res.* 25 (2008) 48–54.
- [366] K.I. Cumming, Emisphere Tech Inc, M.L. Martin, WO 2000048589 A1, 2000.
- [367] Emisphere launches eligen B12(TM), the first oral prescription tablet proven to normalize B12 levels without the need for an injection.(2015, Mar 19). PR NewswireRetrieved from <http://search.proquest.com/docview/1664418682?accountid=11979>.
- [368] M.G. Riley, M.C. Castelli, E.A. Paehler, Subchronic oral toxicity of salcaprozate sodium (SNAC) in Sprague–Dawley and Wistar rats, *Int. J. Toxicol.* 28 (2009) 278–293.
- [369] R.A. Baughman, S.C. Kapoor, R.K. Agarwal, J. Kisicki, F. Catella-Lawson, G.A. FitzGerald, Oral delivery of anticoagulant doses of heparin. A randomized, double-blind, controlled study in humans, *Circulation* 98 (1998) 1610–1615.
- [370] S.A. Mousa, F. Zhang, A. Aljada, S. Chaturvedi, M. Takiuddin, H. Zhang, L. Chi, M.C. Castelli, K. Friedman, M.M. Goldberg, R.J. Linhardt, Pharmacokinetics and pharmacodynamics of oral heparin solid dosage form in healthy human subjects, *J. Clin. Pharmacol.* 47 (2007) 1508–1520.
- [371] Emisphere Technologies Inc, <http://ir.emisphere.com/common/download/sec.cfm?companyid=EMIS&fid=950130-02-2306&cik=8053262002>.
- [372] R.E. Steinert, B. Poller, M.C. Castelli, J. Drewe, C. Beglinger, Oral administration of glucagon-like peptide 1 or peptide YY 3–36 affects food intake in healthy male subjects, *Am. J. Clin. Nutr.* 92 (2010) 810–817.
- [373] Emisphere Technologies, http://files.shareholder.com/downloads/EMIS/905418616x0x260954/FBE7A502-9D62-4565-84CC-EA892B544BE7/EMIS_News_2006_11_8_General_Releases.pdf2006.
- [374] Oramed Ltd., US20110142800 (2011).
- [375] M. Kidron, S. Dinh, Y. Menachem, R. Abbas, B. Variano, M. Goldberg, E. Arbit, H. Bar-On, A novel per-oral insulin formulation: proof of concept study in non-diabetic subjects, *Diabet. Med.* 21 (2004) 354–357.
- [376] M.A. Nauck, J.R. Petrie, G. Sesti, E. Mannucci, J.P. Courreges, M.L. Lindegaard, C.B. Jensen, S.L. Atkin, A phase 2, randomized, dose-finding study of the novel once-weekly human GLP-1 analog, semaglutide, compared with placebo and open-label liraglutide in patients with type 2 diabetes, *Diabetes Care* (2015).

- [377] J. Lau, P. Bloch, L. Schaffer, I. Pettersson, J. Spetzler, J. Kofoed, K. Madsen, L.B. Knudsen, J. McGuire, D.B. Steensgaard, H.M. Strauss, D.X. Gram, S.M. Knudsen, F.S. Nielsen, P. Thygesen, S. Reedtz-Runge, T. Kruse, Discovery of the once-weekly glucagon-like peptide-1 (GLP-1) analogue semaglutide, *J. Med. Chem.* 58 (2015) 7370–7380.
- [378] A. Mullard, Oral GLP1 analogue rounds phase II corner, *Nat. Rev. Drug Discov.* 14 (2015) 227–227.
- [379] Novo Nordisk A/S US20130345134 A1 (2013).
- [380] L.B. Tanko, Y.Z. Bagger, P. Alexandersen, J.P. Devogelaer, J.Y. Reginster, R. Chick, M. Olson, H. Benmammam, L. Mindeholm, M. Azria, C. Christiansen, Safety and efficacy of a novel salmon calcitonin (sCT) technology-based oral formulation in healthy postmenopausal women: acute and 3-month effects on biomarkers of bone turnover, *J. Bone Miner. Res. Off. J. Am. Soc. Bone Miner. Res.* 19 (2004) 1531–1538.
- [381] S.P. Hammerle, L. Mindeholm, A. Launonen, B. Kiese, R. Loeffler, E. Harfst, M. Azria, M. Arnold, M.R. John, The single dose pharmacokinetic profile of a novel oral human parathyroid hormone formulation in healthy postmenopausal women, *Bone* 50 (2012) 965–973.
- [382] S.J. Wu, J.R. Robinson, Transcellular and lipophilic complex-enhanced intestinal absorption of human growth hormone, *Pharm. Res.* 16 (1999) 1266–1272.
- [383] M.A. Karsdal, I. Byrjalsen, B.J. Riis, C. Christiansen, Investigation of the diurnal variation in bone resorption for optimal drug delivery and efficacy in osteoporosis with oral calcitonin, *BMC Clin. Pharmacol.* 8 (2008) 12.
- [384] M.A. Karsdal, I. Byrjalsen, K. Henriksen, B.J. Riis, E.M. Lau, M. Arnold, C. Christiansen, The effect of oral salmon calcitonin delivered with 5-CNAC on bone and cartilage degradation in osteoarthritic patients: a 14-day randomized study, *Osteoarthr. Cartil.* 18 (2010) 150–159.
- [385] M.A. Karsdal, I. Byrjalsen, B.J. Riis, C. Christiansen, Optimizing bioavailability of oral administration of small peptides through pharmacokinetic and pharmacodynamic parameters: the effect of water and timing of meal intake on oral delivery of Salmon Calcitonin, *BMC Clin. Pharmacol.* 8 (2008) 5.
- [386] Y.Z. Bagger, L.B. Tanko, P. Alexandersen, M.A. Karsdal, M. Olson, L. Mindeholm, M. Azria, C. Christiansen, Oral salmon calcitonin induced suppression of urinary collagen type II degradation in postmenopausal women: a new potential treatment of osteoarthritis, *Bone* 37 (2005) 425–430.
- [387] D.H. Manicourt, M. Azria, L. Mindeholm, E.J. Thonar, J.P. Devogelaer, Oral salmon calcitonin reduces Lequesne's algofunctional index scores and decreases urinary and serum levels of biomarkers of joint metabolism in knee osteoarthritis, *Arthritis Rheum.* 54 (2006) 3205–3211.
- [388] C. Behets, J.M. Williams, D. Chappard, J.P. Devogelaer, D.H. Manicourt, Effects of calcitonin on subchondral trabecular bone changes and on osteoarthritic cartilage lesions after acute anterior cruciate ligament deficiency, *J. Bone Miner. Res. Off. J. Am. Soc. Bone Miner. Res.* 19 (2004) 1821–1826.
- [389] M.A. Karsdal, I. Byrjalsen, P. Alexandersen, A. Bihlet, J.R. Andersen, B.J. Riis, A.C. Bay-Jensen, C. Christiansen, Treatment of symptomatic knee osteoarthritis with oral salmon calcitonin: results from two phase 3 trials, *Osteoarthr. Cartil.* 23 (2015) 532–543.
- [390] M.A. Karsdal, K. Henriksen, A.C. Bay-Jensen, B. Molloy, M. Arnold, M.R. John, I. Byrjalsen, M. Azria, B.J. Riis, P. Qvist, C. Christiansen, Lessons learned from the development of oral calcitonin: the first tablet formulation of a protein in phase III clinical trials, *J. Clin. Pharmacol.* 51 (2011) 460–471.
- [391] M.A. Karsdal, I. Byrjalsen, M. Azria, M. Arnold, L. Choi, B.J. Riis, C. Christiansen, Influence of food intake on the bioavailability and efficacy of oral calcitonin, *Br. J. Clin. Pharmacol.* 67 (2009) 413–420.
- [392] R. New, G. Littlewood, P. Guard, I. Browning, P. Hotten, Intestinal delivery of calcitonin in pig, *Int. J. Pharm.* 156 (1997) 1–8.
- [393] H. Yoshiura, Y. Tahara, M. Hashida, N. Kamiya, A. Hirata, T. Fujii, M. Goto, Design and in vivo evaluation of solid-in-oil suspension for oral delivery of human growth hormone, *Biochem. Eng. J.* 41 (2008) 106–110.
- [394] Access Ltd, US7303762 B2 (2007).
- [395] T.P. Broke-Smith, S.D. Luzio, A. Lockett, et al., Repeat-dosing of oral insulin (Capsulin) in persons with type 2 diabetes, Proceedings of 44th European Association for the Study of Diabetes Annual Meeting, Rome, Italy, 2008.
- [396] R.R.C. New, Absorption Enhancers, Access Ltd., 2007
- [397] Access Ltd., US7651995 (2010).
- [398] Access Ltd., US20100056425 (2010).
- [399] W.A. Lee, H.F.L. Lu, P.W. Maffiud, M.T. Botet, P.A. Baldwin, T.A. Benkert, C.K. Klingbeil, The synthesis, characterization and biological testing of a novel class of mucosal permeation enhancers, *J. Control. Release* 22 (1992) 223–237.
- [400] T. Murakami, N. Yata, H. Tamauchi, J. Nakai, M. Yamazaki, A. Kamada, Studies on absorption promoters for rectal delivery preparations. I. Promoting efficacy of enamine derivatives of amino acids for the rectal absorption of beta-lactam antibiotics in rabbits, *Chem. Pharm. Bull.* 29 (1981) 1998–2004.
- [401] T. Nishihata, J.H. Rytting, A. Kamada, T. Higuchi, M. Routh, L. Caldwell, Enhancement of rectal absorption of insulin using salicylates in dogs, *J. Pharm. Pharmacol.* 35 (1983) 148–151.
- [402] Y. Takizawa, H. Kishimoto, T. Kitazato, H. Ishizaka, N. Kamiya, Y. Ito, M. Tomita, M. Hayashi, Characteristics of reversible absorption-enhancing effect of sodium nitroprusside in rat small intestine, *Eur. J. Pharm. Sci.* 49 (2013) 664–670.
- [403] A. Kamada, T. Nishihata, S. Kim, M. Yamamoto, N. Yata, Study of enamine derivatives of phenylglycine as adjuvants for the rectal absorption of insulin, *Chem. Pharm. Bull.* 29 (1981) 2012–2019.
- [404] N. Toshiaki, S. Masao, I. Hideki, K. Akira, Y. Toshihito, K. Ryuzo, S. Motoaki, An effective formulation for an insulin suppository; examination in normal dogs, *Int. J. Pharm.* 38 (1987) 83–90.
- [405] J.A. Moore, S.A. Pletcher, M.J. Ross, Absorption enhancement of growth hormone from the gastrointestinal tract of rats, *Int. J. Pharm.* 34 (1986) 35–43.
- [406] E. Hosny, O. el-Ahmady, H. el-Shattawy, A.e.-M. Nabih, H. el-Damacy, S. Gamal-el-Deen, N. el-Kabbany, Effect of sodium salicylate on insulin rectal absorption in humans, *Arzneimittelforschung* 44 (1994) 611–613.
- [407] N. Toshiaki, O. Yasufumi, I. Hideki, K. Masao, K. Akira, Y. Toshihito, K. Ryuzo, S. Motoaki, Trials of rectal insulin suppositories in healthy humans, *Int. J. Pharm.* 34 (1986) 157–161.
- [408] T. Yagi, N. Hakui, Y. Yamasaki, R. Kawamori, M. Shichiri, H. Abe, S. Kim, M. Miyake, K. Kamikawa, T. Nishihata, A. Kamada, Insulin suppository: enhanced rectal absorption of insulin using an enamine derivative as a new promoter, *J. Pharm. Pharmacol.* 35 (1983) 177–178.
- [409] M. Thanou, J.C. Verhoef, H.E. Junginger, Chitosan and its derivatives as intestinal absorption enhancers, *Adv. Drug Deliv. Rev.* 50 (Suppl. 1) (2001) S91–101.
- [410] S.M. van der Merwe, J.C. Verhoef, J.H. Verheijden, A.F. Kotze, H.E. Junginger, Trimethylated chitosan as polymeric absorption enhancer for improved peroral delivery of peptide drugs, *Eur. J. Pharm. Biopharm.* 58 (2004) 225–235.
- [411] A. Bernkop-Schnurch, Chitosan and its derivatives: potential excipients for peroral peptide delivery systems, *Int. J. Pharm.* 194 (2000) 1–13.
- [412] C. Prego, D. Torres, M.J. Alonso, The potential of chitosan for the oral administration of peptides, *Expert Opin. Drug Deliv.* 2 (2005) 843–854.
- [413] N.G. Schipper, K.M. Varum, P. Artursson, Chitosans as absorption enhancers for poorly absorbable drugs. 1: influence of molecular weight and degree of acetylation on drug transport across human intestinal epithelial (Caco-2) cells, *Pharm. Res.* 13 (1996) 1686–1692.
- [414] H.L. Luessen, B.J. de Leeuw, M.W. Langemeyer, A.B. de Boer, J.C. Verhoef, H.E. Junginger, Mucoadhesive polymers in peroral peptide drug delivery. VI. Carbomer and chitosan improve the intestinal absorption of the peptide drug busserelin in vivo, *Pharm. Res.* 13 (1996) 1668–1672.
- [415] A.F. Kotzé, B.J. de Leeuw, H.L. Lueßen, A.G. de Boer, J.C. Verhoef, H.E. Junginger, Chitosans for enhanced delivery of therapeutic peptides across intestinal epithelia: in vitro evaluation in Caco-2 cell monolayers, *Int. J. Pharm.* 159 (1997) 243–253.
- [416] J. Nunthanid, M. Laungtana-Anan, P. Sriamornsak, S. Limmatvapirat, S. Puttipipatkachorn, L.Y. Lim, E. Khor, Characterization of chitosan acetate as a binder for sustained release tablets, *J. Control. Release* 99 (2004) 15–26.
- [417] J.M. Viljoen, J.H. Steenekamp, A.F. Marais, A.F. Kotze, Effect of moisture content, temperature and exposure time on the physical stability of chitosan powder and tablets, *Drug Dev. Ind. Pharm.* 40 (2014) 730–742.
- [418] M. Kubbinga, M.A. Nguyen, P. Staubach, S. Teerenstra, P. Langguth, The influence of chitosan on the oral bioavailability of acyclovir—a comparative bioavailability study in humans, *Pharm. Res.* 32 (2015) 2241–2249.
- [419] P. Baldrick, The safety of chitosan as a pharmaceutical excipient, *Regul. Toxicol. Pharmacol.* 56 (2010) 290–299.
- [420] G.M. Buys, L.H. du Plessis, A.F. Marais, A.F. Kotze, J.H. Hamman, Direct compression of chitosan: process and formulation factors to improve powder flow and tablet performance, *Curr. Drug Deliv.* 10 (2013) 348–356.
- [421] K.M. Picker-Freyer, D. Brink, Evaluation of powder and tableting properties of chitosan, *AAPS PharmSciTech* 7 (3) (2006) E1–E10.
- [422] K.M. Picker-Freyer, D. Brink, Evaluation of powder and tableting properties of chitosan, *AAPS PharmSciTech* 7 (2006) 75.
- [423] M.M. Thanou, J.C. Verhoef, S.G. Romeijn, J.F. Nagelkerke, F.W. Merkus, H.E. Junginger, Effects of N-trimethyl chitosan chloride, a novel absorption enhancer, on caco-2 intestinal epithelia and the ciliary beat frequency of chicken embryo trachea, *Int. J. Pharm.* 185 (1999) 73–82.
- [424] M.R. Avadi, M.J. Zohuriaan-Mehr, P. Younessi, M. Amini, M.R. Tehrani, A. Shafiee, Optimized synthesis and characterization of N-triethyl chitosan, *J. Bioact. Compat. Polym.* 18 (2003) 469–479.
- [425] M. Thanou, M.T. Nihot, M. Jansen, J.C. Verhoef, H.E. Junginger, Mono-N-carboxymethyl chitosan (MCC), a polyampholytic chitosan derivative, enhances the intestinal absorption of low molecular weight heparin across intestinal epithelia in vitro and in vivo, *J. Pharm. Sci.* 90 (2001) 38–46.
- [426] A.F. Kotzé, H.L. Luessen, B.J. de Leeuw, A.G. de Boer, J.C. Verhoef, H.E. Junginger, Comparison of the effect of different chitosan salts and N-trimethyl chitosan chloride on the permeability of intestinal epithelial cells (Caco-2), *J. Control. Release* 51 (1998) 35–46.
- [427] S.M. van der Merwe, J.C. Verhoef, A.F. Kotze, H.E. Junginger, N-trimethyl chitosan chloride as absorption enhancer in oral peptide drug delivery. Development and characterization of minitab and granule formulations, *Eur. J. Pharm. Biopharm.* 57 (2004) 85–91.
- [428] M. Werle, A. Bernkop-Schnurch, Thiolated chitosans: useful excipients for oral drug delivery, *J. Pharm. Pharmacol.* 60 (2008) 273–281.
- [429] A. Bernkop-Schnurch, C.E. Kast, D. Guggi, Permeation enhancing polymers in oral delivery of hydrophilic macromolecules: thiomers/GSH systems, *J. Control. Release* 93 (2003) 95–103.
- [430] J. Iqbal, G. Shahnaz, G. Perera, F. Hintzen, F. Sarti, A. Bernkop-Schnurch, Thiolated chitosan: development and in vivo evaluation of an oral delivery system for leuprolide, *Eur. J. Pharm. Biopharm.* 80 (2012) 95–102.
- [431] A. Bernkop-Schnurch, Y. Pinter, D. Guggi, H. Kahlbacher, C. Schöffmann, M. Schuh, I. Schmerold, M.D. Del Curto, M. D'Antonio, P. Eposito, C. Huck, The use of thiolated polymers as carrier matrix in oral peptide delivery—proof of concept, *J. Control. Release* 106 (2005) 26–33.
- [432] T.F. Palmberger, J. Hombach, A. Bernkop-Schnurch, Thiolated chitosan: development and in vitro evaluation of an oral delivery system for acyclovir, *Int. J. Pharm.* 348 (2008) 54–60.
- [433] A.H. Kraeland, D. Guggi, A. Bernkop-Schnurch, Oral insulin delivery: the potential of thiolated chitosan-insulin tablets on non-diabetic rats, *J. Control. Release* 95 (2004) 547–555.

- [434] D. Gugli, A.H. Krauland, A. Bernkop-Schnurch, Systemic peptide delivery via the stomach: in vivo evaluation of an oral dosage form for salmon calcitonin, *J. Control. Release* 92 (2003) 125–135.
- [435] A. Rubinstein, B. Tirosh, Mucus gel thickness and turnover in the gastrointestinal tract of the rat: response to cholinergic stimulus and implication for mucoadhesion, *Pharm. Res.* 11 (1994) 794–799.
- [436] S. Khafagy el, M. Morishita, Oral biobug delivery using cell-penetrating peptide, *Adv. Drug Deliv. Rev.* 64 (2012) 531–539.
- [437] M. Pooga, M. Hallbrink, M. Zorko, U. Langel, Cell penetration by transport, *FASEB J.* 12 (1998) 67–77.
- [438] M. Lindgren, U. Langel, Classes and Prediction of Cell-Penetrating Peptides, in: U. Langel (Ed.), *Cell-Penetrating Peptides*, Humana Press 2011, pp. 3–19.
- [439] F. Heitz, M.C. Morris, G. Divita, Twenty years of cell-penetrating peptides: from molecular mechanisms to therapeutics, *Br. J. Pharmacol.* 157 (2009) 195–206.
- [440] Unigene Laboratories Ltd., US6673574B2 (2004).
- [441] Unigene Laboratories Ltd., US6673574 (2004).
- [442] M.E. Lindgren, M.M. Hallbrink, A.M. Elmquist, U. Langel, Passage of cell-penetrating peptides across a human epithelial cell layer in vitro, *Biochem. J.* 377 (2004) 69–76.
- [443] E.-S. Khafagy, M. Morishita, N. Kamei, Y. Eda, Y. Ikono, K. Takayama, Efficiency of cell-penetrating peptides on the nasal and intestinal absorption of therapeutic peptides and proteins, *Int. J. Pharm.* 381 (2009) 49–55.
- [444] N. Kamei, S. Kikuchi, M. Takeda-Morishita, Y. Terasawa, A. Yasuda, S. Yamamoto, N. Ida, R. Nishio, K. Takayama, Determination of the optimal cell-penetrating peptide sequence for intestinal insulin delivery based on molecular orbital analysis with self-organizing maps, *J. Pharm. Sci.* 102 (2013) 469–479.
- [445] S. Khafagy el, R. Iwamae, N. Kamei, M. Takeda-Morishita, Region-dependent role of cell-penetrating peptides in insulin absorption across the rat small intestinal membrane, *AAPS J.* 17 (2015) 1427–1437.
- [446] Toray Industries Inc., US20120065124 (2012).
- [447] S.M. Krug, M. Fromm, D. Gunzel, Two-path impedance spectroscopy for measuring paracellular and transcellular epithelial resistance, *Biophys. J.* 97 (2009) 2202–2211.
- [448] S.M. Krug, M. Amasheh, I. Dittmann, I. Christoffel, M. Fromm, S. Amasheh, Sodium caprate as an enhancer of macromolecule permeation across tricellular tight junctions of intestinal cells, *Biomaterials* 34 (2013) 275–282.
- [449] K. Whitehead, S. Mitragotri, Mechanistic analysis of chemical permeation enhancers for oral drug delivery, *Pharm. Res.* 25 (2008) 1412–1419.
- [450] K. Sugibayashi, Y. Onuki, K. Takayama, Displacement of tight junction proteins from detergent-resistant membrane domains by treatment with sodium caprate, *Eur. J. Pharm. Sci.* 36 (2009) 246–253.
- [451] J. Ye, T. Tsukamoto, A. Sun, S.K. Nigam, A role for intracellular calcium in tight junction reassembly after ATP depletion-repletion, *Am. J. Phys.* 277 (1999) F524–F532.
- [452] A.T. Blikslager, A.J. Moeser, J.L. Gookin, S.L. Jones, J. Odle, Restoration of barrier function in injured intestinal mucosa, *Physiol. Rev.* 87 (2007) 545–564.
- [453] T. Lindmark, T. Nikkila, P. Artursson, Mechanisms of absorption enhancement by medium chain fatty acids in intestinal epithelial Caco-2 cell monolayers, *J. Pharmacol. Exp. Ther.* 275 (1995) 958–964.
- [454] T. Shimazaki, M. Tomita, S. Sadahiro, M. Hayashi, S. Awazu, Absorption-enhancing effects of sodium caprate and palmitoyl carnitine in rat and human colons, *Dig. Dis. Sci.* 43 (1998) 641–645.
- [455] H. Takahashi, T. Shibasaki, K. Takeshita, F. Kaiho, M. Hayashi, The enhancing mechanism of capric acid (C10) from a suppository on rectal drug absorption through a paracellular pathway, *Biol. Pharm. Bull.* 20 (1997) 446–448.
- [456] J. Weyermann, D. Lochmann, A. Zimmer, A practical note on the use of cytotoxicity assays, *Int. J. Pharm.* 288 (2005) 369–376.
- [457] S.H. Welling, L.K. Clemmensen, S.T. Buckley, L. Hovgaard, P.B. Brockhoff, H.H. Refsgaard, In silico modelling of permeation enhancement potency in Caco-2 monolayers based on molecular descriptors and random forest, *Eur. J. Pharm. Biopharm.* 94 (2015) 152–159.
- [458] P.D. Ward, T.K. Tippin, D.R. Thakker, Enhancing paracellular permeability by modulating epithelial tight junctions, *Pharm. Sci. Technol. Today* 3 (2000) 346–358.
- [459] L. Kiss, F.R. Walter, A. Bocsik, S. Veszelka, B. Ozsvári, L.G. Puskas, P. Szabo-Revesz, M.A. Deli, Kinetic analysis of the toxicity of pharmaceutical excipients Cremophor EL and RH40 on endothelial and epithelial cells, *J. Pharm. Sci.* 102 (2013) 1173–1181.
- [460] R.E. Osterberg, N.A. See, Toxicity of excipients—a Food and Drug Administration perspective, *Int. J. Toxicol.* 22 (2003) 377–380.
- [461] L.M. Lichtenberger, Where is the evidence that cyclooxygenase inhibition is the primary cause of nonsteroidal anti-inflammatory drug (NSAID)-induced gastrointestinal injury? Topical injury revisited, *Biochem. Pharmacol.* 61 (2001) 631–637.
- [462] M. Riegler, R. Sedivy, T. Sogukoglu, E. Cosentini, G. Bischof, B. Teleky, W. Feil, R. Schiessel, G. Hamilton, E. Wenzl, Effect of growth factors on epithelial restitution of human colonic mucosa in vitro, *Scand. J. Gastroenterol.* 32 (1997) 925–932.
- [463] P.R. Kvietys, R.D. Specian, M.B. Grisham, P. Tso, Jejunal mucosal injury and restitution: role of hydrolytic products of food digestion, *Am. J. Phys.* 261 (1991) G384–G391.
- [464] R.A. Argenzio, C.K. Henrikson, J.A. Lacos, Restitution of barrier and transport function of porcine colon after acute mucosal injury, *Am. J. Phys.* 255 (1988) G62–G71.
- [465] R. Moore, S. Carlson, J.L. Madara, Rapid barrier restitution in an in vitro model of intestinal epithelial injury, *Lab. Invest.* 60 (1989) 237–244.
- [466] R. Daig, T. Andus, E. Aschenbrenner, W. Falk, J. Scholmerich, V. Gross, Increased interleukin 8 expression in the colon mucosa of patients with inflammatory bowel disease, *Gut* 38 (1996) 216–222.
- [467] A.J. Wilson, K. Byron, P.R. Gibson, Interleukin-8 stimulates the migration of human colonic epithelial cells in vitro, *Clin. Sci. (Lond.)* 97 (1999) 385–390.
- [468] A. Sturm, D.C. Baumgart, J.H. d'Heureuse, A. Hotz, B. Wiedenmann, A.U. Dignass, CXCL8 modulates human intestinal epithelial cells through a CXCR1 dependent pathway, *Cytokine* 29 (2005) 42–48.
- [469] R. Brew, J.S. Erikson, D.C. West, A.R. Kinsella, J. Slavin, S.E. Christmas, Interleukin-8 as an autocrine growth factor for human colon carcinoma cells in vitro, *Cytokine* 12 (2000) 78–85.
- [470] K. Nakanishi, M. Masada, T. Nadai, Effect of pharmaceutical adjuvants on the rectal permeability of drugs. III. Effect of repeated administration and recovery of the permeability, *Chem. Pharm. Bull.* 31 (1983) 4161–4166.
- [471] J.L. Gookin, J.M. Rhoads, R.A. Argenzio, Inducible nitric oxide synthase mediates early epithelial repair of porcine ileum, *Am. J. Physiol. Gastrointest. Liver Physiol.* 283 (2002) G157–G168.
- [472] R. Moore, S. Carlson, J.L. Madara, Villus contraction aids repair of intestinal epithelium after injury, *Am. J. Phys.* 257 (1989) G274–G283.
- [473] A.T. Blikslager, M.C. Roberts, J.M. Rhoads, R.A. Argenzio, Prostaglandins I2 and E2 have a synergistic role in rescuing epithelial barrier function in porcine ileum, *J. Clin. Invest.* 100 (1997) 1928–1933.
- [474] M. Iizuka, S. Konno, Wound healing of intestinal epithelial cells, *World J. Gastroenterol.* 17 (2011) 2161–2171.
- [475] A.B. Cox, L.A. Rawlinson, A.W. Baird, V. Bzik, D.J. Brayden, In vitro interactions between the oral absorption promoter, sodium caprate (C(10)) and *S. typhimurium* in rat intestinal ileal mucosa, *Pharm. Res.* 25 (2008) 114–122.
- [476] K. Sonaje, K.J. Lin, M.T. Tseng, S.P. Wey, F.Y. Su, E.Y. Chuang, C.W. Hsu, C.T. Chen, H.W. Sung, Effects of chitosan-nanoparticle-mediated tight junction opening on the oral absorption of endotoxins, *Biomaterials* 32 (2011) 8712–8721.
- [477] E.J. Nielsen, N. Kamei, M. Takeda-Morishita, Safety of the cell-penetrating peptide penetratin as an oral absorption enhancer, *Biol. Pharm. Bull.* 38 (2015) 144–146.
- [478] M.R. Mainous, E.A. Deitch, in: G. Schlag, H. Redl (Eds.), *Bacterial Translocation In: Pathophysiology of Shock, Sepsis, And Organ Failure*, Springer Science and Business Media 2012, pp. 265–278.
- [479] N.G. Ilback, M. Nyblom, J. Carlfors, B. Fagerlund-Aspenstrom, S. Tavelin, A.W. Glynn, Do surface-active lipids in food increase the intestinal permeability to toxic substances and allergenic agents? *Med. Hypotheses* 63 (2004) 724–730.
- [480] P.R. Kvietys, R.D. Specian, G. Cepinskas, Polyamines attenuate jejunal mucosal injury induced by oleic acid, *Am. J. Phys.* 263 (1992) G224–G229.
- [481] M. Coskun, Intestinal epithelium in inflammatory bowel disease, *Front. Med. (Lausanne)* 1 (2014) 24.
- [482] C. Barmeyer, J.D. Schulzke, M. Fromm, Claudin-related intestinal diseases, *Semin. Cell Dev. Biol.* 42 (2015) 30–38.
- [483] R.S. Lanigan, T.A. Yamarik, Final report on the safety assessment of EDTA, calcium disodium EDTA, diammonium EDTA, dipotassium EDTA, disodium EDTA, TEA-EDTA, tetrasodium EDTA, tripotassium EDTA, trisodium EDTA, HEDTA, and trisodium HEDTA, *Int. J. Toxicol.* 21 (Suppl. 2) (2002) 95–142.
- [484] M.J. Dorfel, O. Huber, A phosphorylation hotspot within the occludin C-terminal domain, *Ann. N. Y. Acad. Sci.* 1257 (2012) 38–44.
- [485] E.I. Tuomanen, S.M. Prasad, J.S. George, A.I. Hoepelman, P. Ibsen, I. Heron, R.M. Starzyk, Reversible opening of the blood-brain barrier by anti-bacterial antibodies, *Proc. Natl. Acad. Sci. U. S. A.* 90 (1993) 7824–7828.
- [486] X. Li, R. Saeki, A. Watari, K. Yagi, M. Kondoh, Tissue distribution and safety evaluation of a claudin-targeting molecule, the C-terminal fragment of Clostridium perfringens enterotoxin, *Eur. J. Pharm. Sci.* 52 (2014) 132–137.
- [487] F. McCartney, S. Waters, S. Bleiel, D.J. Brayden, S. Maher, Design of a histological score card to standardise the assessment of intestinal toxicity of candidate pharmaceutical excipients, 42nd Annual Meeting and Exposition of the CRS, Edinburgh, Scotland (Poster 295), 2015.
- [488] C.S. Schasteen, M.G. Donovan, J.N. Cogburn, A novel in vitro screen to discover agents which increase the absorption of molecules across the intestinal epithelium, *J. Control. Release* 21 (1992) 49–62.
- [489] A. Yamamoto, E. Hayakawa, Y. Kato, A. Nishiura, V.H. Lee, A mechanistic study on enhancement of rectal permeability to insulin in the albino rabbit, *J. Pharmacol. Exp. Ther.* 263 (1992) 25–31.
- [490] K. Sakai, T.M. Kutsuna, T. Nishino, Y. Fujiwara, N. Yata, Contribution of calcium ion sequestration by polyoxyethylated nonionic surfactants to the enhanced colonic absorption of p-aminobenzoic acid, *J. Pharm. Sci.* 75 (1986) 387–390.
- [491] K. Ichikawa, I. Ohata, M. Mitomi, S. Kawamura, H. Maeno, H. Kawata, Rectal absorption of insulin suppositories in rabbits, *J. Pharm. Pharmacol.* 32 (1980) 314–318.
- [492] Y. Lin, Q. Shen, H. Katsumi, N. Okada, T. Fujita, X. Jiang, A. Yamamoto, Effects of Labrasol and other pharmaceutical excipients on the intestinal transport and absorption of rhodamine123, a P-glycoprotein substrate, in rats, *Biol. Pharm. Bull.* 30 (2007) 1301–1307.
- [493] Y. Ito, H. Arai, K. Uchino, K. Iwasaki, N. Shibata, K. Takada, Effect of adsorbents on the absorption of lansoprazole with surfactant, *Int. J. Pharm.* 289 (2005) 69–77.
- [494] Z. Hu, R. Prasad Yv, R. Tawa, T. Konishi, M. Ishida, N. Shibata, K. Takada, Diethyl ether fraction of Labrasol having a stronger absorption enhancing effect on gentamicin than Labrasol itself, *Int. J. Pharm.* 234 (2002) 223–235.
- [495] Y.V. Prasad, S.P. Puthli, S. Eaimtrakarn, M. Ishida, Y. Yoshikawa, N. Shibata, K. Takada, Enhanced intestinal absorption of vancomycin with Labrasol and D-alpha-tocopheryl PEG 1000 succinate in rats, *Int. J. Pharm.* 250 (2003) 181–190.
- [496] Y.V. Rama Prasad, S. Eaimtrakarn, M. Ishida, Y. Kusawake, R. Tawa, Y. Yoshikawa, N. Shibata, K. Takada, Evaluation of oral formulations of gentamicin containing labrasol in beagle dogs, *Int. J. Pharm.* 268 (2003) 13–21.
- [497] D.D. Deshmukh, R. Nagilla, W.R. Ravis, G.V. Betageri, Effect of dodecylmaltoiside (DDM) on uptake of BCS III compounds, tiludronate and cromolyn, in Caco-2 cells and rat intestine model, *Drug Deliv.* 17 (2010) 145–151.
- [498] M. Marusic, T. Zupancic, G. Hribar, R. Komel, G. Anderluh, S. Caserman, The Caco-2 cell culture model enables sensitive detection of enhanced protein permeability in the presence of N-decyl-beta-D-maltopyranoside, *New Biotechnol.* 30 (2013) 507–515.

- [499] T. Sugiyama, A. Yamamoto, Y. Kawabe, T. Uchiyama, Y.S. Quan, S. Muranishi, Effects of various absorption enhancers on the intestinal absorption of water soluble drugs by in vitro Ussing chamber method: correlation with an in situ absorption experiment, *Biol. Pharm. Bull.* 20 (1997) 812–814.
- [500] A. Yamamoto, T. Uchiyama, R. Nishikawa, T. Fujita, S. Muranishi, Effectiveness and toxicity screening of various absorption enhancers in the rat small intestine: effects of absorption enhancers on the intestinal absorption of phenol red and the release of protein and phospholipids from the intestinal membrane, *J. Pharm. Pharmacol.* 48 (1996) 1285–1289.
- [501] T. Uchiyama, A. Yamamoto, H. Hatano, T. Fujita, S. Muranishi, Effectiveness and toxicity screening of various absorption enhancers in the large intestine: intestinal absorption of phenol red and protein and phospholipid release from the intestinal membrane, *Biol. Pharm. Bull.* 19 (1996) 1618–1621.
- [502] Y.S. Quan, K. Hattori, E. Lundborg, T. Fujita, M. Murakami, S. Muranishi, A. Yamamoto, Effectiveness and toxicity screening of various absorption enhancers using Caco-2 cell monolayers, *Biol. Pharm. Bull.* 21 (1998) 615–620.
- [503] T.K. Tipping, D.R. Thakker, Biorelevant refinement of the Caco-2 cell culture model to assess efficacy of paracellular permeability enhancers, *J. Pharm. Sci.* 97 (2008) 1977–1992.
- [504] T. Suzuka, A. Furuya, A. Kamada, T. Nishihata, Effect of phenothiazines, disodium ethylenediaminetetraacetic acid and diethyl maleate on in vitro rat colonic transport of cefmetazole and inulin, *J. Pharmacobiodyn.* 10 (1987) 63–71.
- [505] I. Dos Santos, F. Fawaz, A.M. Lagueny, F. Bonini, Improvement of norfloxacin oral bioavailability by EDTA and sodium caprate, *Int. J. Pharm.* 260 (2003) 1–4.
- [506] E.L. LeCluyse, L.E. Appel, S.C. Sutton, Relationship between drug absorption enhancing activity and membrane perturbing effects of acylcarnitines, *Pharm. Res.* 8 (1991) 84–87.
- [507] N. Doi, M. Tomita, M. Hayashi, Absorption enhancement effect of acylcarnitines through changes in tight junction protein in Caco-2 cell monolayers, *Drug Metab. Pharmacokinet.* 26 (2011) 162–170.
- [508] T. Yang, J.J. Arnold, F. Ahsan, Tetradecylmaltoside (TDM) enhances in vitro and in vivo intestinal absorption of enoxaparin, a low molecular weight heparin, *J. Drug Target.* 13 (2005) 29–38.
- [509] A.F. Kotze, H.L. Luessen, B.J. de Leeuw, B.G. de Boer, J.C. Verhoef, H.E. Junginger, N-trimethyl chitosan chloride as a potential absorption enhancer across mucosal surfaces: in vitro evaluation in intestinal epithelial cells (Caco-2), *Pharm. Res.* 14 (1997) 1197–1202.
- [510] M. Thanou, B.I. Florea, M.W. Langemeyer, J.C. Verhoef, H.E. Junginger, N-trimethylated chitosan chloride (TMC) improves the intestinal permeation of the peptide drug busserelin in vitro (Caco-2 cells) and in vivo (rats), *Pharm. Res.* 17 (2000) 27–31.
- [511] J.H. Hochman, J.A. Fix, E.L. LeCluyse, In vitro and in vivo analysis of the mechanism of absorption enhancement by palmitoylcarnitine, *J. Pharmacol. Exp. Ther.* 269 (1994) 813–822.
- [512] N. Surendran, L.D. Nguyen, A.R. Giuliano, J. Blanchard, Mechanisms of acylcarnitine-mediated enhancement of calcium transport in the Caco-2 cell monolayer model, *J. Pharm. Sci.* 84 (1995) 269–274.
- [513] D.Z. Liu, E.L. LeCluyse, D.R. Thakker, Dodecylphosphocholine-mediated enhancement of paracellular permeability and cytotoxicity in Caco-2 cell monolayers, *J. Pharm. Sci.* 88 (1999) 1161–1168.
- [514] A.C. Chao, M.T. Taylor, P.E. Daddona, M. Broughall, J.A. Fix, Molecular weight-dependent paracellular transport of fluorescent model compounds induced by palmitoylcarnitine chloride across the human intestinal epithelial cell line Caco-2, *J. Drug Target.* 6 (1998) 37–43.
- [515] E.J. Nielsen, S. Yoshida, N. Kamei, R. Iwamae, S. Khafagy el, J. Olsen, U.L. Rahbek, B.L. Pedersen, K. Takayama, M. Takeda-Morishita, In vivo proof of concept of oral insulin delivery based on a co-administration strategy with the cell-penetrating peptide penetratin, *J. Control. Release* 189 (2014) 19–24.
- [516] N. Kamei, Y. Onuki, K. Takayama, M. Takeda-Morishita, Mechanistic study of the uptake/permeation of cell-penetrating peptides across a caco-2 monolayer and their stimulatory effect on epithelial insulin transport, *J. Pharm. Sci.* 102 (2013) 3998–4008.
- [517] X. Zhu, W. Shan, P. Zhang, Y. Jin, S. Guan, T. Fan, Y. Yang, Z. Zhou, Y. Huang, Penetratin derivative-based nanocomplexes for enhanced intestinal insulin delivery, *Mol. Pharm.* 11 (2014) 317–328.
- [518] F.G. Welt, T.C. Woods, E.R. Edelman, Oral heparin prevents neointimal hyperplasia after arterial injury: inhibitory potential depends on type of vascular injury, *Circulation* 104 (2001) 3121–3124.
- [519] M. Yamazaki, S. Itoh, M. Sawanoi, M. Kobayashi, S. Suzuki, T. Komatsu, K. Tanabe, The effect of fatty acids on the rectal absorption of acyclovir in rats, *J. Pharm. Pharmacol.* 42 (1990) 441–443.
- [520] M. Tomita, M. Shiga, M. Hayashi, S. Awazu, Enhancement of colonic drug absorption by the paracellular permeation route, *Pharm. Res.* 5 (1988) 341–346.
- [521] E. Miyamoto, A. Tsuji, T. Yamana, Effects of surfactants on the GI absorption of beta-lactam antibiotics in rats, *J. Pharm. Sci.* 72 (1983) 651–654.
- [522] E.A. Hosny, N.M. Ghilzai, T.A. al-Najar, M.M. Elmazar, Hypoglycemic effect of oral insulin in diabetic rabbits using pH-dependent coated capsules containing sodium salicylate without and with sodium cholate, *Drug Dev. Ind. Pharm.* 24 (1998) 307–311.
- [523] E.A. Hosny, H.I. Al-Shora, M.M. Elmazar, Effect of different bile salts on the relative hypoglycemia of witepsol W35 suppositories containing insulin in diabetic Beagle dogs, *Drug Dev. Ind. Pharm.* 27 (2001) 837–845.
- [524] E. Ziv, M. Kidron, I. Raz, M. Krausz, Y. Blatt, A. Rotman, H. Bar-On, Oral administration of insulin in solid form to nondiabetic and diabetic dogs, *J. Pharm. Sci.* 83 (1994) 792–794.
- [525] M. Sakai, T. Imai, H. Ohtake, H. Azuma, M. Otagiri, Effects of absorption enhancers on the transport of model compounds in Caco-2 cell monolayers: assessment by confocal laser scanning microscopy, *J. Pharm. Sci.* 86 (1997) 779–785.
- [526] M. Kidron, H. Bar-On, E.M. Berry, E. Ziv, The absorption of insulin from various regions of the rat intestine, *Life Sci.* 31 (1982) 2837–2841.
- [527] Y.L. Lo, J.D. Huang, Effects of sodium deoxycholate and sodium caprate on the transport of epirubicin in human intestinal epithelial Caco-2 cell layers and everted gut sacs of rats, *Biochem. Pharmacol.* 59 (2000) 665–672.
- [528] M. Sakai, T. Imai, H. Ohtake, H. Azuma, M. Otagiri, Effects of absorption enhancers on cytoskeletal actin filaments in Caco-2 cell monolayers, *Life Sci.* 63 (1998) 45–54.
- [529] P. Sharma, M.V. Varma, H.P. Chawla, R. Panchagnula, In situ and in vivo efficacy of peroral absorption enhancers in rats and correlation to in vitro mechanistic studies, *Farmaco* 60 (2005) 874–883.
- [530] J.W. Dobbins, H.J. Binder, Effect of bile salts and fatty acids on the colonic absorption of oxalate, *Gastroenterology* 70 (1976) 1096–1100.
- [531] M.W. Gouda, N. Khalafalah, S.A. Khalil, Effect of surfactants on absorption through membranes V: concentration-dependent effect of a bile salt (sodium deoxycholate) on absorption of a poorly absorbable drug, phenolsulfonphthalein, in humans, *J. Pharm. Sci.* 66 (1977) 727–728.
- [532] G.W. Gullikson, W.S. Cline, V. Lorenzsonn, L. Benz, W.A. Olsen, P. Bass, Effects of anionic surfactants on hamster small intestinal membrane structure and function: relationship to surface activity, *Gastroenterology* 73 (1977) 501–511.
- [533] L. Lei, H. Sun, D. Liu, L. Liu, S. Li, Transport of Val-Leu-Pro-Val-Pro in human intestinal epithelial (Caco-2) cell monolayers, *J. Agric. Food Chem.* 56 (2008) 3582–3586.
- [534] S. Feldman, M. Salvino, M. Gibaldi, Physiologic surface-active agents and drug absorption. VII. Effect of sodium deoxycholate on phenol red absorption in the rat, *J. Pharm. Sci.* 59 (1970) 705–707.
- [535] I. Legen, M. Kracun, M. Salobir, J. Kerc, The evaluation of some pharmaceutically acceptable excipients as permeation enhancers for amoxicillin, *Int. J. Pharm.* 308 (2006) 84–89.
- [536] I. Utsumi, K. Kono, Y. Takeuchi, Effect of surfactants on drug absorption. IV. Mechanism of the action of sodium glycocholate on the absorption of benzoylthiamine disulfide in the presence of sodium laurylsulfate and polysorbate 80, *Chem. Pharm. Bull.* 21 (1973) 2161–2167.
- [537] S.Y. Cho, J.S. Kim, H. Li, C. Shim, R.J. Linhardt, Y.S. Kim, Enhancement of paracellular transport of heparin disaccharide across Caco-2 cell monolayers, *Arch. Pharm. Res.* 25 (2002) 86–92.
- [538] U. Werner, T. Kissel, M. Reers, Effects of permeation enhancers on the transport of a peptidomimetic thrombin inhibitor (CRC 220) in a human intestinal cell line (Caco-2), *Pharm. Res.* 13 (1996) 1219–1227.
- [539] T. Kimura, K. Inui, H. Sezaki, Differences in effects on drug absorption between dihydroxy and trihydroxy bile salts, *J. Pharmacobiodyn.* 8 (1985) 578–585.
- [540] K. Kakemi, H. Sezaki, R. Konishi, T. Kimura, M. Murakami, Effect of bile salts on the gastrointestinal absorption of drugs. I, *Chem. Pharm. Bull.* 18 (1970) 275–280.
- [541] T. Kimura, H. Sezaki, K. Kakemi, Effect of bile salts on the gastrointestinal absorption of drugs. IV. Site of intestinal absorption of sodium taurocholate and its consequence on drug absorption in the rat, *Chem. Pharm. Bull.* 20 (1972) 1656–1662.
- [542] P.C. Lerk, H. Sucker, Application of sucrose laurate, a new pharmaceutical excipient, in peroral formulations of cyclosporin A, *Int. J. Pharm.* 92 (1993) 197–202.
- [543] D.J. Brayden, V.A. Bzik, A.L. Lewis, L. Illum, Critical Sorb promotes permeation of flux markers across isolated rat intestinal mucosae and Caco-2 monolayers, *Pharm. Res.* 29 (2012) 2543–2554.
- [544] K. Takahashi, T. Murakami, A. Kamata, R. Yumoto, Y. Higashi, N. Yata, Pharmacokinetic analysis of the absorption enhancing action of decanoic acid and its derivatives in rats, *Pharm. Res.* 11 (1994) 388–392.
- [545] D.J. Hauss, H.Y. Ando, The influence of concentration of two salicylate derivatives on rectal insulin absorption enhancement, *J. Pharm. Pharmacol.* 40 (1988) 659–661.
- [546] H. Ouyang, S.L. Morris-Natschke, K.S. Ishaq, P. Ward, D. Liu, S. Leonard, D.R. Thakker, Structure-activity relationship for enhancement of paracellular permeability across Caco-2 cell monolayers by 3-alkylamido-2-alkoxypropylphosphocholines, *J. Med. Chem.* 45 (2002) 2857–2866.
- [547] E.J. van Hoogdalem, A.T. Wackwitz, A.G. de Boer, D.D. Breimer, 3-Amino-1-hydroxypropylidene-1,1-diphosphonate (APD): a novel enhancer of rectal cefoxitin absorption in rats, *J. Pharm. Pharmacol.* 41 (1989) 339–341.
- [548] T. Nishihata, J.H. Rytting, T. Higuchi, Enhanced rectal absorption of theophylline, lidocaine, cefmetazole, and levodopa by several adjuvants, *J. Pharm. Sci.* 71 (1982) 865–868.
- [549] W. Zhang, M.A. Parniak, S.G. Sarafianos, P.E. Empey, L.C. Rohan, In vitro transport characteristics of EFdA, a novel nucleoside reverse transcriptase inhibitor using Caco-2 and MDCKII cell monolayers, *Eur. J. Pharmacol.* 732 (2014) 86–95.
- [550] A. Leone-Bay, M. Sato, D. Paton, A.H. Hunt, D. Sarubbi, M. Carozza, J. Chou, J. McDonough, R.A. Baughman, Oral delivery of biologically active parathyroid hormone, *Pharm. Res.* 18 (2001) 964–970.
- [551] T. Nishihata, J.H. Rytting, A. Kamada, T. Higuchi, Enhanced intestinal absorption of insulin in rats in the presence of sodium 5-methoxysalicylate, *Diabetes* 30 (1981) 1065–1067.
- [552] S. Yoshioka, L. Caldwell, T. Higuchi, Enhanced rectal bioavailability of polypeptides using sodium 5-methoxysalicylate as an absorption promoter, *J. Pharm. Sci.* 71 (1982) 593–594.
- [553] L. Caldwell, T. Nishihata, J. Fix, S. Selk, R. Cargill, C.R. Gardner, T. Higuchi, Absorption-promoting adjuvants: animal studies on their effects on rectal drug absorption, *Methods Find. Exp. Clin. Pharmacol.* 6 (1984) 503–507.
- [554] T. Nishihata, J.H. Rytting, T. Higuchi, L. Caldwell, Enhanced rectal absorption of insulin and heparin in rats in the presence of non-surfactant adjuvants, *J. Pharm. Pharmacol.* 33 (1981) 334–335.
- [555] T. Nishihata, J.H. Rytting, T. Higuchi, L.J. Caldwell, S.J. Selk, Enhancement of rectal absorption of water-soluble antibiotics in dogs, *Int. J. Pharm.* 21 (1984) 239–248.

- [556] G.E. Peters, L.E. Hutchinson, R. Hyde, C. McMartin, S.B. Metcalfe, Effects of sodium 5-methoxysalicylate on macromolecule absorption and mucosal morphology in a vascularly perfused rat gut preparation in vivo, *J. Pharm. Sci.* 76 (1987) 857–861.
- [557] K. Suni, N. Toshiaki, K. Shoji, O. Yasufumi, K. Akira, Y. Toshihito, K. Ryuzo, S. Motoaki, Effect of gelatin in the formulation on rectal insulin absorption in the presence of enamine in normal rats and depancreatized dogs, *Int. J. Pharm.* 21 (1984) 179–186.
- [558] S. Kim, A. Kamada, T. Higuchi, T. Nishihata, Effect of enamine derivatives on the rectal absorption of insulin in dogs and rabbits, *J. Pharm. Pharmacol.* 35 (1983) 100–103.
- [559] R.H. Engel, S.J. Riggi, Intestinal absorption of heparin facilitated by sulfated or sulfonated surfactants, *J. Pharm. Sci.* 58 (1969) 706–709.
- [560] W. Chen, Z. Lu, A. Viljoen, J. Hamman, Intestinal drug transport enhancement by Aloe vera, *Planta Med.* 75 (2009) 587–595.
- [561] F.G. Chambers, S.S. Koshiy, R.F. Saidi, D.P. Clark, R.D. Moore, C.L. Sears, Bacteroides fragilis toxin exhibits polar activity on monolayers of human intestinal epithelial cells (T84 cells) in vitro, *Infect. Immun.* 65 (1997) 3561–3570.
- [562] A. Guzman, R. Garcia, Effects of fatty ethers and stearic acid on the gastrointestinal absorption of insulin, *P. R. Health Sci. J.* 9 (1990) 155–159.
- [563] S. Hirata, T. Futaguchi, K. Tamura, A. Odaguchi, Tanaka, rectal absorption of Gly1-alpha1(-18) adrenocorticotropin amide, *Chem. Pharm. Bull.* 26 (1978) 1061–1065.
- [564] W.W. Davis, R.R. Pfeiffer, J.F. Quay, Normal and promoted gastrointestinal absorption of water-soluble substances. I. Induced rapidly reversible hyperabsorptive state in the canine fundic stomach pouch, *J. Pharm. Sci.* 59 (1970) 960–963.
- [565] J.A. Galloway, M.A. Root, New forms of insulin, *Diabetes* 21 (1972) 637–648.
- [566] T. Okada, A. Nara, S. Matsunaga, N. Fusetani, M. Shimizu, Assessment of the marine toxins by monitoring the integrity of human intestinal Caco-2 cell monolayers, *Toxicol. In Vitro* 14 (2000) 219–226.
- [567] B.R. Stoll, H.R. Leipold, S. Milstein, D.A. Edwards, A mechanistic analysis of carrier-mediated oral delivery of protein therapeutics, *J. Control. Release* 64 (2000) 217–228.
- [568] G.M. Mlynek, L.J. Calvo, J.R. Robinson, Carrier-enhanced human growth hormone absorption across isolated rabbit intestinal tissue, *Int. J. Pharm.* 197 (2000) 13–21.
- [569] A. Leone-Bay, D.R. Paton, J.J. Weidner, The development of delivery agents that facilitate the oral absorption of macromolecular drugs, *Med. Res. Rev.* 20 (2000) 169–186.
- [570] S.J. Wu, J.R. Robinson, Transport of human growth hormone across Caco-2 cells with novel delivery agents: evidence for P-glycoprotein involvement, *J. Control. Release* 62 (1999) 171–177.
- [571] S.J. Milstein, H. Leipold, D. Sarubbi, A. Leone-Bay, G.M. Mlynek, J.R. Robinson, M. Kasimova, E. Freire, Partially unfolded proteins efficiently penetrate cell membranes—implications for oral drug delivery, *J. Control. Release* 53 (1998) 259–267.
- [572] E. Toutiou, M. Donbrow, A. Rubinstein, Effective intestinal absorption of insulin in diabetic rats using a new formulation approach, *J. Pharm. Pharmacol.* 32 (1980) 108–110.
- [573] H. Bar-On, E.M. Berry, A. Eldor, M. Kidron, D. Lichtenberg, E. Ziv, Enteral administration of insulin in the rat, *Br. J. Pharmacol.* 73 (1981) 21–24.
- [574] A. Rubinstein, E. Rubinstein, E. Toitou, M. Donbrow, Increase of the intestinal absorption of gentamicin and amikacin by a nonionic surfactant, *Antimicrob. Agents Chemother.* 19 (1981) 696–699.
- [575] M. Kidron, A. Eldor, D. Lichtenberg, E. Toutiou, E. Ziv, H. Bar-On, Enteral administration of heparin, *Thromb. Res.* 16 (1979) 833–835.
- [576] G. Borchard, H.L. Lueßen, A.G. de Boer, J.C. Verhoef, C.-M. Lehr, H.E. Junginger, The potential of mucoadhesive polymers in enhancing intestinal peptide drug absorption. III: effects of chitosan-glutamate and carbomer on epithelial tight junctions in vitro, *J. Control. Release* 39 (1996) 131–138.
- [577] S. Michael, M. Thole, R. Dillmann, A. Fahr, J. Drewe, G. Fricker, Improvement of intestinal peptide absorption by a synthetic bile acid derivative, cholylosarcosine, *Eur. J. Pharm. Sci.* 10 (2000) 133–140.
- [578] A. Nusrat, M. Giry, J.R. Turner, S.P. Colgan, C.A. Parkos, D. Carnes, E. Lemichez, P. Boquet, J.L. Madara, Rho protein regulates tight junctions and perijunctional actin organization in polarized epithelia, *Proc. Natl. Acad. Sci. U. S. A.* 92 (1995) 10629–10633.
- [579] G. Hecht, C. Pothoulakis, J.T. LaMont, J.L. Madara, Clostridium difficile toxin A perturbs cytoskeletal structure and tight junction permeability of cultured human intestinal epithelial monolayers, *J. Clin. Invest.* 82 (1988) 1516–1524.
- [580] A. Nusrat, C. von Eichel-Streiber, J.R. Turner, P. Verkade, J.L. Madara, C.A. Parkos, Clostridium difficile toxins disrupt epithelial barrier function by altering membrane microdomain localization of tight junction proteins, *Infect. Immun.* 69 (2001) 1329–1336.
- [581] J.L. Madara, D. Barenberg, S. Carlson, Effects of cytochalasin D on occluding junctions of intestinal absorptive cells: further evidence that the cytoskeleton may influence paracellular permeability and junctional charge selectivity, *J. Cell Biol.* 102 (1986) 2125–2136.
- [582] Z. Shao, Y. Li, T. Chermak, A.K. Mitra, Cyclodextrins as mucosal absorption promoters of insulin. II. Effects of beta-cyclodextrin derivatives on alpha-chymotryptic degradation and enteral absorption of insulin in rats, *Pharm. Res.* 11 (1994) 1174–1179.
- [583] M. Tomita, M. Hayashi, S. Awazu, Absorption-enhancing mechanism of sodium caprate and decanoylcarnitine in Caco-2 cells, *J. Pharmacol. Exp. Ther.* 272 (1995) 739–743.
- [584] T.A. Aguirre, M. Rosa, S.S. Guterres, A.R. Pohlmann, I. Coulter, D.J. Brayden, Investigation of coco-glucoside as a novel intestinal permeation enhancer in rat models, *Eur. J. Pharm. Biopharm.* 88 (2014) 856–865.
- [585] K. Nakanishi, M. Masada, T. Nadai, Effect of nonsteroidal anti-inflammatory drugs on the absorption of macromolecular drugs in rat rectum, *Chem. Pharm. Bull.* 34 (1986) 2628–2631.
- [586] M. Miyake, T. Nishihata, A. Nagano, Y. Kyobashi, A. Kamada, Rectal absorption of [Asu1,7]-eel calcitonin in rats, *Chem. Pharm. Bull.* 33 (1985) 740–745.
- [587] T. Nishihata, M. Miyake, H. Takahata, A. Kamada, The effect of adjuvants on the colonie absorption of cefmetazole and [Asu1,7]-eel calcitonin in rats: concentration dependent absorption pathways, *Int. J. Pharm.* 33 (1986) 89–97.
- [588] T. Nishihata, M. Miyake, A. Kamada, Study of the mechanism behind adjuvant action of diethylethoxymethylene malonate enhancing the rectal absorption of cefmetazole and lysozyme, *J. Pharmacobiodyn.* 7 (1984) 607–613.
- [589] M.R. Avadi, A. Jalali, A.M. Sadeghi, K. Shamimi, K.H. Bayati, E. Nahid, A.R. Dehpour, M. Rafiee-Tehrani, Diethyl methyl chitosan as an intestinal paracellular enhancer: ex vivo and in vivo studies, *Int. J. Pharm.* 293 (2005) 83–89.
- [590] T. Suzuki, H. Hara, Diffructose anhydride III and sodium caprate activate paracellular transport via different intracellular events in Caco-2 cells, *Life Sci.* 79 (2006) 401–410.
- [591] T. Uchida, Y. Toida, S. Sakakibara, Y. Miyanaga, H. Tanaka, M. Nishikata, K. Tazuya, N. Yasuda, K. Matsuyama, Preparation and characterization of insulin-loaded acrylic hydrogels containing absorption enhancers, *Chem. Pharm. Bull.* 49 (2001) 1261–1266.
- [592] Y. Watanabe, Y. Matsumoto, M. Seki, M. Takase, M. Matsumoto, Absorption enhancement of polypeptide drugs by cyclodextrins. I. Enhanced rectal absorption of insulin from hollow-type suppositories containing insulin and cyclodextrins in rabbits, *Chem. Pharm. Bull.* 40 (1992) 3042–3047.
- [593] S.N. Malik, D.H. Canaham, M.W. Gouda, Effect of surfactants on absorption through membranes III: effects of dioctyl sodium sulfosuccinate and poloxalene on absorption of a poorly absorbable drug, phenolsulfonphthalein, in rats, *J. Pharm. Sci.* 64 (1975) 987–990.
- [594] N. Khalafallah, M.W. Gouda, S.A. Khalil, Effect of surfactants on absorption through membranes IV: effects of dioctyl sodium sulfosuccinate on absorption of a poorly absorbable drug, phenolsulfonphthalein, in humans, *J. Pharm. Sci.* 64 (1975) 991–994.
- [595] T. Nishihata, C.S. Lee, B.T. Nghiem, T. Higuchi, Possible mechanism behind the adjuvant action of phosphate derivatives on rectal absorption of cefoxitin in rats and dogs, *J. Pharm. Sci.* 73 (1984) 1523–1525.
- [596] M. Mishima, A. Nagatomi, T. Yamakita, Y. Miura, O. Tsuzuki, Promotion of rectal absorption of sodium ampicillin by disodium glycyrrhetic acid 3 beta-O-monohemiphthalate in rats, *Biol. Pharm. Bull.* 18 (1995) 566–570.
- [597] R. Gerhard, G. Schmidt, F. Hofmann, K. Aktories, Activation of Rho GTPases by Escherichia coli cytotoxic necrotizing factor 1 increases intestinal permeability in Caco-2 cells, *Infect. Immun.* 66 (1998) 5125–5131.
- [598] G.T. Knipp, N.F. Ho, C.L. Barsuhn, R.T. Borchardt, Paracellular diffusion in Caco-2 cell monolayers: effect of perturbation on the transport of hydrophilic compounds that vary in charge and size, *J. Pharm. Sci.* 86 (1997) 1105–1110.
- [599] M. Miyake, T. Nishihata, N. Wada, E. Takeshima, A. Kamada, Rectal absorption of lysozyme and heparin in rabbits in the presence of non-surfactant adjuvants, *Chem. Pharm. Bull.* 32 (1984) 2020–2025.
- [600] T. Nishihata, S. Kim, S. Morishita, A. Kamada, N. Yata, T. Higuchi, Adjuvant effects of glyceryl esters of acetoacetic acid on rectal absorption of insulin and inulin in rabbits, *J. Pharm. Sci.* 72 (1983) 280–285.
- [601] N.A. Motlekar, K.S. Srivenugopal, M.S. Wachtel, B.B. Youan, Evaluation of the Oral Bioavailability of Low Molecular Weight Heparin Formulated With Glycyrrhetic Acid as Permeation Enhancer, *Drug Dev. Res.* 67 (2006) 166–174.
- [602] T. Matsumura, Y. Jin, Y. Kabumoto, Y. Takegahara, K. Oguma, W.I. Lencer, Y. Fujinaga, The HA proteins of botulinum toxin disrupt intestinal epithelial intercellular junctions to increase toxin absorption, *Cell. Microbiol.* 10 (2008) 355–364.
- [603] J.F. Liang, V.C. Yang, Insulin-cell penetrating peptide hybrids with improved intestinal absorption efficiency, *Biochem. Biophys. Res. Commun.* 335 (2005) 734–738.
- [604] A. Watari, M. Hashigawa, K. Yagi, M. Kondoh, Homoharringtonine increases intestinal epithelial permeability by modulating specific claudin isoforms in Caco-2 cell monolayers, *Eur. J. Pharm. Biopharm.* 89 (2015) 232–238.
- [605] G. Sandri, S. Rossi, F. Ferrari, M.C. Bonferoni, N. Zerrouk, C. Caramella, Mucoadhesive and penetration enhancement properties of three grades of hyaluronic acid using porcine buccal and vaginal tissue, Caco-2 cell lines, and rat jejunum, *J. Pharm. Pharmacol.* 56 (2004) 1083–1090.
- [606] T. Nishihata, T. Higuchi, Ortho-hydroxybenzoates may act at the protein fraction to enhance membrane permeability, *Biochim. Biophys. Acta* 775 (1984) 269–271.
- [607] A. Youakim, M. Ahdieh, Interferon-gamma decreases barrier function in T84 cells by reducing ZO-1 levels and disrupting apical actin, *Am. J. Phys.* 276 (1999) G1279–G1288.
- [608] J.L. Madara, J. Stafford, Interferon-gamma directly affects barrier function of cultured intestinal epithelial monolayers, *J. Clin. Invest.* 83 (1989) 724–727.
- [609] M. Bruewer, A. Luegering, T. Kucharzik, C.A. Parkos, J.L. Madara, A.M. Hopkins, A. Nusrat, Proinflammatory cytokines disrupt epithelial barrier function by apoptosis-independent mechanisms, *J. Immunol.* 171 (2003) 6164–6172.
- [610] R.B. Adams, S.M. Planchon, J.K. Roche, IFN-gamma modulation of epithelial barrier function. Time course, reversibility, and site of cytokine binding, *J. Immunol.* 150 (1993) 2356–2363.
- [611] K.B. Madden, L. Whitman, C. Sullivan, W.C. Gause, J.F. Urban Jr., I.M. Katona, F.D. Finkelman, T. Shea-Donohue, Role of STAT6 and mast cells in IL-4- and IL-13-induced alterations in murine intestinal epithelial cell function, *J. Immunol.* 169 (2002) 4417–4422.
- [612] R. Al-Sadi, D. Ye, K. Dokladny, T.Y. Ma, Mechanism of IL-1beta-induced increase in intestinal epithelial tight junction permeability, *J. Immunol.* 180 (2008) 5653–5661.

- [613] R.M. Al-Sadi, T.Y. Ma, IL-1 β causes an increase in intestinal epithelial tight junction permeability, *J. Immunol.* 178 (2007) 4641–4649.
- [614] T. Matysiak-Budnik, A. Thomas-Collignon, F. Megraud, M. Heyman, Alterations of epithelial permeability by *Helicobacter* and IL-1 β in vitro: protective effect of rebamipide, *Dig. Dis. Sci.* 46 (2001) 1558–1566.
- [615] P.J. Ceponis, F. Botelho, C.D. Richards, D.M. McKay, Interleukins 4 and 13 increase intestinal epithelial permeability by a phosphatidylinositol 3-kinase pathway. Lack of evidence for STAT 6 involvement, *J. Biol. Chem.* 275 (2000) 29132–29137.
- [616] M.C. Berin, P.C. Yang, L. Ciok, S. Waserman, M.H. Perdue, Role for IL-4 in macromolecular transport across human intestinal epithelium, *Am. J. Phys.* 276 (1999) C1046–C1052.
- [617] Y. Tazuke, R.A. Drongowski, D.H. Teitelbaum, A.G. Coran, Interleukin-6 changes tight junction permeability and intracellular phospholipid content in a human enterocyte cell culture model, *Pediatr. Surg. Int.* 19 (2003) 321–325.
- [618] N.A. Motlekar, K.S. Srivenugopal, M.S. Wachtel, B.B. Youan, Modulation of gastrointestinal permeability of low-molecular-weight heparin by L-arginine: in-vivo and in-vitro evaluation, *J. Pharm. Pharmacol.* 58 (2006) 591–598.
- [619] M. Murakami, K. Takada, S. Muranishi, Promoting effect of Azone on intestinal absorption of poorly absorbable drugs in rats, *Int. J. Pharm.* 31 (1986) 231–238.
- [620] Muranishi, Absorption barriers and absorption promoters in the intestine, in: D.D. Breimer, P. Speiser (Eds.), *Topics in Pharmaceutical Science*, Elsevier B.V., Amsterdam 1987, pp. 445–455.
- [621] B.S. Ramakrishna, M. Mathan, V.I. Mathan, Alteration of colonic absorption by long-chain unsaturated fatty acids. Influence of hydroxylation and degree of unsaturation, *Scand. J. Gastroenterol.* 29 (1994) 54–58.
- [622] T. Sawai, R. Lampman, Y. Hua, B. Segura, R.A. Drongowski, A.G. Coran, C.M. Harmon, Lysophosphatidylcholine alters enterocyte monolayer permeability via a protein kinase C/Ca²⁺ mechanism, *Pediatr. Surg. Int.* 18 (2002) 591–594.
- [623] L. Hovgaard, H. Brøndsted, H.M. Nielsen, Drug delivery studies in Caco-2 monolayers. II. Absorption enhancer effects of lysophosphatidylcholines, *Int. J. Pharm.* 114 (1995) 141–149.
- [624] C. Tagesson, L. Franzen, G. Dahl, B. Westrom, Lysophosphatidylcholine increases rat ileal permeability to macromolecules, *Gut* 26 (1985) 369–377.
- [625] P.-Y. Yeh, M.M. Berenson, W.S. Samowitz, P. Kopečková, J. Kopecek, Site-specific drug delivery and penetration enhancement in the gastrointestinal tract, *J. Control. Release* 36 (1995) 109–124.
- [626] M. Sekine, K. Sasahara, K. Hasegawa, R. Okada, S. Awazu, Improvement of bioavailability of poorly absorbed drugs. V. Effect of surfactants on the promoting effect of medium chain glyceride for the rectal absorption of beta-lactam antibiotics in rats and dogs, *J. Pharmacobiodyn.* 8 (1985) 653–660.
- [627] P.D. Lundin, M. Bojrup, H. Ljusberg-Wahren, B.R. Westrom, S. Lundin, Enhancing effects of monohexanoin and two other medium-chain glyceride vehicles on intestinal absorption of desmopressin (DDAVP), *J. Pharmacol. Exp. Ther.* 282 (1997) 585–590.
- [628] P. Liu, P. Davis, H. Liu, T.R. Krishnan, Evaluation of cytotoxicity and absorption enhancing effects of melittin – a novel absorption enhancer, *Eur. J. Pharm. Biopharm.* 48 (1999) 85–87.
- [629] S. Maher, L. Feighery, D.J. Brayden, S. McClean, Melittin as a permeability enhancer II: in vitro investigations in human mucus secreting intestinal monolayers and rat colonic mucosae, *Pharm. Res.* 24 (2007) 1346–1356.
- [630] N. Yata, N. Sugihara, R. Yamajo, T. Murakami, Y. Higashi, H. Kimata, K. Nakayama, T. Kuzuki, O. Tanaka, Enhanced rectal absorption of beta-lactam antibiotics in rat by monodesmosides isolated from pericarp of *Sapindus mukurossi* (Enmei-hi), *J. Pharmacobiodyn.* 8 (1985) 1041–1047.
- [631] N. Yata, W.M. Wu, R. Yamajo, T. Murakami, Y. Higashi, T. Higuchi, Enhanced rectal absorption of sodium ampicillin by N-acyl derivatives of collagen peptide in rabbits and rats, *J. Pharm. Sci.* 74 (1985) 1058–1061.
- [632] W.M. Wu, T. Murakami, Y. Higashi, N. Yata, Enhancement of the rectal absorption of sodium ampicillin by N-acylamino acids in rats, *J. Pharm. Sci.* 76 (1987) 508–512.
- [633] S. Somasekharan, R. Brandt, T. Iwamoto, J.M. Tomich, B.D. Schultz, Epithelial barrier modulation by a channel forming peptide, *J. Membr. Biol.* 222 (2008) 17–30.
- [634] M. Murakami, K. Takada, T. Fujii, S. Muranishi, Intestinal absorption enhanced by unsaturated fatty acids: inhibitory effect of sulphydryl modifiers, *Biochim. Biophys. Acta* 939 (1988) 238–246.
- [635] G. Fetihi, F. Habib, N. Okada, T. Fujita, M. Attia, A. Yamamoto, Nitric oxide donors can enhance the intestinal transport and absorption of insulin and [Asu(1,7)]-eel calcitonin in rats, *J. Control. Release* 106 (2005) 287–297.
- [636] G. Fetihi, F. Habib, H. Katsumi, N. Okada, T. Fujita, M. Attia, A. Yamamoto, Excellent absorption enhancing characteristics of NO donors for improving the intestinal absorption of poorly absorbable compound compared with conventional absorption enhancers, *Drug Metab. Pharmacokinet.* 21 (2006) 222–229.
- [637] N. Utoguchi, Y. Watanabe, T. Shida, M. Matsumoto, Nitric oxide donors enhance rectal absorption of macromolecules in rabbits, *Pharm. Res.* 15 (1998) 870–876.
- [638] M. Thanou, S. Henderson, A. Kydonieus, C. Elson, N-sulfonato-N,O-carboxymethylchitosan: a novel polymeric absorption enhancer for the oral delivery of macromolecules, *J. Control. Release* 117 (2007) 171–178.
- [639] F. Mahmud, O.C. Jeon, T.A. Al-Hilal, S. Kwon, V.C. Yang, D.S. Lee, Y. Byun, Absorption Mechanism of a Physical Complex of Monomeric Insulin and Deoxycholyll-lysyl-methylester in the Small Intestine, *Mol. Pharm.* 12 (2015) 1911–1920.
- [640] M. Maresca, R. Mahfoud, A. Pfohl-Leschkowicz, J. Fantini, The mycotoxin ochratoxin A alters intestinal barrier and absorption functions but has no effect on chloride secretion, *Toxicol. Appl. Pharmacol.* 176 (2001) 54–63.
- [641] J. McLaughlin, P.J. Padfield, J.P. Burt, C.A. O'Neill, Ochratoxin A increases permeability through tight junctions by removal of specific claudin isoforms, *Am. J. Physiol. Cell Physiol.* 287 (2004) C1412–C1417.
- [642] M. Murakami, H. Yoshikawa, K. Takada, S. Muranishi, Effect of oleic Acid vesicles on intestinal absorption of carboxyfluorescein in rats, *Pharm. Res.* 3 (1986) 35–40.
- [643] J. McLaughlin, D. Lambert, P.J. Padfield, J.P. Burt, C.A. O'Neill, The mycotoxin patulin, modulates tight junctions in caco-2 cells, *Toxicol. In Vitro* 23 (2009) 83–89.
- [644] H.M. Mohan, D. Collins, S. Maher, E.G. Walsh, D.C. Winter, P.J. O'Brien, D.J. Brayden, A.W. Baird, The mycotoxin patulin increases colonic epithelial permeability in vitro, *Food Chem. Toxicol.* 50 (2012) 4097–4102.
- [645] G. Imanidis, K.C. Hartner, N.A. Mazer, Intestinal permeation and metabolism of a model peptide (leuprolide) and mechanisms of permeation enhancement by non-ionic surfactants, *Int. J. Pharm.* 120 (1995) 41–50.
- [646] E. Touitou, M. Donbrow, Promoted rectal absorption of insulin: formulative parameters involved in the absorption from hydrophilic bases, *Int. J. Pharm.* 15 (1983) 13–24.
- [647] J. Nakamura, S. Takada, S. Ueda, T. Hamaura, A. Yamamoto, T. Kimura, H. Sezaki, Assessment of pharmaceutical excipient-induced gastrointestinal mucosal damage in rats in vivo by measuring the permeation of phenolsulfonphthalein, *Chem. Pharm. Bull.* 33 (1985) 3527–3529.
- [648] A. Kaneda, K. Nishimura, S. Muranishi, H. Sezaki, Mechanism of drug absorption from micellar solution. II. Effect of polysorbate 80 on the absorption of micelle-free drugs, *Chem. Pharm. Bull.* 22 (1974) 523–528.
- [649] J. Drewe, G. Fricker, J. Vonderscher, C. Beglinger, Enteral absorption of octreotide: absorption enhancement by polyoxyethylene-24-cholesterol ether, *Br. J. Pharmacol.* 108 (1993) 298–303.
- [650] K. Takada, Y. Furuya, H. Yoshikawa, S. Muranishi, T. Yasumura, T. Oka, Increased systemic availability of cyclosporin A by formulation design: pharmacokinetic consideration of its transport, *Int. J. Pharm.* 44 (1988) 107–116.
- [651] C.W. Marano, L.A. Garulacan, N. Ginanni, J.M. Mullen, Phorbol ester treatment increases paracellular permeability across IEC-18 gastrointestinal epithelium in vitro, *Dig. Dis. Sci.* 46 (2001) 1490–1499.
- [652] M.C. Berin, M.G. Buell, Phorbol myristate acetate ex vivo model of enhanced colonic epithelial permeability. Reactive oxygen metabolite and protease independence, *Dig. Dis. Sci.* 40 (1995) 2268–2279.
- [653] G. Hecht, B. Robinson, A. Koutsouris, Reversible disassembly of an intestinal epithelial monolayer by prolonged exposure to phorbol ester, *Am. J. Phys.* 266 (1994) G214–G221.
- [654] W.F. Stenson, R.A. Easom, T.E. Riehl, J. Turk, Regulation of paracellular permeability in Caco-2 cell monolayers by protein kinase C, *Am. J. Phys.* 265 (1993) G955–G962.
- [655] U. Fagerholm, B. Sjostrom, J. Sroka-Markovic, A. Wijk, M. Svensson, H. Lennernas, The effect of a drug-delivery system consisting of soybean phosphatidyl choline and medium-chain monoacylglycerol on the intestinal permeability of hexarelin in the rat, *J. Pharm. Pharmacol.* 50 (1998) 467–473.
- [656] N. Kamei, M. Morishita, J. Ehara, K. Takayama, Permeation characteristics of oligo-arginine through intestinal epithelium and its usefulness for intestinal peptide drug delivery, *J. Control. Release* 131 (2008) 94–99.
- [657] M. Morishita, N. Kamei, J. Ehara, K. Isowa, K. Takayama, A novel approach using functional peptides for efficient intestinal absorption of insulin, *J. Control. Release* 118 (2007) 177–184.
- [658] T. Yamaki, M. Uchida, Y. Kuwahara, Y. Shimazaki, K. Ohtake, M. Kimura, H. Uchida, J. Kobayashi, M. Ogihara, Y. Morimoto, H. Natsume, Effect of poly-L-arginine on intestinal absorption of hydrophilic macromolecules in rats, *Biol. Pharm. Bull.* 36 (2013) 496–500.
- [659] G. Ranaldi, I. Marigliano, I. Vespignani, G. Perozzi, Y. Sambuy, The effect of chitosan and other polycations on tight junction permeability in the human intestinal Caco-2 cell line(1), *J. Nutr. Biochem.* 13 (2002) 157–167.
- [660] W.C. Yen, V.H. Lee, Role of Na⁺ in the asymmetric paracellular transport of 4-phenylazobenzoyloxycarbonyl-L-Pro-L-Leu-Gly-L-Pro-D-Arg across rabbit colonic segments and Caco-2 cell monolayers, *J. Pharmacol. Exp. Ther.* 275 (1995) 114–119.
- [661] A.C. Chao, J.V. Nguyen, M. Broughall, J. Recchia, C.R. Kensil, P.E. Daddona, J.A. Fix, Enhancement of intestinal model compound transport by DS-1, a modified Quilaja saponin, *J. Pharm. Sci.* 87 (1998) 1395–1399.
- [662] L. Jiang, X. Long, Q. Meng, Rhamnolipids enhance epithelial permeability in Caco-2 monolayers, *Int. J. Pharm.* 446 (2013) 130–135.
- [663] K. Tsutsumi, S.K. Li, R.V. Hymas, C.L. Teng, L.G. Tillman, G.E. Hardee, W.I. Higuchi, N.F. Ho, Systematic studies on the paracellular permeation of model permeants and oligonucleotides in the rat small intestine with chenodeoxycholate as enhancer, *J. Pharm. Sci.* 97 (2008) 350–367.
- [664] B.J. Aungst, N.J. Rogers, E. Shefter, Comparison of nasal, rectal, buccal, sublingual and intramuscular insulin efficacy and the effects of a bile salt absorption promoter, *J. Pharmacol. Exp. Ther.* 244 (1988) 23–27.
- [665] Y. Kimura, Y. Hosoda, M. Shima, S. Adachi, R. Matsuno, Physico-chemical Properties of Fatty Acids for Assessing the Threshold Concentration to Enhance the Absorption of a Hydrophilic Substance, *Biosci. Biotechnol. Biochem.* 62 (1998) 443–447.
- [666] Y. Kimura, Y. Hosoda, M. Yamaguchi, H. Nagano, M. Shima, S. Adachi, R. Matsuno, Effects of medium-chain fatty acids on intracellular calcium levels and the cytoskeleton in human intestinal (Caco-2) cell monolayers, *Biosci. Biotechnol. Biochem.* 65 (2001) 743–751.
- [667] M. Miyake, N. Kamada, Y. Oka, T. Mukai, T. Minami, H. Toguchi, M. Odami, K. Ogawara, K. Higaki, T. Kimura, Development of suppository formulation safely improving rectal absorption of rebamipide, a poorly absorbable drug, by utilizing sodium laurate and taurine, *J. Control. Release* 99 (2004) 63–71.
- [668] L. Stanzani, G. Mascellani, G.P. Corbelli, P. Bianchini, Rectal absorption of some glycosaminoglycan sulphates and heparin in rats, *J. Pharm. Pharmacol.* 33 (1981) 783–786.
- [669] A.L. Salzman, M.J. Menconi, N. Unno, R.M. Ezzell, D.M. Casey, P.K. Gonzalez, M.P. Fink, Nitric oxide dilates tight junctions and depletes ATP in cultured Caco-2BBE intestinal epithelial monolayers, *Am. J. Phys.* 268 (1995) G361–G373.

- [670] S.S. Davis, W.R. Burnham, P. Wilson, J. O'Brien, Use of adjuvants for enhancement of rectal absorption of cefoxitin in humans, *Antimicrob. Agents Chemother.* 28 (1985) 211–215.
- [671] E.A. Hosny, Z.M. Al-Marzouki, M.E. Metwally, M.Y. Souaida, A.R. Alshaik, Evaluation of efficiency of insulin suppository formulations containing sodium salicylate or sodium cholate in insulin dependent diabetic patients, *Boll. Chim. Farm.* 142 (2003) 361–366.
- [672] M.A. Hurni, A.B. Noach, M.C. Blom-Roosemalen, A.G. de Boer, J.F. Nagelkerke, D.D. Breimer, Permeability enhancement in Caco-2 cell monolayers by sodium salicylate and sodium taurodihydrofusidate: assessment of effect-reversibility and imaging of transepithelial transport routes by confocal laser scanning microscopy, *J. Pharmacol. Exp. Ther.* 267 (1993) 942–950.
- [673] A.R. Chopra, R. Kommagani, P. Saha, J.-F. Louet, C. Salazar, J. Song, J. Jeong, M. Finegold, B. Viollet, F. DeMayo, L. Chan, D.D. Moore, B.W. O'Malley, Cellular energy depletion resets whole-body energy by promoting coactivator-mediated dietary fuel absorption, *Cell Metab.* 13 (2011) 35–43.
- [674] J.A. Fix, P.S. Leppert, P.A. Porter, L.J. Caldwell, Influence of ionic strength on rectal absorption of gentamicin sulfate in the presence and absence of sodium salicylate, *J. Pharm. Sci.* 72 (1983) 1134–1137.
- [675] S. Lundin, N. Pantzar, L. Hedin, B.R. Weström, Intestinal absorption enhancement by sodium taurodihydrofusidate of a peptide hormone analogue (dDAVP) and a macromolecule (BSA) in vitro and in vivo, *Int. J. Pharm.* 59 (1990) 263–269.
- [676] E.J. van Hoogdalem, C.D. Heijligers-Feijen, R.A. Mathot, A.T. Wackwitz, J.B. van Bree, J.C. Verhoef, A.G. de Boer, D.D. Breimer, Rectal absorption enhancement of cefoxitin and desglycinamide arginine vasopressin by sodium tauro-24,25-dihydrofusidate in conscious rats, *J. Pharmacol. Exp. Ther.* 251 (1989) 741–744.
- [677] E.J. van Hoogdalem, C.D. Heijligers-Feijen, J.C. Verhoef, A.G. de Boer, D.D. Breimer, Absorption enhancement of rectally infused insulin by sodium tauro-24,25-dihydrofusidate (STDHF) in rats, *Pharm. Res.* 7 (1990) 180–183.
- [678] R.W. Freel, M. Hatch, D.L. Earnest, A.M. Goldner, Role of tight-junctional pathways in bile salt-induced increases in colonic permeability, *Am. J. Phys.* 245 (1983) G816–G823.
- [679] S. Feldman, M. Gibaldi, Physiologic surface-active agents and drug absorption II. Comparison of the effect of sodium taurodeoxycholate and ethylenediaminetetraacetic acid on salicylamide and salicylate transfer across the everted rat small intestine, *J. Pharm. Sci.* 58 (1969) 967–970.
- [680] N.E. Osman, B. Westrom, Q. Wang, L. Persson, B. Karlsson, Spermine affects intestinal in vitro permeability to different-sized molecules in rats, *Comp. Biochem. Physiol. C Pharmacol. Toxicol. Endocrinol.* 120 (1998) 211–216.
- [681] M. Miyake, T. Minami, H. Toguchi, M. Odomi, K. Ogawara, K. Higaki, T. Kimura, Importance of bile acids for novel oral absorption system containing polyamines to improve intestinal absorption, *J. Control. Release* 115 (2006) 130–133.
- [682] R.E. Herman, E.G. Makienko, M.G. Prieve, M. Fuller, M.E. Houston Jr., P.H. Johnson, Phage display screening of epithelial cell monolayers treated with EGTA: identification of peptide FDFWTP that modulates tight junction activity, *J. Biomol. Screen.* 12 (2007) 1092–1101.
- [683] T.Y. Ma, G.K. Iwamoto, N.T. Hoa, V. Akotia, A. Pedram, M.A. Boivin, H.M. Said, TNF- α -induced increase in intestinal epithelial tight junction permeability requires NF- κ B activation, *Am. J. Physiol. Gastrointest. Liver Physiol.* 286 (2004) G367–G376.
- [684] A.H. Gitter, K. Bendfeldt, H. Schmitz, J.D. Schulzke, C.J. Bentzel, M. Fromm, Epithelial barrier defects in HT-29/B6 colonic cell monolayers induced by tumor necrosis factor- α , *Ann. N. Y. Acad. Sci.* 915 (2000) 193–203.
- [685] P. Rodriguez, M. Heyman, C. Candalh, M.A. Blaton, C. Bouchaud, Tumour necrosis factor- α induces morphological and functional alterations of intestinal HT29 cl.19A cell monolayers, *Cytokine* 7 (1995) 441–448.
- [686] T.Y. Ma, M.A. Boivin, D. Ye, A. Pedram, H.M. Said, Mechanism of TNF- α modulation of Caco-2 intestinal epithelial tight junction barrier: role of myosin light-chain kinase protein expression, *Am. J. Physiol. Gastrointest. Liver Physiol.* 288 (2005) G422–G430.
- [687] S. Khafagy el, M. Morishita, N. Ida, R. Nishio, K. Isowa, K. Takayama, Structural requirements of penetratin absorption enhancement efficiency for insulin delivery, *J. Control. Release* 143 (2010) 302–310.
- [688] P. Younessi, M.R. Avadi, K. Shamimi, A.M. Sadeghi, L. Moezi, E. Nahid, K. Bayati, A.R. Dehpour, M. Rafiee-Tehrani, Preparation and ex vivo evaluation of TEC as an absorption enhancer for poorly absorbable compounds in colon specific drug delivery, *Acta Pharm. (Zagreb, Croatia)* 54 (2004) 339–345.
- [689] E. Papini, B. Satin, N. Norais, M. de Bernard, J.L. Telford, R. Rappuoli, C. Montecucco, Selective increase of the permeability of polarized epithelial cell monolayers by *Helicobacter pylori* vacuolating toxin, *J. Clin. Invest.* 102 (1998) 813–820.
- [690] K. Kowari, I. Hirotsawa, H. Kurai, N. Utoguchi, M. Fujii, Y. Watanabe, Pharmacokinetics and pharmacodynamics of human chorionic gonadotropin (hCG) after rectal administration of hollow-type suppositories containing hCG, *Biol. Pharm. Bull.* 25 (2002) 678–681.
- [691] T. Murakami, K. Ohoku, R. Yumoto, M. Yoshii, M. Une, T. Kuramoto, T. Hoshita, N. Yata, Enhancing effect of 5 α -cyprinol sulfate on mucosal membrane permeability to sodium ampicillin in rats, *Eur. J. Pharm. Biopharm.* 49 (2000) 111–117.